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**Report prepared for the WHO annual consultation
on the composition of influenza vaccine for the
Northern Hemisphere 2016-2017**

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Northern Hemisphere Vaccine Recommendation Meeting for 2016-17 influenza season.

WHO CC, London: Final Report for WHO meeting: 22-24 February 2016.

Influenza Activity in the WHO European Region, October 2015 – February 2016

Weekly reporting on influenza activity started in week 40/2015. The course of the 2015-2016 season, to week 04/2016, has seen levels of transmission increasing slowly following week 51/2014 (**Figure E1-A**) in the countries of southern, western and northern Europe and west Asia with no particular pattern in the spread of influenza across the Region (**Figure E2**).

The proportions of respiratory specimens from patients presenting with influenza-like illness (ILI) or acute respiratory infection (ARI) symptoms that have tested positive for influenza virus have been increasing with the overall positivity rate in specimens from sentinel sources for week 04/2016 being 45%, indicating increased influenza activity in the WHO European Region as a whole. Increasing trends in ILI and ARI, with high and very high intensity of influenza activity, have been reported for a number of countries in southern, eastern and northern European and western and central Asian parts of the Region, while countries in the western European part of the Region have, so far, experienced low and medium intensity of influenza activity (**Figure E2**).

For the whole WHO European Region, over 33,000 influenza detections have been reported to the European Surveillance System (TESSy) - with type A detections (89%) predominating over type B (11%) (**Table E1**). Within type A, the A(H1N1)pdm09 subtype (89%) has predominated over A(H3N2) viruses (11%) (**Table E1**) and this pattern is evident for both sentinel and non-sentinel influenza virus detections (**Figure E1-B & C**). For type B viruses, of those for which lineage-determination was performed, 85% were of the Victoria lineage (**Table E1**). All countries in the Region have reported influenza type A virus detections to outnumber those of influenza B; however, the proportions of influenza B detections have been considerably greater in countries in the western European part of the Region compared to those in other sub-regions, yielding a type A:B ratio of 60%:40%. While influenza A(H1N1)pdm09 detections have greatly outnumbered A(H3N2) detections across the Region, it is notable that countries in western Asia which have accounted for 1527/2351 (65%) of A(H3N2) detections (**Table E1 and Figures E2 & E3**).

Since week 40/2014, seven countries (Finland, France, Ireland, Romania, Spain, Sweden and the United Kingdom) have reported a total of 1219 laboratory-confirmed hospitalized influenza cases (727 cases from ICUs and 492 from other hospital wards) (**Figure E4-A**). Among cases admitted to ICUs influenza type A detections (95%) have predominated over type B (5%). Within type A, the A(H1N1)pdm09 subtype (95%) has predominated over A(H3N2) viruses (5%). However, a greater proportion of influenza B detections (28%) was observed in all other hospital wards, with A(H1N1)pdm09 being the dominant subtype (97%) among influenza A detections. For hospitalized patients where age was reported, the group aged 15-64 years accounted for the greatest number of cases (**Figure E4-B**). A(H1N1)pdm09 viruses were dominant in all age groups.

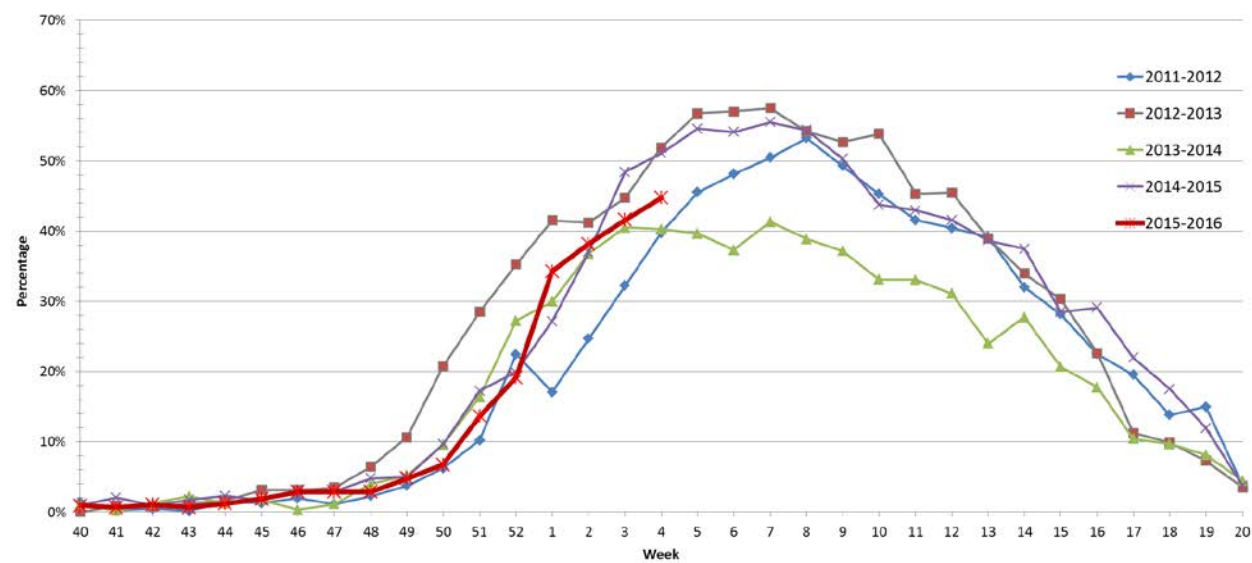
Since week 52/2015, several European countries with sentinel surveillance systems for severe acute respiratory infection (SARI) have reported increasing numbers of cases associated with A(H1N1)pdm09 infection. These cases occur mainly in the groups aged 15–29 and 30–64 (**Figure E4-C**).

Since week 40/2014, TESSy has received reports indicating that only two A(H1N1)pdm09 influenza viruses out of 466 A(H1N1)pdm09, 55 A(H3N2) and 47 type B viruses assessed for neuraminidase inhibitor susceptibility showed highly reduced inhibition (HRI) by oseltamivir associated with NA-H275Y amino acid substitution; all others showed molecular and/or phenotypic evidence for normal inhibition (NI) by neuraminidase inhibitors (NAI).

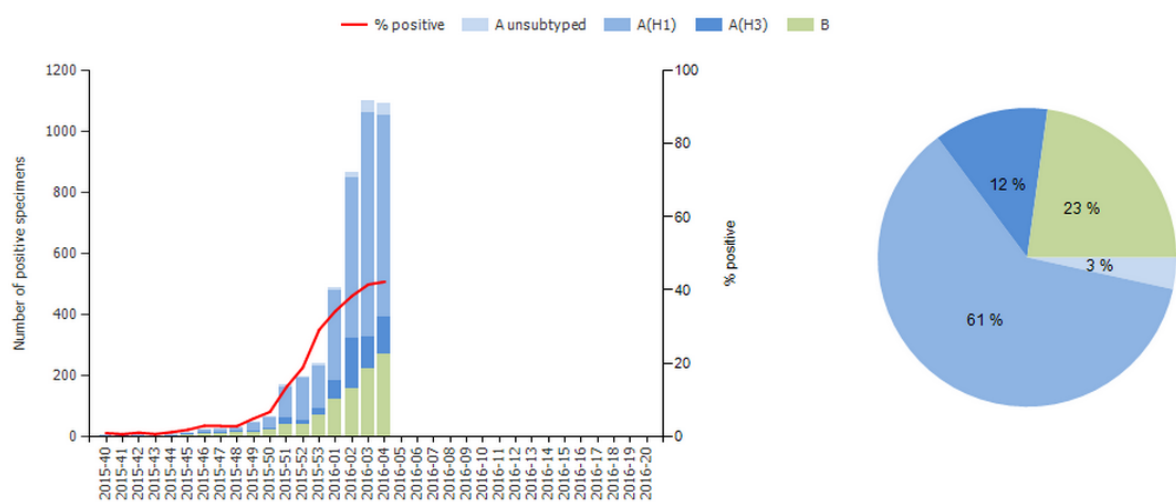
Although a number of countries have reported increased severity of disease associated with influenza infection and some deaths, as of week 04/2016 there has been no indication of an increase in excess mortality due to influenza reported by the European project for monitoring excess mortality for public health action (EuroMOMO – <http://www.euromomo.eu>). Data for this reporting were submitted by 15 countries: Belgium, Denmark, Estonia, Finland, France, Greece, Hungary, Ireland, the Netherlands, Norway, Portugal, Spain, Switzerland, Sweden and the UK.

Figure E1. Influenza Activity in the WHO European Region 2015-2016 (weeks 40/2015-04/2015)

A. Percentage of sentinel ILI/ARI specimens testing positive for influenza by season



B. Sentinel Influenza Detections



C. Non-sentinel Influenza Detections

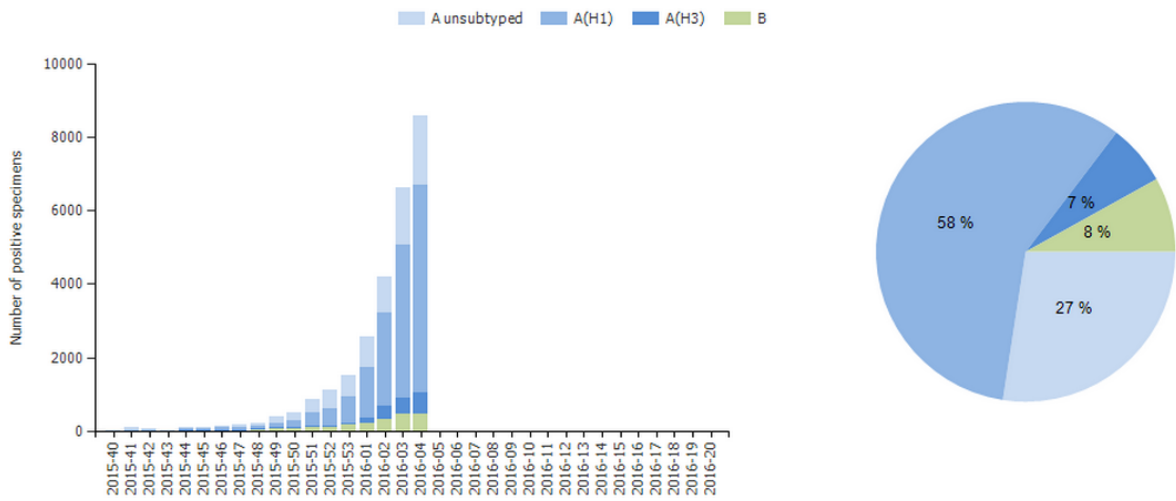


Figure E2. Development of the 2015-2016 Influenza Season in the WHO European Region

Intensity of Influenza Activity and Dominant Virus (sub)type

Year	2015															2016			
Country	Week	40	41	42	43	44	45	46	47	48	49	50	51	52	53	1	2	3	4
Iceland		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	AH1	-
Ireland		-	-	-	-	-	-	-	-	-	-	-	-	-	-	B/AH1	B/AH1	AH1	AH1
Portugal		-	-	-	-	-	-	-	-	-	-	-	AH1	AH1	AH1	AH1	AH1	AH1	AH1
Spain		-	-	-	-	-	-	-	-	-	-	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1
United Kingdom (Northern Ireland)		-	B	AH3	A	A	-	AH1	AH1	A	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1
United Kingdom (Wales)		-	-	-	-	-	-	AH1	-	-	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1
United Kingdom (Scotland)		-	-	-	-	A	A	A	A	A	A	A	A	A	A	A	A	A	A
United Kingdom (England)		-	-	-	-	-	-	-	-	-	-	A	-	A	A	A	A	A	A
France		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Belgium		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	B	B	B/AH1
Netherlands		-	-	-	-	-	-	-	-	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	A
Luxembourg		-	-	-	-	-	-	-	-	-	-	B	-	-	-	B	-	-	B
Switzerland		-	A	-	A	A/B	-	A	A	A/B	A	A/B	A/B	A/B	A/B	A	B/AH1	B/AH1	B/AH1
Germany		-	-	-	-	-	-	-	-	-	-	AH1	-	-	-	AH1	AH1	AH1	AH1
Denmark		-	-	-	-	-	-	-	-	-	-	-	-	-	AH1	AH1	AH1	AH1	AH1
Norway		-	-	-	-	A	A	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1
Italy		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Austria		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Malta		-	-	-	-	-	-	-	-	-	-	-	-	AH1	AH1	AH1	AH1	AH1	AH1
Slovenia		-	-	-	-	-	-	-	-	-	-	A	A	AH3	AH3	AH3	AH3	AH3	AH3
Sweden		-	-	-	-	-	-	-	-	A	A	A	A	A	A	A	A	AH1	AH1
Croatia		-	-	-	-	-	-	-	-	-	-	-	AH1	-	-	-	AH1	AH1	AH1
Czech Republic		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	AH1
Bosnia and Herzegovina		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	AH1	-
Slovakia		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Albania		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	AH1	AH1	-
Hungary		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Poland		-	-	AH1	-	A	A	-	A	A	A	-	A	A	A	A	A	A	AH1
Montenegro		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Serbia		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
The former Yugoslav Republic of Macedonia		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Greece		-	-	-	-	-	-	-	-	-	-	-	-	-	-	AH1	AH1	AH1	AH1
Lithuania		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	AH1	AH1
Bulgaria		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	AH1	AH1
Latvia		-	-	-	-	-	-	-	-	-	-	-	-	A	AH1	AH1	A	A	AH1
Romania		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	AH1	AH1
Estonia		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	B/AH1	-
Finland		-	-	-	-	-	-	-	AH1H3	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1
Belarus		-	-	-	-	-	-	-	-	-	-	-	-	AH1	AH1	AH1	AH1	AH1	AH1
Republic of Moldova		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	AH1	AH1
Ukraine		-	-	-	-	-	-	-	-	-	-	-	B/AH1	AH1	AH1	AH1	AH1	AH1	AH1
Cyprus		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Israel		-	-	-	-	-	-	-	-	-	-	-	B/AH1	B/AH1	B/AH1	B/AH1	B/AH1	B/AH1	B/AH1
Turkey		-	-	-	-	-	-	-	-	-	AH1	AH1H3	AH1	AH1H3	AH1	AH1H3	AH1	AH1H3	AH1H3
Georgia		-	-	-	-	-	-	-	-	AH1	-	-	-	-	-	AH1	AH1	AH1	AH1
Armenia		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Azerbaijan		-	-	-	-	-	-	-	-	-	-	-	-	-	-	AH3	-	-	-
Uzbekistan		-	-	-	-	-	-	-	-	-	-	AH1	-	-	-	AH1	-	AH1	AH1
Kazakhstan		-	-	-	-	-	-	-	-	-	-	AH1	AH1	AH1	AH1	AH1	AH1	-	AH1
Russian Federation		-	-	-	-	-	-	-	-	-	-	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1
Tajikistan		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Kyrgyzstan		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-

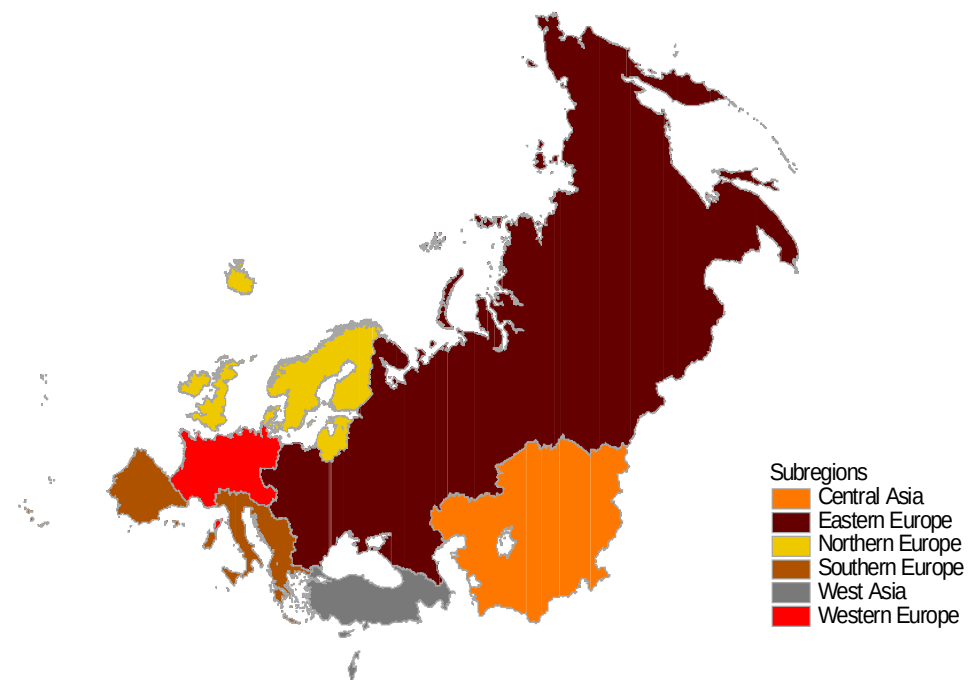
WEST

EAST



* Baseline influenza activity is the level at which clinical influenza activity remains throughout the summer and most of the winter.

Table E1. Distribution of Influenza Detections in the WHO European Region¹: 2015-2016 Season (weeks 40/2015-04/2016)

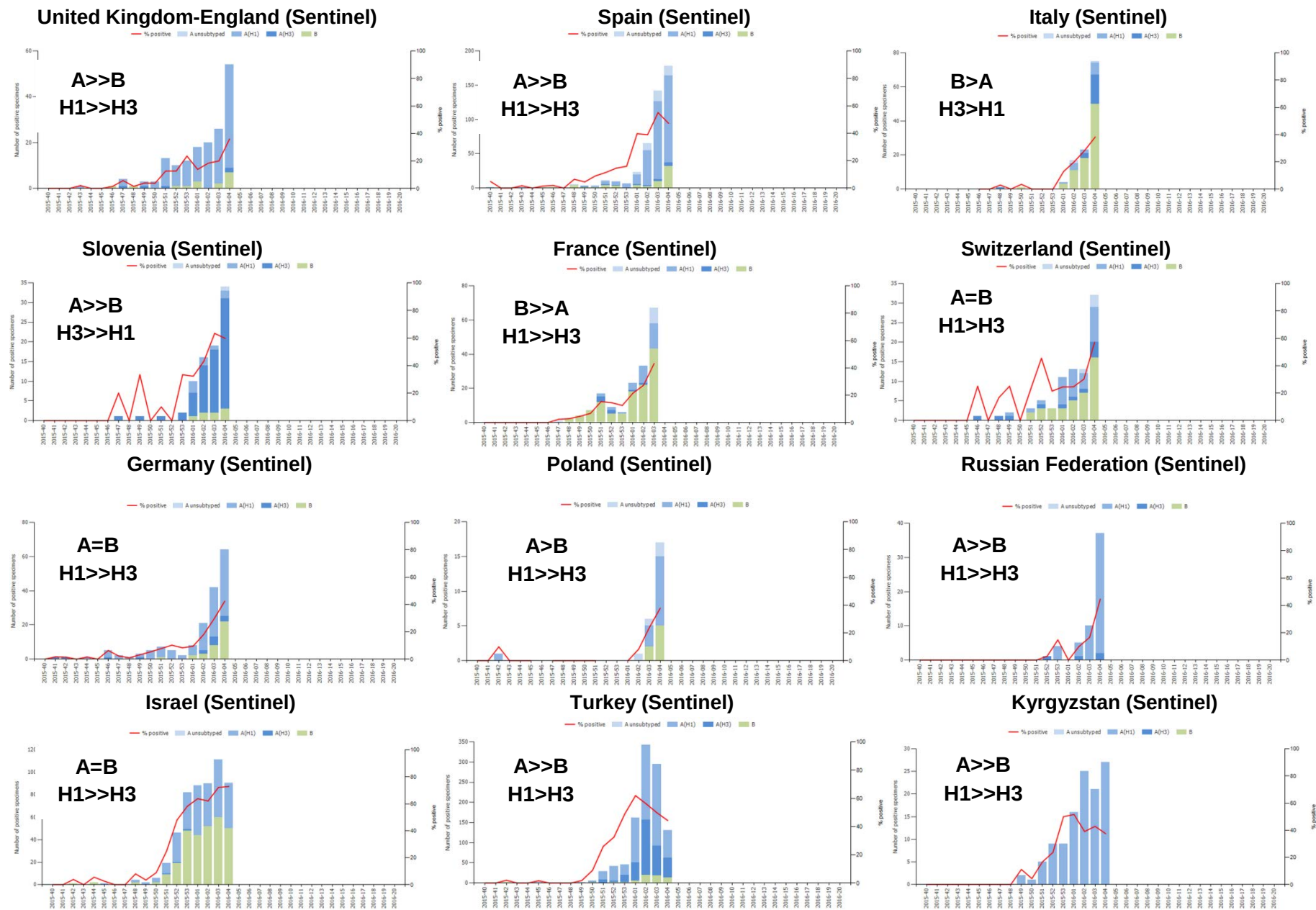


Influenza Detections*	Southern Europe		Western Europe		Eastern Europe		Northern Europe		West Asia		Central Asia		WHO European Region	
Influenza A	3355	91%	1789	60%	8104	98%	10001	89%	5926	89%	455	100%	29630	89%
Influenza A subtyped	2308		966		7464		4117		5899		352		21106	
<i>A (H1N1)pdm09</i>	2069	90%	836	87%	7207	97%	3959	96%	4372	74%	312	89%	18755	89%
<i>A (H3N2)</i>	239	10%	130	13%	257	3%	158	4%	1527	26%	40	11%	2351	11%
Influenza B	319	9%	1177	40%	135	2%	1232	11%	764	11%	0	0%	3627	11%
B lineage determined	67		232		22		185		0		0		506	
<i>B/Yamagata lineage</i>	6	9%	14	6%	6	27%	49	26%	0		0		75	15%
<i>B/Victoria lineage</i>	61	91%	218	94%	16	73%	136	74%	0		0		431	85%
Total	3674		2966		8239		11233		6690		455		33257	

*combined ILI, ARI sentinel and non-sentinel respiratory specimens positive for influenza

¹ WHO European Region Member States grouped into sub-regions according to United Nations geographic composition. Available from: <http://unstats.un.org/UNSD/METHODS/M49/M49REGIN.HTM>

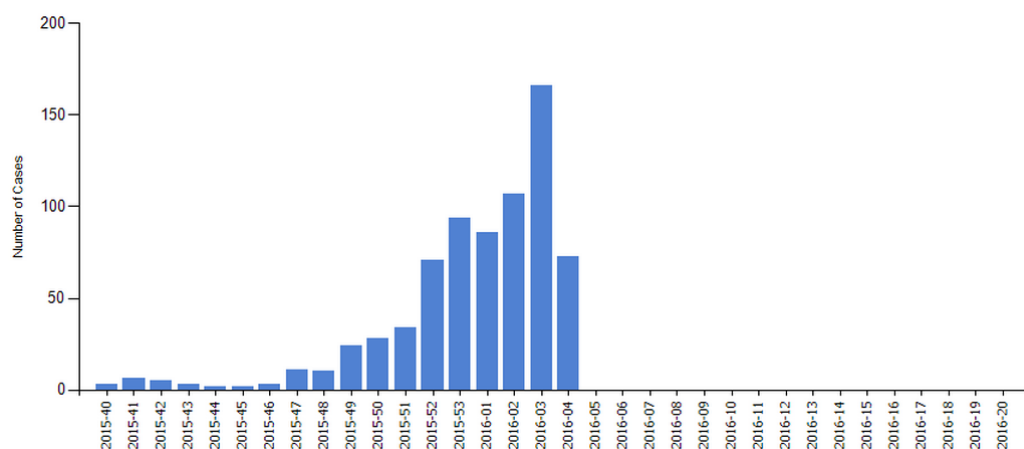
Figure E3. Patterns of Influenza Type and Subtype Circulation in the WHO European Region



= represents similar ratio; > represents a ratio of at least 1.5:1; >> represents a ratio of ≥5

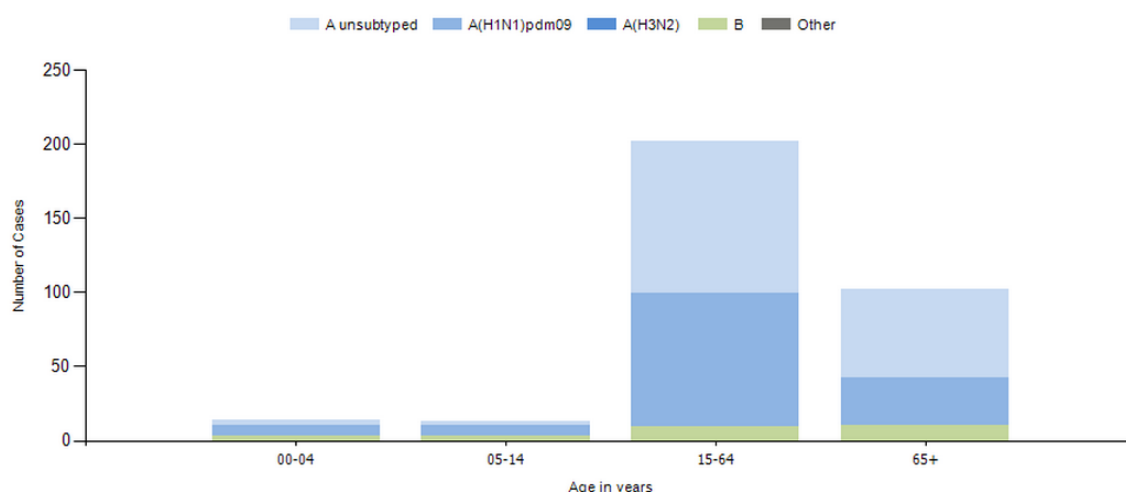
Figure E4. Hospitalized influenza cases

A. Laboratory-confirmed cases admitted to ICU by week*



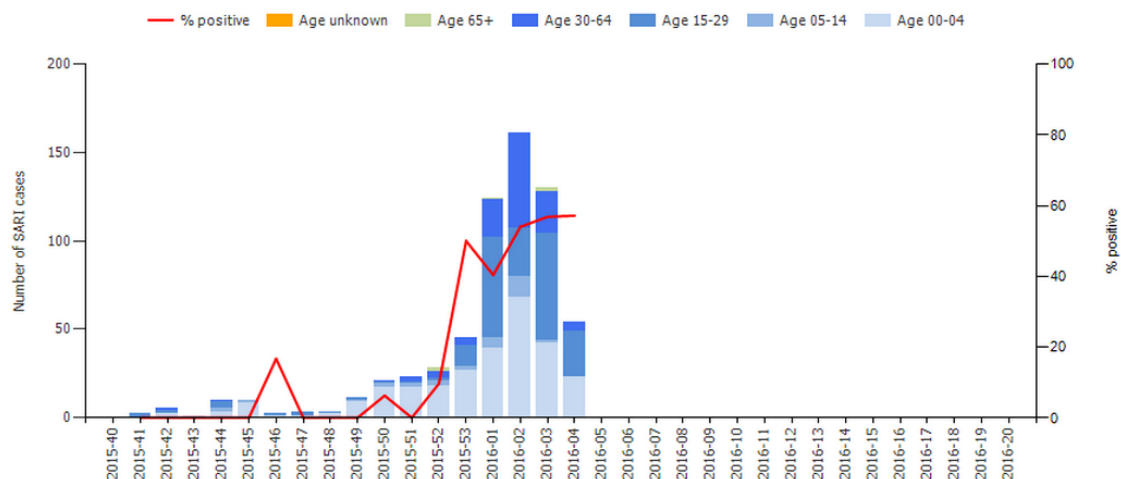
*Finland, France, Ireland, Romania, Spain, Sweden, and the United Kingdom

B. Distribution of virus (sub)type by age group in laboratory-confirmed hospitalized cases (ICU)*



*Finland, France, Ireland, Romania, Spain, Sweden, and the United Kingdom

C. SARI cases by age group (years) and percentage of specimens positive for influenza at sentinel hospitals*



* Data reported for Armenia

Influenza specimens received at the Francis Crick Institute, Mill Hill Laboratory.

Specimens, viruses or clinical samples, with collection dates after 2015-08-31 have been received from 45 countries in Europe, Africa, the Middle East, the Indian Ocean, Central Asia and the Far East. Nearly 90% of the samples received were type A viruses, with A(H1N1)pdm09 viruses outnumbering A(H3N2) viruses at a ratio of over 4:1 (**Table 1**). Of the type B viruses, viruses of the B/Victoria lineage predominated over those of the B/Yamagata lineage at a ratio of over 3:1. These proportions are generally similar to those reported within the WHO European Region (**Figures E1B & E1C, Table E1**). **Table 2** shows the isolation rates from specimens received with collection dates after 2015-08-31. Both viruses and clinical samples were recovered efficiently. For H3N2 viruses, efficient recovery was obtained as assessed by detection of virus neuraminidase activity using the MUNANA sialidase assay, but a sizable proportion (> 50%) of these could not be assessed by HI assay due to HA titres being low (<4 HA units in the presence of 20nM oseltamivir). **Table 3** summarises the reactivity in HI assays of A(H1N1)pdm09, A(H3N2) and viruses of the B/Victoria and B/Yamagata lineages with post-infection ferret antisera raised against vaccine viruses and reference viruses; for A(H3N2) viruses, the reactivity of test viruses in plaque reduction neutralisation assays (PNRA) is also summarised. Numbers and percentages of viruses showing ≤ 2 -, 4- and ≥ 8 -fold reductions in titres relative to the titres of the antisera for their respective homologous reference/vaccine viruses are shown. The antiserum raised against the cell culture-propagated A/Switzerland/9715293/2013 had a low homologous titre and relative titres for the test viruses could only be estimated as ≤ 2 -fold or ≥ 4 -fold reduced compared with the titre of this antiserum for the homologous virus.

Table 1. Summary of clinical samples and virus isolates received, with collection dates since 2015-08-31, by country*

MONTH	TOTAL RECEIVED	A		H1N1pdm09		H3N2		B		B Victoria lineage		B Yamagata lineage	
Country		Number received	Number propagated	Number received	Number propagated ¹	Number received	Number propagated ²	Number received	Number propagated	Number received	Number propagated ¹	Number received	Number propagated ¹
2015													
SEPTEMBER													
Cameroon	9			3	in process	3	in process	3	in process				
Egypt	6			6	in process								
Ghana	6			2	in process	3	in process					1	in process
Iceland	1			1	1								
Iran	1					1	1	0					
Madagascar	6			6	in process								
Oman	4					4	in process						
Poland	1					1	in process						
Saudi Arabia	27	6	0	12	9	9	2	7					
Spain	1			1	in process								
Sweden	1			1	in process								
United Kingdom	3					3	3	0					
Zambia	3								3	in process			
OCTOBER													
Austria	1			1	1								
Belgium	1			1	1								
Cameroon	10			2	in process	3	in process		4	in process		1	in process
Egypt	0												
France	2			2	2								
Germany	2			1	1							1	1
Ghana	2			1	in process							1	1
Iran	5			3	3	1	0	1				1	1
Italy	2			1	in process							1	in process
Jordan	2			2	in process								
Latvia	1			1	1								
Madagascar	5			5	in process								
Netherlands	1			1	1								
Norway	11			8	5	2	2	0		1	1		
Oman	9			3	in process	6	in process						
Portugal	1			1	1								
Romania	1			1	1								
Slovenia	2	1	0	1	1								
Spain	2			2	in process								
Sweden	1					1	in process						
Tunisia	2			1	in process	1	in process						
Turkey	2			1	in process	1	in process						
United Kingdom	8			7	7					1	1		
Zambia	6					3	in process		3	in process			
NOVEMBER													
Austria	9			5	5	4	4	0					
Belgium	11			2	in process	1	0	1		6	in process	2	2
Cameroon	11	2	in process	4	in process	3	in process					2	in process
Croatia	1			1	in process								
Denmark	1			1	1								
Egypt	4			4	in process								
Estonia	3			3	in process								
Finland	4			2	2	2	1	1					
France	5			1	in process	1	1	0		3	in process		
Georgia	2			2	1								
Germany	14			11	11	2	0	2		1	1		
Ghana	4			2	1	2	0	1					
Iran	8			6	3							2	2
Italy	2					1	in process		1	in process			
Jordan	4			3	in process	1	in process						
Kazakhstan	8			6	in process	2	0	1					
Madagascar	8			8	in process								
Netherlands	7			5	in process								
Norway	11			7	7	1	1	0		2	in process	2	2
Oman	13			7	in process				6	in process			
Poland	1	1	in process										
Portugal	8			8	7								
Russia	2			2	2								
Slovenia	3					2	0	2	1	in process			
Spain	4	2	0	2	in process								
Sweden	11			7	7	4	4	0					
Switzerland	3					3	in process						
Tunisia	2			2	in process								
Ukraine	5			4	in process					1	1		
United Kingdom	7			7	5								
Zambia	7	2	in process			5	in process						

MONTH	TOTAL RECEIVED	A		H1N1pdm09		H3N2			B		B Victoria lineage		B Yamagata lineage	
Country		Number received	Number propagated	Number received	Number propagated ¹	Number received	Number propagated ²		Number received	Number propagated	Number received	Number propagated ¹	Number received	Number propagated ¹
DECEMBER														
Algeria	8			8	in process									
Austria	3					3	2	1						
Cameroon	4			2	2	1	0	1			1	0		
Croatia	5			4	in process	1	in process							
Czech Republic	2			2	2									
Denmark	7			7	7									
Egypt	8			8	in process									
Estonia	6			1	0				1	0	4	in process		
Finland	8			7	6	1	1	0						
France	24			7	in process	2	2	0			15	in process		
Georgia	6			6	in process									
Germany	21			17	17	3	3	0			1	1		
Ghana	5			2	2	2	0	2			1	in process		
Greece	6			5	in process	1	0	1						
Hong Kong	1					1	0	1						
Iceland	3			1	1	1	0	1			1	1		
Ireland	4			2	2	1	0	1			1	1		
Israel	24			15	in process						9	9		
Italy	8			3	in process	5	2	3						
Jordan	7			7	in process									
Kazakhstan	18			18	in process									
Latvia	3			3	3									
Madagascar	7			7	in process									
Moldova	2			2	in process									
Netherlands	5			5	in process									
Norway	5			3	3	2	1	1						
Oman	1			1	in process									
Poland	14	10	in process	3	3						1	0		
Portugal	14			10	7	3	2	1					1	1
Romania	1					1	0	1						
Russia	29			25	24	4	in process							
Slovenia	5					5	1	3						
Spain	14	1	in process	7	6	2	0	1			4	4		
Switzerland	10			3	in process	1	0	1			4	in process	2	2
Turkey	70	1	in process	55	in process	13	in process		1	in process				
Ukraine	44	3	in process	40	in process						1	1		
United Kingdom	3			3	3									
Zambia	3	3	in process											
2016														
JANUARY														
Algeria	2			1	1	1	0	0						
Armenia	40			40	in process									
Croatia	2										2	in process		
Czech Republic	3			3	3									
Egypt	3			3	in process									
Estonia	3			2	0						1	1		
Georgia	2			2	1									
Germany	10			7	7						3	3		
Greece	25			25	in process									
Hong Kong	9			3	3	1	0	1			3	3	2	2
Hungary	7			4	in process						3	in process		
Iceland	6			5	5								1	1
Ireland	10			9	9						1	1		
Italy	1			1	1									
Jordan	27			26	in process	1	in process							
Moldova	18			17	in process				1	in process				
Netherlands	3			3	in process									
Portugal	6			6	6									
Romania	8			7	7	1	0	1						
Russia	2			2	in process									
Slovenia	8			3	3	3	0	3	2	0				
Spain	2			2	2									
Switzerland	2			1	1	1	1	0						
Tunisia	3					3	in process							
Turkey	66			51	in process	11	in process		4	in process				
Ukraine	8	1	0	7	3									
45 Countries	1015	33	0	709	227	151	34	40	30	0	72	30	20	15
					69.9%		14.9%				7.1%		2.0%	
		88.0%							12.0%					

1. Propagated to sufficient titre to perform HI assay (the totalled number does not include any from batches that are in process)

2. Propagated to sufficient titre to perform HI assay in presence of 20nM oseltamivir (the totalled number does not include any from batches that are in process)

Numbers in red indicate viruses recovered but with insufficient HA titre to permit HI assay

* As of 2015-02-11

Table 2. Isolation rates for specimens collected after 2015-08-31

	Clinical sample			Virus isolate		
	Number processed	Number propagated ¹	Percent recovered	Number processed	Number propagated ¹	Percent recovered
H1	103	55	53%	260	256	98%
H3	34	6 ² 22 ³	82% (18%) ⁴	80	40 ² 37 ³	96% (50%) ⁴
B Victoria	8	5	63%	48	48	100%
B Yamagata	4	4	100%	11	11	100%

1. Propagated to sufficient HA titre to perform HI assay

2. Propagated to sufficient HA titre in the presence of 20nM oseltamivir to allow HI assay

3. Viruses recovered (NA activity by MUNANA-based assay) but with insufficient HA titre to allow HI assay

4. The percentages for viruses that were assayed by HI are shown in parentheses

Table 3. Summary of isolates with collection dates after 2015-08-31 showing reduction in reactivity with antisera raised against reference viruses in HI & plaque-reduction microneutralisation assays

Virus type/subtype	Number tested	≤2-fold ^a	4-fold ^a	≥8-fold ^a
H1 ¹	307	289 (94.1%)	17 (5.5%)	1 (0.3%)
H3 ^{2a}	46	0	6 (13.0%)	40 (87.0%)
H3 ^{2b}	46	33 (69.6%)	14 (30.4%) ⁶	
H3 ^{2c}	46	27 (58.7%)	14 (30.4%)	5 (10.9%)
H3 ^{2d}	46	15 (32.6%)	10 (21.7%)	21 (45.7%)
H3 ^{3a}	47	30 (63.8%)	14 (29.8%)	3 (6.4%)
H3 ^{3b}	47	9 (19.1%)	9 (19.1%)	29 (61.7%)
H3 ^{3c}	47	1 (2.1%)	0	46 (97.9%)
B/Victoria ⁴	51	43 (84.3%)	8 (15.7%)	0
B/Yamagata ⁵	15	13 (86.7%)	2 (13.3%)	0

a. Reduction in titre, compared to the homologous titre with post-infection ferret antisera indicated below:

1. Compared to homologous titres for ferret antisera raised against vaccine virus A/California/7/2009

2a. Compared to homologous titres for ferret antisera raised against egg-propagated A/Switzerland/9715293/2013 (genetic subgroup 3C.3a)

2b. Compared to homologous titres for ferret antisera raised against cell-propagated A/Switzerland/9715293/2013 (genetic subgroup 3C.3a)

2c. Compared to homologous titres for ferret antisera raised against cell-propagated A/Hong Kong/5738/2014 (genetic subgroup 3C.2a)

2d. Compared to homologous titres for ferret antisera raised against egg-propagated A/Hong Kong/4801/2015 (genetic subgroup 3C.2a)

3a. Compared to homologous titres for ferret antisera raised against cell-propagated A/Switzerland/9715293/2013 (genetic subgroup 3C.3a) in plaque-reduction neutralisation assays

3b. Compared to homologous titres for ferret antisera raised against egg-propagated A/Hong Kong/4801/2014 (genetic subgroup 3C.2a [NYK]) in plaque-reduction neutralisation assays

3c. Compared to homologous titres for ferret antisera raised against egg-propagated A/Netherlands/525/2014 (genetic subgroup 3C.3b) in plaque-reduction neutralisation assays

4. Compared to homologous titres for ferret antisera raised against at least one of three tissue-culture grown B/Brisbane/60/2008-like equivalents (B/Paris/1762/2008, B/Hong Kong/514/2009 & B/Odessa/3886/2010)

5. Compared to homologous titres for ferret antisera raised against vaccine virus B/Phuket/3073/2013

6. Unable to differentiate fold drop in titre for most viruses as it falls outside the dynamic range of the antiserum

Influenza A(H1N1)pdm09 virus analysis.

Influenza A(H1N1)pdm09 viruses were received from NICs in 44 countries or administrative regions in four WHO Regions: African (5: Algeria, Cameroon, Egypt, Ghana and Madagascar), Eastern Mediterranean (5: Iran, Jordan, Oman, Saudi Arabia and Tunisia), European (33 countries) and Western Pacific (1: Hong Kong SAR). **Table 4** summarises the antigenic (HI) results for viruses, recovered from clinical samples with collection dates after 2015-08-31, that were propagated successfully at the Francis Crick Institute.

The vast majority of H1N1 viruses collected after 2015-08-31 were antigenically similar to the vaccine virus (**Table 4**). **Tables 5-1 to 5-13** show the results of HI assays of H1N1 viruses carried out since our September 2015 report. Test viruses in each table are sorted by date of collection and viruses with collection dates after 2015-08-31 are shown below red lines in **Tables 5-4 to 5-13**. A summary of the results for viruses with collection dates after 2015-08-31 is shown in **Table 5-14**. Three different antisera raised against the vaccine virus A/California/7/2009 were used: in **Tables 5-1 to 5-3** antiserum F05/14 was used, in **Tables 5-4 to 5-9** antiserum F01/15 was used, and in **Tables 5-10 to 5-13** an antiserum kindly provided by NIBSC was used (NIB F05/14).

Over 99% of test viruses were recognised by the antisera raised against the A/California/7/2009 vaccine virus at titres within 4-fold of the titres for the homologous virus, and over 90% within 2-fold of the homologous titre. The antiserum used for the majority of samples, NIB F05/14, recognised 99% of the test viruses collected since 2015-08-31 at a titre within 2-fold of the homologous titre. With the exception of the antiserum raised against A/Christchurch/16/2010 the other antisera also recognised the test viruses very well, with 98 to 100% of viruses being recognised at titres within 4-fold of the homologous titres for the antisera, and 81 to 99% at titres within 2-fold of the homologous titres (**Table 5-14**). The antiserum raised against A/Christchurch/16/2010 recognised only half (51%) of the test viruses collected since 2015-08-31 at a titre within 4-fold of the titre for the homologous virus and only 6% at a titre within 2-fold of the homologous titre. Of the test viruses for which the genetic group or sub-group were known (see below) 19% fell into genetic group 6B, 74% in genetic subgroup 6B.1 and 7% in genetic subgroup 6B.2.

To complement the antigenic analysis by HI, virus neutralisation tests were carried out by a plaque reduction neutralisation assay (PRNA) for a subset of test viruses. The results of virus neutralisation assays performed using MDCK cells are shown in **Tables 6-1 and 6-2**. The genetic groups and noteworthy sequence features in the HA are marked. Absolute values of plaque reduction caused by a particular antiserum are given, extrapolated from digitally scanned 96-well plates, and these values have also been rounded to the nearest multiple in a 2-fold dilution series as used in HI assay. A total of 35 test viruses were analysed, all with collection dates after 2015-08-31. All test viruses were recognised well by the panel of antisera. Notably, only one virus (A/Norway/2774/2015) was recognised at a lower titre than the homologous titre by the antiserum raised against A/California/7/2009. The antiserum raised against the egg-propagated cultivar of A/Slovenia/2903/2015, a reference virus in genetic subgroup 6B.1, had a high neutralising titre for the homologous

virus and recognised the cell culture-propagated cultivar of A/Slovenia/2903/2015 at a titre of 2466 (**Table 6-1**) compared with the titre for the egg-propagated cultivar of 10240 (**Table 6-2**). The test viruses were all recognised at titres of 5120 or above by this antiserum. The antiserum raised against A/Slovenia/2903/2015 recognised A/California/7/2009 and A/St. Petersburg/27/12011 at titres within 4-fold of the homologous titre, but both A/Bayern/69/2009 and A/South Africa/3626/2013 were recognised less well. A/Bayern/69/2009 carries an amino acid substitution G155E in HA1 and A/South Africa/3626/2013 is polymorphic at residue 127 of HA1 (D127X). A/Bayern/69/2009 is recognised only poorly by all the heterologous antisera. It is notable that A/Norway/2774/2015, recognised generally less well than the other test viruses by the panel of antisera, carries the substitutions R113K, D122E, V152T, K163Q and V173I in HA1 and it falls in the genetic subgroup 6B.2. Overall, 26 viruses (74%) analysed in PRNA fell into genetic subgroup 6B.1, eight (23%) fell into genetic group 6B but only one (3%) fell in genetic subgroup 6B.2.

Figure 1 shows a phylogenetic tree for the HA genes of representative H1N1 viruses. Over the last five years the HA genes have evolved and eight genetic groups have been designated, with A/California/7/2009 representing group 1, and viruses in group 6 have formed clusters designated groups 6A, 6B and 6C. Viruses collected after 2015-08-31 fell into genetic group 6B and in two emerging subgroups, 6B.1 and 6B.2 (**Figure 1**). Subgroup 6B and subgroups 6B.1 and 6B.2 are defined by the following amino acid substitutions in **HA1** and **HA2**:

- **P83S, D97N, S185T, S203T K163Q, A256T, K283E and I321V** in **HA1**, and **E47K, S124N** and **E172K** in **HA2** compared with A/California/7/2009 define **genetic group 6B**, e.g. A/South Africa/3626/2013.
- **S84N, S162N (+CHO)** and **I216T** in **HA1** define **subgroup 6B.1** within group 6B, e.g. A/Slovenia/2903/2015
- **V152T** and **V173I** in **HA1** and **E164G** and **D174E** in **HA2** define **subgroup 6B.2** within group 6B, e.g. A/England/377/2015.

Other minor groupings are detected in subgroup 6B. These include a group represented by viruses from Lisbon with the amino acid substitutions **A48P, S69P, D274N** in **HA1** and **L38Q** in **HA2** and another group can be defined by the substitution **K311T** in **HA1** with representatives from Georgia and Madagascar.

An amino acid sequence alignment of the HAs for representative viruses is shown in **Figure 2**; the vaccine virus is shown in red, reference viruses are coloured blue, glycosylation motifs are highlighted, the genetic group/subgroup is shown in green and the HA1/HA2 proteolytic-processing site is highlighted.

Figure 3 illustrates the location of amino acid residues on an H1-HA structure defining genetic subgroup 6B and the two major subgroups 6B.1 and 6B.2.

Figure 4 shows a phylogenetic tree for NA genes of representative H1N1 viruses. The HA and NA phylogenetic trees are generally congruent.

Figure 5 shows a NA amino acid alignment for a selected group of these viruses, annotated as **Figure 2**.

Antiviral resistance.

Phenotypic testing to assess susceptibility to oseltamivir and zanamivir was performed on 222 viruses with collection dates after 2015-08-31. All viruses tested showed normal inhibition (NI) to both neuraminidase inhibitors (**Table 15**).

Conclusion.

Although new genetic subgroups have emerged in recent months, the A(H1N1)pdm09 viruses received were antigenically similar to the vaccine virus A/California/7/2009 and retain susceptibility to oseltamivir and zanamivir.

Table 4. Summary of influenza A(H1N1)pdm09 viruses analysed, collected after 2015-08-31

MONTH	H1N1pdm09				
	Country	Number received ¹	Number propagated ²	Fold reduction in HI titre	
				≤2 fold ³	≥8 fold ³
SEPTEMBER					
Iceland	1	1	1		
Madagascar	6	in process	4		
Saudi Arabia	12	9	6	3	
OCTOBER					
Austria	1	1	1		
Belgium	1	1	1		
France	2	2	2		
Germany	1	1	1	1	
Iran	3	3	3		
Latvia	1	1	1		
Madagascar	5	in process	1		
Netherlands	1	1	1	1	
Norway	8	5	5		
Portugal	1	1	1		
Romania	1	1	1		
Slovenia	1	1	1	1	
United Kingdom	7	7	6	1	
NOVEMBER					
Austria	5	5	5		
Denmark	1	1	1		
Finland	2	2	2		
Georgia	2	1	1		
Germany	11	11	5	5	1
Ghana	2	1	1		
Iran	6	3	3		
Kazakhstan	6	in process	1		
Netherlands	5	in process	3		
Norway	7	7	5	2	
Portugal	8	7	7		
Russia	2	2	2		
Sweden	7	7	7		
Ukraine	4	in process	1		
United Kingdom	7	5	5		
DECEMBER					
Algeria	8	in process	3		
Cameroon	2	2	2		
Croatia	4	in process	3		
Czech Republic	2	2	2		
Denmark	7	7	7		
Finland	7	6	6		
France	7	in process	5		
Georgia	6	in process	5		
Germany	17	17	17		
Ghana	2	2	2		
Iceland	1	1	1		
Ireland	2	2	2		
Israel	15	in process	13		
Italy	3	in process	2		
Kazakhstan	18	in process	2		
Latvia	3	3	3		
Madagascar	7	in process	1		
Moldova	2	in process	1		
Norway	3	3	3		
Poland	3	3	3		
Portugal	10	7	7		
Russia	25	24	24		
Spain	7	6	6		
Switzerland	3	in process	1		
Turkey	55	in process	1		
Ukraine	40	in process	20	3	
United Kingdom	3	3	3		
JANUARY					
Algeria	1	1	1		
Armenia	40	in process	1		
Czech Republic	3	3	3		
Georgia	2	1	1		
Germany	7	7	7		
Greece	25	in process	2		
Hong Kong	3	3	3	1	
Iceland	5	5	5		
Ireland	9	9	9		
Italy	1	1	1		
Portugal	6	6	6		
Romania	7	7	7		
Russia	2	in process	1		
Slovenia	3	3	3		
Spain	2	2	2		
Switzerland	1	1	1		
Turkey	51	in process	6		
Ukraine	7	3	3		
40 Countries					
	564	227	289	17	1
			94.1%	5.6%	0.3%

1. Numbers shown in red indicate that not all of the specimens received had been propagated at the time this report was prepared

2. Propagated to sufficient titre to perform HI assays

3. Reduction in HI titre with ferret antiserum raised against A/California/7/2009 (egg grown vaccine virus)

Table 5-1. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2015-09-09)

Viruses	Other information	Passage history	Collection date	Passage history	Haemagglutination inhibition titre								
					Post-infection ferret antisera								
					A/Cal	A/Bayern	A/Lviv	A/Chch	A/Astrak	A/St. P	A/St. P	A/HK	A/Sth Afr
					7/09	69/09	N6/09	16/10	1/11	27/11	100/11	5659/12	3626/13
					Egg	MDCK	MDCK	Egg	MDCK	Egg	Egg	MDCK	Egg
	Ferret number				F05/14 ¹	F09/15 ¹	F14/13 ¹	F15/14 ¹	F22/13 ¹	F26/14 ¹	F24/11 ¹	F30/12 ¹	F3/14 ¹
	Genetic group				1	1	1	4	5	6	7	6A	6B
REFERENCE VIRUSES													
A/California/7/2009	1		2009-04-09	E1/E3	1280	1280	640	320	320	320	320	320	320
A/Bayern/69/2009	1		2009-07-01	MDCK5/MDCK1	160	640	320	80	80	40	80	80	80
A/Lviv/N6/2009	1		2009-10-27	MDCK4/SIAT1/MDCK3	320	1280	1280	320	160	160	160	320	80
A/Christchurch/16/2010	4		2010-07-12	E1/E3	1280	2560	1280	5120	2560	1280	5120	2560	1280
A/Astrakhan/1/2011	5		2011-02-28	MDCK1/MDCK5	1280	1280	640	1280	2560	1280	2560	2560	1280
A/St. Petersburg/27/2011	6		2011-02-14	E1/E3	1280	1280	1280	640	2560	1280	5120	2560	1280
A/St. Petersburg/100/2011	7		2011-03-14	E1/E3	1280	1280	1280	1280	2560	1280	5120	5120	1280
A/Hong Kong/5659/2012	6A		2012-05-21	MDCK4/MDCK2	320	320	160	320	640	320	1280	1280	640
A/South Africa/3626/2013	6B		2013-06-06	E1/E3	1280	1280	1280	640	1280	640	2560	1280	1280
TEST VIRUSES													
A/Hungary/419/2015	6B		2015-02-09	MDCK3/MDCK1	640	640	320	320	640	640	1280	1280	1280
A/Paris/1453/2015	6B		2015-03-10	MDCK1/MDCK1	1280	1280	640	640	1280	640	2560	2560	1280
A/Nord Pas de Calais/1462/2015	6B		2015-03-10	MDCK1/MDCK1	1280	1280	640	1280	1280	640	5120	2560	2560
A/Picardie/1479/2015	6B		2015-03-12	MDCK1/MDCK1	1280	1280	640	1280	2560	1280	2560	2560	2560
A/Brisbane/35/2015	6B		2015-03-12	MDCK3/MDCK1	640	640	640	640	1280	640	1280	1280	1280
A/South Australia/22/2015	6B		2015-03-12	E2/E1	1280	640	640	640	1280	640	2560	1280	1280
A/Alsace/1504/2015	6B		2015-03-14	MDCK1/MDCK1	1280	1280	640	640	1280	1280	2560	2560	1280
A/Guyane/1748/2015	6B		2015-03-14	MDCK1/MDCK1	1280	1280	640	1280	2560	1280	5120	2560	2560
A/Alsace/1532/2015	6B		2015-03-17	MDCK1/MDCK1	1280	1280	640	1280	2560	1280	5120	2560	2560
A/Guyane/1752/2015	6B		2015-03-21	MDCK1/MDCK1	1280	1280	640	640	2560	1280	2560	2560	1280
A/Istanbul/1384/2015	6B		2015-04-02	MDCK1/MDCK1	2560	1280	640	1280	2560	1280	5120	5120	2560
A/Champagne Ardenne/1666/2015	6B		2015-04-02	MDCK1/MDCK1	1280	1280	640	1280	2560	1280	5120	2560	2560
A/Guyane/1759/2015	6B		2015-04-09	MDCK1/MDCK1	1280	1280	640	1280	2560	1280	2560	2560	1280
A/Norway/2330/2015	6B		2015-04-29	MDCK1/MDCK1	1280	640	640	640	1280	640	2560	2560	1280
A/Norway/2227/2015	6B		2015-05-06	MDCK1/MDCK1	1280	1280	640	1280	2560	1280	5120	5120	2560
A/South Africa/R2439/2015	6B		2015-05-18	Ex/E1	1280	1280	1280	1280	1280	1280	2560	2560	1280
A/South Africa/R2659/2015	6B		2015-05-20	E1/E1	1280	640	1280	640	1280	640	2560	2560	1280
A/Norway/2443/2015	6B		2015-05-27	MDCK1/MDCK1	1280	1280	1280	1280	2560	1280	5120	5120	2560
A/South Africa/R2977/2015	6B		2015-06-05	E1/E1	640	640	320	640	1280	320	1280	1280	640
A/South Africa/R3001/2015	6B		2015-06-05	E2/E1	640	640	640	640	1280	640	2560	2560	1280
A/South Africa/R2991/2015	6B		2015-06-05	E2/E1	640	640	640	640	1280	640	2560	1280	1280
A/South Africa/R3086/2015	6B		2015-06-08	E2/E1	1280	1280	1280	640	1280	640	2560	2560	1280
A/South Africa/R3127/2015	6B		2015-06-08	E1/E1	1280	2560	1280	1280	2560	1280	5120	5120	2560
A/South Africa/R3664/2015	6B		2015-06-22	E2/E2	640	640	640	640	1280	640	1280	1280	640

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) ¹ < = <40

Vaccine

Sequences in phylogenetic trees

Table 5-2. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2015-09-16)

Viruses	Haemagglutination inhibition titre												
	Post-infection ferret antisera												
	Other information	Passage history	Collection date	Passage history	A/Cal	A/Bayern	A/Lviv	A/Chch	A/Astrak	A/St. P	A/St. P	A/HK	A/Sth Afr
					7/09	69/09	N6/09	16/10	1/11	27/11	100/11	5659/12	3626/13
					Egg	MDCK	MDCK	Egg	MDCK	Egg	Egg	MDCK	Egg
F05/14 ⁻¹					F09/15 ⁻¹	F14/13 ⁻¹	F15/14 ⁻¹	F22/13 ⁻¹	F26/14 ⁻¹	F24/11 ⁻¹	F30/12 ⁻¹	F3/14 ⁻¹	
Genetic group	1	1	1	4	5	6	7	6A	6B				
REFERENCE VIRUSES													
A/California/7/2009	1	2009-04-09	E1/E3	640	640	640	160	160	160	320	160	160	
A/Bayern/69/2009	1	2009-07-01	MDCK5/MDCK1	160	640	320	80	80	40	40	80	80	G155E
A/Lviv/N6/2009	1	2009-10-27	MDCK4/SIAT1/MDCK3	320	640	640	160	80	80	80	160	80	G155E>G, D222G
A/Christchurch/16/2010	4	2010-07-12	E1/E3	640	640	1280	2560	1280	640	1280	2560	640	
A/Astrakhan/1/2011	5	2011-02-28	MDCK1/MDCK5	640	320	320	320	1280	640	1280	2560	640	
A/St. Petersburg/27/2011	6	2011-02-14	E1/E3	2560	1280	1280	640	2560	1280	5120	2560	1280	
A/St. Petersburg/100/2011	7	2011-03-14	E1/E3	1280	640	640	640	1280	640	2560	2560	640	
A/Hong Kong/5659/2012	6A	2012-05-21	MDCK4/MDCK2	160	160	80	80	320	160	640	640	320	
A/South Africa/3626/2013	6B	2013-06-06	E1/E3	320	320	320	320	640	320	640	640	640	
TEST VIRUSES													
A/Lorraine/1590/2015		unknown	MDCK1/MDCK1	1280	640	640	640	1280	640	2560	2560	1280	
A/Oman/555/2015		2015-01-03	SIAT1	640	640	640	640	1280	640	2560	1280	640	
A/Mozambique/IR418/2015	6B	2015-01-21	MDCK1	640	640	320	320	640	320	1280	1280	640	
A/Yokohama/50/2015		2015-02-02	MDCK3/MDCK1	640	320	160	320	1280	320	1280	1280	640	
A/Mozambique/IR467/2015	6B	2015-02-13	SIAT1	320	320	320	320	640	320	1280	640	320	
A/Paris/1686/2015	6B	2015-02-16	MDCK1/MDCK1	320	320	160	320	640	320	640	640	320	
A/Mozambique/IR495/2015	6B	2015-02-19	MDCK1	320	640	320	320	640	320	640	640	640	
A/Iceland/39/2015	6B	2015-02-28	MDCKx/MDCK1	640	320	320	640	1280	640	1280	1280	1280	
A/Mozambique/IR543/2015	6B	2015-03-03	MDCK1	640	640	320	640	1280	640	2560	1280	1280	
A/Oman/1287/2015	6B	2015-03-12	MDCK1	640	640	640	640	1280	640	2560	2560	1280	
A/Oman/1706/2015	6B	2015-03-29	MDCK1	640	640	640	640	640	320	1280	1280	640	
A/Oman/1873/2015	6B	2015-03-30	MDCK1	1280	640	640	640	1280	640	2560	2560	1280	
A/Oman/1846/2015	6B	2015-04-01	MDCK1	640	320	320	320	640	320	1280	1280	1280	
A/Oman/1850/2015	6B	2015-04-01	MDCK1	640	640	320	640	1280	640	2560	1280	1280	
A/Jordan/11784/2015		2015-04-28	MDCK1/MDCK1	640	320	320	320	640	320	1280	1280	640	
A/Jordan/11796/2015	6B	2015-04-28	MDCK1/MDCK1	160	320	320	320	640	160	640	640	320	
A/Jordan/11819/2015	6B	2015-04-28	MDCK1/MDCK1	160	320	320	320	640	160	640	640	320	
A/Jordan/11817/2015	6B	2015-04-28	MDCK1/MDCK1	160	640	320	320	640	320	640	640	640	
A/Jordan/11838/2015	6B	2015-04-29	MDCK1/MDCK1	160	320	160	320	320	160	640	640	320	
A/Jordan/11862/2015		2015-04-29	MDCK1/MDCK1	320	320	160	320	640	320	1280	1280	640	
A/Jordan/11902/2015	6B	2015-05-04	MDCK1/MDCK1	320	160	160	320	640	320	1280	640	640	
A/Jordan/11917/2015	6B	2015-05-05	MDCK1/MDCK1	640	320	160	320	640	320	1280	1280	640	
A/Jordan/30323/2015		2015-05-07	MDCK1/MDCK1	320	320	160	320	640	320	640	640	640	
A/Jordan/11969/2015		2015-05-12	MDCK1/MDCK1	320	160	160	160	640	160	640	640	320	
A/Jordan/30347/2015		2015-05-19	MDCK1/MDCK1	160	640	640	320	640	320	640	640	640	
A/South Africa/R3168/2015	6B	2015-06-03	MDCK1	640	640	640	640	1280	640	2560	2560	1280	
A/South Africa/R3171/2015	6B	2015-06-09	MDCK1	640	320	320	320	640	640	1280	1280	640	
A/Jordan/20636/2015	6B	2015-07-27	MDCK1/MDCK2	1280	640	320	640	2560	640	5120	2560	1280	

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) ¹ < = <40

Vaccine

Sequences in phylogenetic trees

Table 5-3. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2015-09-22)

Viruses	Other information	Passage history	Collection date	Passage history	Haemagglutination inhibition titre									
					Post-infection ferret antisera									
					A/Cal	A/Bayern	A/Lviv	A/Chch	A/Astrak	A/St. P	A/St. P	A/HK	A/Sth Afr	
					7/09	69/09	N6/09	16/10	1/11	27/11	100/11	5659/12	3626/13	
		Ferret number			Egg	MDCK	MDCK	Egg	MDCK	Egg	Egg	MDCK	Egg	
		Genetic group			F05/14 ¹	F09/15 ¹	F14/13 ¹	F15/14 ¹	F22/13 ¹	F26/14 ¹	F24/11 ¹	F30/12 ¹	F3/14 ¹	
					1	1	1	4	5	6	7	6A	6B	
REFERENCE VIRUSES														
A/California/7/2009		1	2009-04-09	E1/E3	1280	1280	1280	320	320	320	320	160	320	
A/Bayern/68/2009		1	2009-07-01	MDCK5/MDCK1	160	320	320	80	80	40	80	80	80	G155E
A/Lviv/16/2009		1	2009-10-27	MDCK4/SIAT1/MDCK3	320	640	1280	320	160	160	160	160	80	G155E>G, D222G
A/Christchurch/16/2010		4	2010-07-12	E1/E3	1280	640	1280	2560	2560	640	2560	2560	1280	
A/Astrakhan/1/2011		5	2011-02-28	MDCK1/MDCK5	1280	320	640	640	2560	640	2560	2560	1280	
A/St. Petersburg/27/2011		6	2011-02-14	E1/E3	1280	640	1280	640	2560	1280	2560	2560	1280	
A/St. Petersburg/100/2011		7	2011-03-14	E1/E3	1280	640	1280	640	2560	1280	5120	2560	1280	
A/Hong Kong/5659/2012		6A	2012-05-21	MDCK4/MDCK2	320	160	80	160	320	160	640	640	320	
A/South Africa/3626/2013		6B	2013-06-06	E1/E3	640	640	640	320	1280	640	1280	1280	640	
TEST VIRUSES														
A/Florida/62/2014		6B	2014-10-28	C3/MDCK1	1280	1280	640	640	1280	640	2560	1280	1280	
A/Florida/62/2014		6B	2014-10-28	E4/E1	320	320	320	320	640	320	640	640	320	
A/Laos/1187/2014		6B	2014-11-18	E3/E1	640	640	640	640	640	640	1280	1280	1280	
A/Oman/259/2015		6B	2015-01-16	MDCK2	640	640	320	640	1280	640	1280	1280	1280	
A/Oman/342/2015		6B	2015-01-20	MDCK1	640	640	640	640	1280	640	2560	1280	1280	
A/Oman/424/2015		6B	2015-01-24	MDCK2	640	640	640	320	320	160	320	320	320	
A/Yokohama/50/2015		6B	2015-02-02	MDCK3/MDCK1	640	320	160	320	1280	320	1280	1280	640	
A/Oman/562/2015		6B	2015-02-03	MDCK1	1280	640	640	640	1280	640	2560	2560	1280	
A/Oman/654/2015		6B	2015-02-07	MDCK1	1280	640	320	640	1280	640	2560	2560	1280	
A/Oman/835/2015		6B	2015-02-18	MDCK2	1280	1280	640	640	1280	640	5120	2560	1280	
A/Oman/837/2015		6B	2015-02-18	MDCK1	640	320	320	640	1280	320	1280	1280	640	
A/Oman/838/2015		6B	2015-02-18	MDCK2	640	640	320	640	1280	640	2560	1280	1280	
A/St. Petersburg/61/2015		6B	2015-02-20	E2/E1/E1	80	160	80	80	80	40	80	80	80	
A/Oman/962/2015		6B	2015-03-01	MDCK1	1280	1280	640	640	1280	640	2560	1280	1280	
A/Oman/1026/2015		6B	2015-03-04	MDCK2	640	640	320	320	640	640	1280	640	640	
A/Jordan/20336/2015		6B	2015-04-08	MDCK1/MDCK1	320	640	320	320	640	320	1280	640	640	
A/Jordan/11341/2015		6B	2015-04-14	MDCK1/MDCK1	320	320	320	320	640	320	640	640	640	
A/Jordan/11345/2015		6B	2015-04-14	MDCK1/MDCK1	1280	1280	640	640	1280	640	2560	2560	1280	
A/Jordan/11392/2015		6B	2015-04-15	MDCK1/MDCK1	320	640	320	320	640	320	640	640	640	
A/Jordan/11438/2015		6B	2015-04-16	MDCK1/MDCK1	640	640	320	640	1280	640	1280	1280	1280	
A/Jordan/11446/2015		6B	2015-04-16	MDCK1/MDCK1	640	640	320	640	1280	640	1280	1280	1280	
A/Bangladesh/5381/2015		6B	2015-04-18	C1/C1/MDCK1	320	320	160	80	160	80	80	160	80	
A/Jordan/20421/2015		6B	2015-04-18	MDCK1/MDCK1	320	640	320	320	640	320	1280	640	640	
A/Jordan/11482/2015		6B	2015-04-19	MDCK1/MDCK1	160	320	320	320	640	160	320	320	320	
A/Jordan/11490/2015		6B	2015-04-19	MDCK1/MDCK1	160	320	320	320	640	320	640	320	640	
A/Jordan/11494/2015		6B	2015-04-19	MDCK1/MDCK1	640	1280	640	640	1280	640	2560	1280	1280	
A/Jordan/11496/2015		6B	2015-04-19	MDCK1/MDCK1	640	640	640	640	1280	640	1280	1280	1280	
A/Jordan/11529/2015		6B	2015-04-20	MDCK1/MDCK1	160	320	320	320	640	320	640	320	640	
A/Jordan/11560/2015		6B	2015-04-21	MDCK1/MDCK1	80	320	160	160	320	320	320	320	320	
A/Jordan/11566/2015		6B	2015-04-21	MDCK1/MDCK1	160	320	320	320	640	320	320	1280	320	
A/Jordan/11573/2015		6B	2015-04-21	MDCK1/MDCK1	640	640	640	640	1280	640	2560	1280	1280	
A/Jordan/11574/2015		6B	2015-04-21	MDCK1/MDCK1	640	640	640	640	1280	640	2560	1280	1280	
A/Jordan/11582/2015		6B	2015-04-22	MDCK1/MDCK1	320	640	320	320	640	320	640	640	320	
A/Jordan/11597/2015		6B	2015-04-22	MDCK1/MDCK1	160	640	320	320	640	320	320	320	320	
A/Jordan/11598/2015		6B	2015-04-22	MDCK1/MDCK1	80	320	320	160	320	160	320	320	160	
A/Jordan/11681/2015		6B	2015-04-26	MDCK1/MDCK1	160	320	320	320	640	320	320	640	320	
A/Jordan/11697/2015		6B	2015-04-26	MDCK1/MDCK1	640	320	320	640	1280	640	2560	1280	1280	
A/Jordan/11730/2015		6B	2015-04-26	MDCK1/MDCK1	320	320	320	320	1280	640	1280	1280	640	
A/Jordan/11731/2015		6B	2015-04-26	MDCK1/MDCK1	640	640	320	640	1280	640	2560	1280	1280	
A/Jordan/11749/2015		6B	2015-04-27	MDCK1/MDCK1	320	320	320	320	640	640	640	640	320	
A/Jordan/30250/2015		6B	2015-04-27	MDCK1/MDCK1	640	320	320	320	640	640	1280	1280	640	
A/Peru/06/2015		6B	2015-04-29	C2/C1/MDCK1	40	40	40	<	<	<	<	<	<	G155S
A/Norway/2335/2015		6B	2015-05-05	MDCK1	640	320	160	320	1280	640	1280	1280	640	
A/Bangladesh/3002/2015		6B	2015-05-07	E3/E1	640	640	640	320	640	320	1280	1280	640	
A/Bangladesh/3002/2015		6B	2015-05-07	C1/C1/MDCK1	640	320	320	320	1280	320	1280	1281	640	
A/Norway/2320/2015		6B	2015-05-13	MDCK1	1280	1280	640	640	2560	1280	2560	2560	2560	
A/Bangladesh/4005/2015		6B	2015-05-14	C1/C1/MDCK1	80	80	80	80	160	40	320	320	320	
A/Zambia/01-00092/2015		6B	2015-05-31	MDCK2/MDCK1	640	640	320	640	1280	640	2560	2560	1280	
A/Zambia/01-00096/2015		6B	2015-06-14	MDCK2/MDCK1	640	640	320	640	1280	640	1280	1280	1280	
A/Zambia/04-00264/2015		6B	2015-06-19	MDCK2/MDCK1	640	640	320	640	1280	640	2560	2560	1280	
A/Zambia/06-00038/2015		6B	2015-06-22	SIAT1	640	320	160	320	640	320	1280	1280	1280	
A/Zambia/04-00277/2015		6B	2015-06-23	MDCK2/SIAT1	640	1280	320	640	1280	640	2560	2560	1280	
A/Zambia/04-00276/2015		6B	2015-06-23	MDCK2/SIAT1	1280	1280	640	640	1280	640	2560	2560	2560	
A/Zambia/06-00039/2015		6B	2015-06-23	SIAT1	640	640	320	640	1280	640	2560	2560	1280	
A/Zambia/06-00043/2015		6B	2015-06-24	SIAT1	1280	1280	640	640	1280	640	2560	2560	1280	
A/Zambia/04-00280/2015		6B	2015-06-29	MDCK2/SIAT1	640	640	320	640	1280	320	1280	2560	1280	
A/Zambia/01-00118/2015		6B	2015-07-28	SIAT1	640	640	160	320	640	320	1280	1280	640	

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) ¹ < = <40

Vaccine

Sequences in phylogenetic trees

Table 5-4. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2015-09-29, 2015-10-15, 2015-11-18)

Viruses	Other information	Passage history	Collection date	Passage history	Haemagglutination inhibition titre									
					Post-infection ferret antisera									
					A/Cal	A/Bayern	A/Lviv	A/Chch	A/Astrak	A/St. P	A/St. P	A/HK	A/Sth Afr	
					7/09	69/09	N6/09	16/10	1/11	27/11	100/11	5659/12	3626/13	
					Egg	MDCK	MDCK	Egg	MDCK	Egg	Egg	MDCK	Egg	
		Ferret number			F01/15 ¹	F09/15 ¹	F14/13 ¹	F15/14 ¹	F22/13 ¹	F26/14 ¹	F24/11 ¹	F30/12 ¹	F3/14 ¹	
		Genetic group			1	1	1	4	5	6	7	6A	6B	
REFERENCE VIRUSES														
A/California/7/2009		1	2009-04-09	E1/E3	640	640	640	160	160	160	160	160	160	
A/Bayern/69/2009		1	2009-07-01	MDCK5/MDCK1	80	320	320	80	40	40	40	40	40	G155E
A/Lviv/N6/2009		1	2009-10-27	MDCK4/SIAT1/MDCK3	320	640	1280	160	80	80	80	160	80	G155E>G, D222G
A/Christchurch/16/2010		4	2010-07-12	E1/E3	1280	1280	1280	2560	1280	640	2560	2560	1280	
A/Astrakhan/1/2011		5	2011-02-28	MDCK1/MDCK5	640	640	640	640	1280	640	2560	2560	1280	
A/St. Petersburg/27/2011		6	2011-02-14	E1/E3	640	640	640	320	1280	640	2560	1280	640	
A/St. Petersburg/100/2011		7	2011-03-14	E1/E3	640	640	640	640	1280	640	2560	1280	640	
A/Hong Kong/5659/2012		6A	2012-05-21	MDCK4/MDCK1	80	160	80	80	320	160	640	640	320	
A/South Africa/3626/2013		6B	2013-06-06	E1/E3	320	320	640	320	640	320	640	640	640	
TEST VIRUSES														
A/Serbia/805/2015		6B	2015-02-03	MDCK2	2560	1280	1280	2560	5120	2560	5120	5120	2560	
A/Zambia/04-00231/2015		6B	2015-05-21	MDCK2/MDCK1	320	320	160	320	640	320	1280	640	640	
A/England/336/2015		6B.1	2015-10-03	SIAT1/MDCK1	320	320	320	320	640	320	1280	1280	640	S162N, I216T
A/England/343/2015		6B.1	2015-10-05	MDCK1/MDCK1	320	640	320	320	640	640	1280	1280	1280	S162N, I216T
A/England/341/2015		6B	2015-10-05	SIAT1/MDCK1	640	1280	640	640	1280	640	2560	2560	1280	
A/England/350/2015		6B.1	2015-10-06	SIAT1/MDCK1	160	320	320	320	640	320	1280	640	640	S162N, I216T
A/England/353/2015			2015-10-16	SIAT1/MDCK1	320	320	320	320	1280	640	2560	1280	1280	
A/England/351/2015		6B.1	2015-10-19	SIAT1/MDCK1	320	320	320	320	640	320	1280	1280	640	S162N, I216T

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) ¹ < = <40

Vaccine

Sequences in phylogenetic trees

Table 5-5. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2015-11-25, 2015-11-27, 2015-12-02)

Viruses	Other information	Passage history	Collection date	Passage history	Haemagglutination inhibition titre								
					Post infection ferret antisera								
					A/Cal	A/Bayern	A/Lviv	A/Chch	A/Astrak	A/St. P	A/St. P	A/HK	A/Sth Afr
					7/09	69/09	N6/09	16/10	1/11	27/11	100/11	5659/12	3626/13
					Egg	MDCK	MDCK	Egg	MDCK	Egg	Egg	MDCK	Egg
		Ferret number			F01/15 ¹	F09/15 ¹	F14/13 ¹	F15/14 ¹	F22/13 ¹	F26/14 ¹	F24/11 ¹	F30/12 ¹	F3/14 ¹
		Genetic group			1	1	1	4	5	6	7	6A	6B
REFERENCE VIRUSES													
A/California/7/2009		1	2009-04-09	E1/E3	640	640	640	160	160	160	160	160	160
A/Bayern/69/2009		1	2009-07-01	MDCK5/MDCK1	160	320	320	80	40	40	40	80	40
A/Lviv/N6/2009		1	2009-10-27	MDCK4/SIAT1/MDCK3	640	1280	640	160	80	80	80	160	80
A/Christchurch/16/2010		4	2010-07-12	E1/E3	1280	1280	1280	2560	1280	640	2560	2560	1280
A/Astrakhan/1/2011		5	2011-02-28	MDCK1/MDCK5	640	640	640	640	1280	640	2560	2560	1280
A/St. Petersburg/27/2011		6	2011-02-14	E1/E3	640	640	640	320	1280	640	2560	1280	640
A/St. Petersburg/100/2011		7	2011-03-14	E1/E3	640	640	640	640	1280	640	2560	1280	640
A/Hong Kong/5659/2012		6A	2012-05-21	MDCK4/MDCK1	80	160	80	80	320	160	640	640	320
A/South Africa/3626/2013		6B	2013-06-06	E1/E3	320	640	320	320	640	640	1280	640	640
TEST VIRUSES													
A/Kaliningrad/44/2015		6B	unknown	E2/E1	160	320	160	160	320	320	640	640	320
A/St Petersburg/302/2015		6B	unknown	E2/E1	320	640	320	320	1280	640	1280	1280	640
A/Saudi Arabia/4009/2015		6B.1	2015-09-21	MDCK1	160	320	160	320	640	320	1280	640	640
A/Saudi Arabia/4001/2015		6B.1	2015-09-21	SIAT2	320	640	320	640	1280	640	2560	1280	1280
A/Saudi Arabia/78085/2015		6B	2015-09-22	SIAT1	160	320	320	320	640	320	1280	640	640
A/Saudi Arabia/78093/2015		6B.1	2015-09-22	SIAT2	320	1280	320	640	1280	640	2560	2560	1280
A/Saudi Arabia/78169/2015		6B.1	2015-09-22	SIAT2	640	1280	640	1280	2560	640	5120	2560	2560
A/Saudi Arabia/78649/2015			2015-09-24	MDCK1	320	320	320	640	640	640	1280	1280	640
A/Saudi Arabia/78524/2015		6B.1	2015-09-24	MDCK1	160	320	320	320	640	320	1280	1280	640
A/Saudi Arabia/78558/2015		6B	2015-09-24	MDCK1	320	640	640	640	1280	640	2560	1280	1280
A/Saudi Arabia/78741/2015		6B.1	2015-09-25	MDCK2	320	640	320	640	1280	320	2560	1280	1280

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) ¹ < = <40

Vaccine

Sequences in phylogenetic trees

Table 5-6. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2015-12-08)

Viruses	Other information	Passage history	Collection date	Passage history	Haemagglutination inhibition titre								
					Post-infection ferret antisera								
					A/Cal	A/Bayern	A/Lviv	A/Chch	A/Astrak	A/St. P	A/St. P	A/HK	A/Sth Afr
					7/09	69/09	N6/09	16/10	1/11	27/11	100/11	5659/12	3626/13
					Egg	MDCK	MDCK	Egg	MDCK	Egg	Egg	MDCK	Egg
Ferret number	F01/15 ⁻¹	F09/15 ⁻¹	F34/09C4 ⁻¹	F15/14 ⁻¹	F22/13 ⁻¹	F26/14 ⁻¹	F24/11 ⁻¹	F30/12 ⁻¹	F3/14 ⁻¹				
Genetic group	1	1	1	4	5	6	7	6A	6B				
REFERENCE VIRUSES													
A/California/7/2009	1	2009-04-09	E1/E3	640	640	640	160	160	160	160	160	160	G155E
A/Bayern/69/2009	1	2009-07-01	MDCK5/MDCK1	80	320	160	80	40	40	40	40	40	
A/Lviv/N6/2009	1	2009-10-27	MDCK4/S1/MDCK2	640	1280	640	320	160	160	160	320	160	G155E>G, D222G
A/Christchurch/16/2010	4	2010-07-12	E1/E3	640	1280	1280	2560	1280	320	2560	1280	1280	
A/Astrakhan/1/2011	5	2011-02-28	MDCK1/MDCK5	640	640	2560	640	1280	640	2560	2560	1280	
A/St. Petersburg/27/2011	6	2011-02-14	E1/E3	640	640	2560	640	1280	640	2560	2560	1280	
A/St. Petersburg/100/2011	7	2011-03-14	E1/E2	640	640	160	640	1280	640	2560	1280	640	
A/Hong Kong/5659/2012	6A	2012-05-21	MDCK4/MDCK1	160	160	640	160	320	160	640	640	320	
A/South Africa/3626/2013	6B	2013-06-06	E1/E3	320	320	320	320	640	320	640	640	640	
TEST VIRUSES													
A/Norway/2624/2015	6B	2015-10-20	MDCK1/MDCK1	640	640	1280	640	1280	640	2560	2560	1280	S162N, I216T
A/Norway/2626/2015	6B.1	2015-10-22	MDCK1/MDCK1	320	320	640	320	1280	640	2560	1280	1280	
A/Norway/2628/2015	6B	2015-10-23	MDCK1/MDCK1	320	640	1280	640	1280	640	2560	2560	1280	
A/Norway/2633/2015	6B	2015-10-27	MDCK1/MDCK1	640	640	1280	640	1280	640	2560	2560	1280	
A/Norway/2634/2015	6B	2015-10-27	MDCK1/MDCK1	320	640	1280	640	1280	640	2560	1280	1280	
A/Norway/2646/2015	6B	2015-11-03	MDCK1/MDCK1	320	640	640	320	1280	320	1280	1280	1280	
A/Norway/2647/2015	6B	2015-11-03	MDCK1/MDCK1	320	640	1280	640	2560	640	2560	2560	1280	
A/Norway/2650/2015	6B.1	2015-11-03	MDCK1/MDCK1	320	640	1280	640	1280	640	2560	2560	1280	
A/Norway/2685/2015		2015-11-13	MDCK1	160	320	640	320	640	320	1280	1280	640	

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) ¹ <= <40

Vaccine

Sequences in phylogenetic trees

Table 5-7. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2015-12-15 + 2015-12-18)

Viruses	Other information	Passage history	Collection date	Passage history	Haemagglutination inhibition titre								
					Post-infection ferret antisera								
					A/Cal 7/09	A/Bayern 69/09	A/Lviv N6/09	A/Chch 16/10	A/Astrak 1/11	A/St. P 27/11	A/St. P 100/11	A/HK 5659/12	A/Sth Afr 3626/13
					Egg F01/15 ¹	MDCK F09/15 ¹	MDCK F14/13 ¹	Egg F15/14 ¹	MDCK F22/13 ¹	Egg F26/14 ¹	Egg F24/11 ¹	MDCK F30/12 ¹	Egg F3/14 ¹
		Genetic group			1	1	1	4	5	6	7	6A	6B
REFERENCE VIRUSES													
A/California/7/2009		1	2009-04-09	EP1/E3	1280	640	640	160	160	160	320	160	160
A/Bayern/69/2009		1	2009-07-01	MDCK5/MDCK1	40	160	80	40	<	<	<	<	40
A/Lviv/N6/2009		1	2009-10-27	MDCK4/S1/MDCK3	320	640	640	160	80	80	80	160	80
A/Christchurch/16/2010		4	2010-07-12	E1/E3	1280	640	1280	2560	1280	640	1280	1280	640
A/Astrakhan/1/2011		5	2011-02-28	MDCK1/MDCK5	320	640	320	320	1280	640	1280	1280	640
A/St. Petersburg/27/2011		6	2011-02-14	E1/E3	640	640	640	320	1280	640	2560	1280	640
A/St. Petersburg/100/2011		7	2011-03-14	E1/E3	640	640	640	320	1280	640	2560	1280	640
A/Hong Kong/5659/2012		6A	2012-05-21	MDCK4/MDCK1	80	80	160	80	320	160	640	640	160
A/South Africa/3626/2013		6B	2013-06-06	E1/E3	320	320	320	160	320	320	640	640	640
TEST VIRUSES													
A/Norway/2592/2015		6B	2015-08-25	MDCK1/MDCK3	320	640	320	320	1280	320	1280	1280	1280
A/Slovenia/2903/2015		6B.1	2015-10-12	MDCKx/MDCK1	320	640	320	640	1280	640	5120	2560	2560
A/Berlin/166/2015		6B.2	2015-10-21	C2/MDCK1	320	320	320	320	640	320	1280	1280	640
A/Norway/2674/2015		6B	2015-11-04	MDCK2	320	640	320	320	1280	640	1280	1280	1280
A/Berlin/167/2015		6B.1	2015-11-13	C2/MDCK1	320	640	320	640	1280	640	1280	1280	1280
A/Sachsen/102/2015		6B.1	2015-11-13	C2/MDCK1	320	320	320	320	640	320	1280	1280	640
A/Bayern/147/2015		6B.1	2015-11-19	C2/MDCK1	320	640	320	640	1280	640	1280	1280	640
A/Hamburg/24/2015		6B.1	2015-11-23	C2/MDCK1	320	640	320	640	1280	640	1280	1280	1280
A/Nordrhein-Westfalen/102/2015		6B.1	2015-11-23	C2/MDCK1	320	640	320	640	1280	640	1280	1280	1280
A/Bayern/146/2015		6B.1	2015-11-30	C2/MDCK1	160	320	320	320	640	320	1280	1280	640

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) ¹ < = <40

Vaccine

Sequences in phylogenetic trees

Table 5-8. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2015-12-22)

Viruses	Other information	Passage history	Collection date	Passage history	Haemagglutination inhibition titre								
					Post-infection ferret antisera								
					A/Cal	A/Bayern	A/Lviv	A/Chch	A/Astrak	A/St. P	A/St. P	A/HK	A/Sth Afr
					7/09	69/09	N6/09	16/10	1/11	27/11	100/11	5659/12	3626/13
					Egg	MDCK	MDCK	Egg	MDCK	Egg	Egg	MDCK	Egg
		Ferret number			F01/15 ¹	F09/15 ¹	F14/13 ¹	F15/14 ¹	F22/13 ¹	F26/14 ¹	F24/11 ¹	F30/12 ¹	F3/14 ¹
		Genetic group			1	1	1	4	5	6	7	6A	6B
REFERENCE VIRUSES													
A/California/7/2009		1	2009-04-09	EP1/E3	640	640	640	160	160	160	160	80	160
A/Bayern/69/2009		1	2009-07-01	MDCK5/MDCK1	40	160	80	40	<	<	<	<	40
A/Lviv/N6/2009		1	2009-10-27	MDCK4/S1/MDCK2	320	640	640	160	80	80	80	160	80
A/Christchurch/16/2010		4	2010-07-12	E1/E3	1280	1280	1280	2560	1280	640	2560	1280	1280
A/Astrakhan/1/2011		5	2011-02-28	MDCK1/MDCK5	640	1280	640	640	1280	640	2560	2560	1280
A/St. Petersburg/27/2011		6	2011-02-14	E1/E3	640	640	640	320	1280	640	2560	1280	640
A/St. Petersburg/100/2011		7	2011-03-14	E1/E2	640	640	640	640	1280	640	2560	1280	640
A/Hong Kong/5659/2012		6A	2012-05-21	MDCK4/MDCK1	160	160	160	160	640	160	1280	640	320
A/South Africa/3626/2013		6B	2013-06-06	E1/E3	640	640	640	320	640	320	1280	640	640
TEST VIRUSES													
A/Bremen/25/2015		6B	2015-11-09	C2/MDCK1	320	640	320	640	1280	320	2560	1280	640
A/Berlin/169/2015		6B	2015-11-09	C2/MDCK1	640	640	640	640	1280	640	2560	1280	1280
A/Berlin/168/2015		6B.1	2015-11-10	C2/MDCK1	320	640	320	320	1280	320	1280	1280	1280
A/Nordrhein-Westfalen/103/2015		6B.1	2015-11-13	C2/MDCK1	320	640	320	640	1280	640	1280	1280	1280

G155E

G155E>G, D222G

S162N, I216T

S162N, I216T

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) ¹ < = <40

Vaccine

Sequences in phylogenetic trees

Table 5-9. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2016-01-13)

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
				Post-infection ferret antisera									
				A/Cal	A/Bayern	A/Lviv	A/Chch	A/Astrak	A/St. P	A/St. P	A/HK	A/Sth Afr	
				7/09 Egg F01/15 ¹	69/09 MDCK F09/15 ¹	N6/09 MDCK F14/13 ¹	16/10 Egg F15/14 ¹	1/11 MDCK F22/13 ¹	27/11 Egg F26/14 ¹	100/11 Egg F24/11 ¹	5659/12 MDCK F30/12 ¹	3626/13 Egg F3/14 ¹	
	Passage history Ferret number Genetic group			1	1	1	4	5	6	7	6A	6B	
REFERENCE VIRUSES													
A/California/7/2009	1	2009-04-09	E1/E3	640	640	640	160	160	160	160	160	160	
A/Bayern/69/2009	1	2009-07-01	MDCK5/MDCK1	80	160	160	80	40	<	40	40	40	G155E
A/Lviv/16/2009	1	2009-10-27	MDCK4/SIAT1/MDCK3	640	1280	640	160	80	80	80	160	80	G155E>G, D222G
A/Christchurch/16/2010	4	2010-07-12	E1/E3	1280	2560	2560	5120	2560	1280	2560	2560	2560	
A/Astrakhan/1/2011	5	2011-02-28	MDCK1/MDCK5	640	640	640	640	2560	640	2560	2560	1280	
A/St. Petersburg/27/2011	6	2011-02-14	E1/E3	1280	1280	640	640	1280	640	2560	2560	1280	
A/St. Petersburg/100/2011	7	2011-03-14	E1/E2	1280	640	640	640	2560	640	2560	2560	1280	
A/Hong Kong/5659/2012	6A	2012-05-21	MDCK4/MDCK1	160	320	160	160	640	320	640	1280	640	
A/South Africa/3626/2013	6B	2013-06-06	E1/E3	320	320	320	320	640	320	640	640	640	
TEST VIRUSES													
A/Lisboa/29/20015	6B.1	2015-10-03	MDCK1/MDCK1	640	1280	640	640	1280	640	2560	2560	1280	S162N, I216T
A/Iran/112039/2015	6B.1	2015-10-05	SIAT1/MDCK1	320	640	640	640	1280	640	2560	1280	1280	S162N, I216T
A/Iran/109635/2015	6B.1	2015-10-08	SIAT1/MDCK1	320	640	320	640	1280	640	2560	1280	1280	S162N, I216T
A/Iran/1359/2015	6B.1	2015-10-10	SIAT1/MDCK1	320	640	320	640	1280	640	2560	1280	1280	S162N, I216T
A/Belgium/2015G0741/2015	6B.1	2015-10-20	SIAT1/MDCK1	320	640	320	320	640	320	2560	1280	640	S162N, I216T
A/England/357/2015	6B.2	2015-10-26	SIAT1/MDCK1	320	640	160	640	1280	640	2560	1280	1280	V152T, V173I
A/Latvia/11-044910/2015	6B.1	2015-10-26	C2/MDCK1	320	640	320	640	1280	640	2560	1280	1280	S162N, I216T
A/Netherlands/2905/2015	6B.1	2015-10-28	MDCK1/MDCK1	160	320	160	320	640	320	1280	1280	640	S162N, I216T
A/Lisboa/30/20015	6B	2015-11-08	MDCK1/MDCK1	320	640	320	640	1280	640	2560	1280	1280	
A/Finland/541/2015	6B.1	2015-11-09	MDCK1/MDCK1	320	640	640	640	1280	640	2560	1280	1280	S162N, I216T
A/Netherlands/2915/2015	6B.1	2015-11-10	MDCK1/MDCK1	320	640	320	640	1280	640	2560	1280	1280	S162N, I216T
A/Netherlands/2917/2015	6B.1	2015-11-12	MDCK1/MDCK1	320	640	320	640	1280	640	2560	2560	1280	S162N, I216T
A/England/358/2015	6B	2015-11-16	SIAT1/MDCK1	320	640	320	640	640	320	1280	1280	1280	
A/England/364/2015	6B.1	2015-11-17	SIAT1/MDCK1	320	640	160	640	640	640	2560	1280	1280	S162N, I216T
A/England/365/2015	6B.1	2015-11-17	SIAT1/MDCK1	320	640	320	640	1280	640	2560	1280	1280	S162N, I216T
A/Lisboa/31/20015	6B	2015-11-18	MDCK1/MDCK1	320	320	160	320	640	320	1280	1280	1280	
A/Lisboa/32/20015	6B.1	2015-11-19	SIAT2/MDCK1	320	640	640	640	1280	640	2560	2560	1280	S162N, I216T
A/England/370/2015	6B.1	2015-11-20	SIAT1/MDCK1	320	640	320	640	1280	640	2560	1280	1280	S162N, I216T
A/Lisboa/33/20015	6B	2015-11-25	SIAT1/MDCK1	320	640	320	640	1280	640	2560	1280	1280	
A/St Petersburg/349/2015	6B.1	2015-11-25	C1/MDCK1	320	640	320	640	1280	640	2560	1280	1280	S162N, I216T
A/Finland/546/2015	6B.1	2015-11-26	MDCK1/MDCK1	320	640	640	640	1280	640	2560	1280	1280	S162N, I216T
A/Lisboa/34/20015	6B	2015-11-27	SIAT2/MDCK1	640	1280	640	640	1280	640	5120	2560	1280	
A/Iran/117007/2015	6B.1	2015-11-27	E1/E1	320	1280	640	640	1280	640	2560	2560	1280	S162N, I216T
A/Iran/1337/2015	6B.1	2015-11-28	SIAT1/MDCK1	320	640	320	640	1280	640	2560	1280	1280	S162N, I216T
A/Iran/116675/2015	6B.1	2015-11-28	E1/E1	320	1280	640	640	1280	640	2560	2560	1280	S162N, I216T
A/Iran/116675/2015	6B.1	2015-11-28	E2/E1	640	1280	1280	1280	2560	1280	5120	2560	1280	S162N, I216T
A/Netherlands/2925/2015	6B.1	2015-11-29	MDCK1/MDCK1	320	640	320	640	1280	640	2560	1280	1280	S162N, I216T
A/Finland/550/2015	6B.1	2015-12-04	MDCK1/MDCK1	320	640	640	640	1280	640	2560	2560	1280	S162N, I216T
A/England/375/2015	6B.2	2015-12-04	MDCK1/MDCK1	320	640	640	640	1280	640	2560	1280	1280	V152T, V173I
A/Finland/553/2015	6B.1	2015-12-06	MDCK1/MDCK1	320	640	640	640	1280	640	2560	1280	1280	S162N, I216T
A/England/372/2015	6B.1	2015-12-07	SIAT1/MDCK1	320	640	320	640	1280	640	2560	1280	1280	S162N, I216T
A/England/377/2015	6B.2	2015-12-10	SIAT1/MDCK1	320	640	640	640	1280	640	2560	2560	1280	V152T, V173I
A/Latvia/12-037196/2015	6B.1	2015-12-15	C2/MDCK1	320	640	320	640	640	640	2560	1280	640	S162N, I216T
A/Latvia/12-042353/2015	6B.1	2015-12-16	SIAT1/MDCK1	320	640	320	640	640	640	2560	1280	1280	S162N, I216T
A/Latvia/12-063862/2015	6B.1	2015-12-29	SIAT1/MDCK1	320	640	320	640	640	320	2560	1280	640	S162N, I216T

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) ¹ < = <40

Vaccine

Sequences in phylogenetic trees

Table 5-10. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2016-01-19)

Haemagglutination inhibition titre													
Viruses	Other information	Collection date	Passage history	Post-infection ferret antisera									
				A/Cal 7/09	A/Bayern 69/09	A/Lviv N6/09	A/Chch 16/10	A/Astrak 1/11	A/St. P 27/11	A/St. P 100/11	A/HK 5659/12	A/Sth Afr 3626/13	
				Egg	MDCK	MDCK	Egg	MDCK	Egg	Egg	MDCK	Egg	
				NIB F05/14 ¹	F09/15 ¹	F14/13 ¹	F15/14 ¹	F22/13 ¹	F26/14 ¹	F24/11 ¹	F30/12 ¹	F3/14 ¹	
				1	1	1	4	5	6	7	6A	6B	
REFERENCE VIRUSES													
A/California/7/2009	NIBSC	1	2009-04-09	E4/E1	640	320	320	320	640	320	1280	640	640
A/Bayern/69/2009		1	2009-07-01	MDCK5/MDCK1	<	320	160	40	40	<	40	40	40
A/Lviv/N6/2009		1	2009-10-27	MDCK4/SIAT1/MDCK3	80	1280	1280	160	80	80	160	160	80
A/Christchurch/16/2010		4	2010-07-12	E1/E3	1280	1280	1280	2560	1280	640	2560	2560	1280
A/Astrakhan/1/2011		5	2011-02-28	MDCK1/MDCK5	1280	1280	640	640	1280	640	2560	2560	1280
A/St. Petersburg/27/2011		6	2011-02-14	E1/E3	1280	1280	640	640	1280	640	2560	2560	1280
A/St. Petersburg/100/2011		7	2011-03-14	E1/E3	1280	640	640	640	1280	640	2560	2560	1280
A/Hong Kong/5659/2012		6A	2012-05-21	MDCK4/MDCK1	320	160	160	160	320	160	640	640	320
A/South Africa/3626/2013		6B	2013-06-06	E1/E3	640	640	640	320	640	640	1280	640	1280
TEST VIRUSES													
A/Maevatanana/4197/2015		6B	2015-09-22	MDCK1/MDCK1	1280	640	320	640	1280	640	2560	2560	1280
A/Maevatanana/4189/2015		6B	2015-09-24	MDCK2/MDCK1	1280	640	320	640	1280	640	2560	1280	1280
A/Antsirabe/4186/2015		6B	2015-09-29	MDCK1/MDCK1	1280	640	640	640	1280	640	2560	1280	1280
A/Maevatanana/4191/2015		6B	2015-09-29	MDCK2/MDCK1	1280	1280	640	1280	1280	1280	5120	2560	1280
A/Desgenettes/2318/2015		6B.1	2015-10-02	MDCK2/MDCK1	1280	320	320	320	640	320	1280	1280	640
A/Grenoble/2322/2015		6B.1	2015-10-02	MDCK2/MDCK1	1280	640	640	640	1280	640	2560	2560	1280
A/Austria/883648/2015		6B.1	2015-10-08	SIAT1/MDCK1	640	320	160	320	640	320	1280	640	640
A/Slovenia/2903/2015		6B.1	2015-10-26	E2	1280	1280	640	640	1280	1280	2560	2560	1280
A/Slovenia/2903/2015	clone 56	6B.1	2015-10-26	E3	1280	640	320	640	1280	640	2560	2560	1280
A/Ghana/DILI-15-1165/2015		6B	2015-11-03	Cx/MDCK1	1280	320	320	640	1280	640	1280	1280	1280
A/Austria/888727/2015		6B.1	2015-11-08	SIAT1/MDCK1	640	320	160	320	640	320	1280	1280	640
A/Austria/888621/2015		6B.1	2015-11-09	SIAT1/MDCK1	640	320	320	320	640	320	1280	1280	640
A/Austria/889649/2015		6B.1	2015-11-13	SIAT1/MDCK1	640	320	320	320	640	320	1280	1280	640
A/Austria/889721/2015		6B.1	2015-11-16	SIAT1/MDCK1	640	320	160	320	320	320	1280	640	640
A/Bayern/148/2015		6B	2015-11-24	C2/MDCK1	1280	640	320	320	1280	320	1280	1280	1280
A/Austria/891161/2015		6B.1	2015-11-24	SIAT1/MDCK1	1280	640	320	320	640	320	1280	1280	1280
A/Baden-Wuerttemberg/253/2015		6B.1	2015-12-01	C2/MDCK1	640	640	320	320	1280	640	1280	1280	640
A/Sachsen-Anhalt/91/2015		6B.1	2015-12-05	C2/MDCK1	640	320	320	320	640	320	1280	1280	640
A/Berlin/171/2015		6B.1	2015-12-07	C2/MDCK1	1280	640	320	640	1280	640	2560	1280	1280
A/Bremen/26/2015		6B.1	2015-12-07	C2/MDCK1	1280	640	320	640	1280	640	2560	1280	1280
A/Berlin/172/2015			2015-12-08	C2/MDCK1	1280	640	320	320	1280	640	1280	1280	1280
A/Ghana/DILI-15-1281/2015		6B	2015-12-08	Cx/MDCK1	1280	640	320	640	1280	640	2560	1280	1280
A/Lyon/9.12.R09/2015		6B.1	2015-12-09	MDCK2/MDCK1	1280	640	320	640	1280	640	2560	1280	1280
A/Baden-Wuerttemberg/254/2015		6B.1	2015-12-09	C2/MDCK1	640	320	320	320	640	320	1280	1280	640
A/Clermont Ferrand/2611/2015		6B.1	2015-12-12	MDCK2/MDCK1	1280	640	320	640	1280	640	2560	1280	1280
A/Bremen/27/2015		6B.1	2015-12-14	C2/MDCK1	1280	640	320	640	1280	640	2560	1280	1280
A/Mecklenburg-Vorpommern/21/2015			2015-12-15	C2/MDCK1	1280	640	320	640	1280	640	2560	1280	1280
A/Berlin/174/2015			2015-12-16	C2/MDCK1	1280	640	320	320	1280	640	1280	1280	1280
A/Berlin/175/2015			2015-12-18	C2/MDCK1	1280	640	320	640	1280	640	2560	2560	1280
A/Sachsen/104/2015		6B.1	2015-12-18	C2/MDCK1	1280	640	320	640	1280	640	2560	1280	1280
A/Mecklenburg-Vorpommern/20/2015			2015-12-20	C2/MDCK1	1280	640	320	640	1280	640	2560	2560	1280
A/Mecklenburg-Vorpommern/19/2015		6B.1	2015-12-21	C2/MDCK1	1280	640	320	640	1280	640	2560	1280	640
A/Berlin/176/2015		6B.1	2015-12-22	C2/MDCK1	640	320	160	320	640	320	1280	1280	640

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

¹ < = <40

Vaccine

Sequences in phylogenetic trees

Table 5-11. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2016-01-26)

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
				Post-infection ferret antisera									
				A/Cal 7/09 Egg NIB F05/14 ¹	A/Bayern 69/09 MDCK F09/15 ¹	A/Lviv N6/09 MDCK F14/13 ¹	A/Chch 16/10 Egg F15/14 ¹	A/Astrak 1/11 MDCK F22/13 ¹	A/St. P 27/11 Egg F26/14 ¹	A/St. P 100/11 Egg F24/11 ¹	A/HK 5659/12 MDCK F30/12 ¹	A/Sth Afr 3626/13 Egg F3/14 ¹	
				1	1	1	4	5	6	7	6A	6B	
REFERENCE VIRUSES													
A/California/7/2009	1	2009-04-09	E4/E2	640									
A/Bayern/69/2009	1	2009-07-01	MDCK5/MDCK1	<	320	160	80	40	<	40	40	40	
A/Lviv/N6/2009	1	2009-10-27	MDCK4/SIAT1/MDCK3	80	640	640	160	80	80	80	160	80	
A/Christchurch/16/2010	4	2010-07-12	E1/E3	1280	1280	1280	2560	1280	640	2560	1280	1280	
A/Astrakhan/1/2011	5	2011-02-28	MDCK1/MDCK5	1280	640	320	640	1280	640	1280	2560	1280	
A/St. Petersburg/27/2011	6	2011-02-14	E1/E3	1280	640	640	640	1280	640	1280	1280	1280	
A/St. Petersburg/100/2011	7	2011-03-14	E1/E3	1280	640	640	640	1280	640	2560	1280	1280	
A/Hong Kong/5659/2012	6A	2012-05-21	MDCK4/MDCK1	320	160	80	160	320	160	640	640	320	
A/South Africa/3626/2013	6B	2013-06-06	E1/E3	640	320	320	320	640	320	640	640	640	
TEST VIRUSES													
A/Iceland/53/2015	6B.1	2015-09-30	MDCK0/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	S162N, I216T
A/Shanghai-Putuo/SWL1860/2015	6B.2	2015-10-29	MDCK4/MDCK2	320	160	160	160	320	160	640	320	320	V152T, V173I
A/Scotland/P2/2015	6B.1	2015-11-19	MDCK1	640	320	320	640	640	320	1280	1280	640	S162N, I216T
A/Denmark/62/2015	6B.1	2015-12-04	SIAT1/MDCK2//MDCK1	640	320	160	320	640	320	1280	640	640	S162N, I216T
A/Denmark/46/2015		2015-12-07	SIAT1/MDCK2//MDCK1	320	160	80	160	320	160	640	320	320	
A/Antsirabe/5056/2015	6B	2015-12-08	MDCK1	1280	640	640	640	1280	640	2560	1280	1280	
A/Denmark/49/2015	6B.1	2015-12-09	SIAT2/MDCK2/MDCK1	640	320	160	320	320	320	640	640	640	S162N, I216T
A/Denmark/50/2015	6B.1	2015-12-10	SIAT1/MDCK2//MDCK1	640	160	160	320	640	320	640	640	640	S162N, I216T
A/Denmark/51/2015		2015-12-10	SIAT1/MDCK2//MDCK1	640	640	320	640	1280	640	1280	1280	1280	
A/Kazakhstan/4203/2015	6B.1	2015-12-13	MDCK1	1280	640	320	640	1280	640	2560	1280	1280	S162N, I216T
A/Israel/Q-469/2015		2015-12-15	C1/MDCK1	320	320	320	640	640	320	1280	1280	640	
A/Israel/Q-504/2015	6B.2	2015-12-15	C1/MDCK1	640	640	320	640	1280	320	1280	1280	640	V152T, V173I
A/Israel/Q-482/2015	6B.1	2015-12-15	C2/MDCK1	160	80	80	160	160	80	320	320	320	S162N, I216T
A/Israel/Q-509/2015		2015-12-15	C2/MDCK1	640	320	160	320	640	320	1280	640	640	
A/Kazakhstan/4227/2015	6B.1	2015-12-16	MDCK1	1280	640	320	640	1280	640	1280	1280	1280	S162N, I216T
A/Israel/Q-525/2015	6B.2	2015-12-16	C2/MDCK1	320	320	160	320	640	320	640	640	640	V152T, V173I
A/Navarra/1850/2015		2015-12-17	SIAT1/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	
A/Israel/A-6615/2015	6B.1	2015-12-20	C1/MDCK1	1280	640	320	640	1280	640	2560	2560	1280	S162N, I216T
A/Switzerland/16034279/2015	6B.1	2015-12-23	MDCK1	1280	640	320	640	1280	640	2560	2560	1280	S162N, I216T
A/Georgia/1469/2015	6B	2015-12-24	MDCK1	1280	640	320	640	1280	640	2560	1280	1280	
A/Madrid/1859/2015	6B.1	2015-12-24	SIAT1/MDCK1	1280	640	320	640	1280	640	1280	1280	1280	S162N, I216T
A/Georgia/4/2016	6B	2015-12-28	MDCK1	1280	640	640	640	1280	640	2560	1280	1280	
A/Iceland/55/2015	6B.1	2015-12-28	MDCK0/MDCK1	1280	640	320	640	1280	640	1280	1280	1280	S162N, I216T
A/Georgia/1479/2015	6B	2015-12-29	MDCK1	1280	640	320	640	1280	640	2560	1280	1280	
A/Israel/Q-657/2015	6B.2	2015-12-29	C1/MDCK1	1280	1280	640	1280	1280	640	2560	2560	2560	V152T, V173I
A/Israel/Q-688/2015		2015-12-29	C1/MDCK1	320	320	320	320	640	320	640	640	640	
A/Georgia/5/2015	6B	2015-12-30	MDCK1	1280	640	320	640	1280	640	2560	1280	1280	
A/Baleares/16036/2015		2015-12-30	SIAT1/MDCK1	1280	640	320	640	1280	640	1280	1280	1280	
A/Poland/7474/2015	6B.1	2015-12-31	MDCK4/MDCK1	640	640	320	640	1280	640	2560	1280	1280	S162N, I216T
A/Poland/7474/2015	6B.1	2015-12-31	SIAT4/MDCK1	2560	1280	640	1280	1280	640	2560	2560	2560	S162N, I216T
A/Navarra/26/2016	6B.1	2016-01-03	SIAT1/MDCK1	1280	640	320	640	1280	640	1280	1280	1280	S162N, I216T
A/Switzerland/16078014/2016	6B.1	2016-01-04	MDCK1	1280	640	320	640	1280	640	2560	1280	1280	S162N, I216T
A/Hong Kong/88/2016	6B.2	2016-01-04	MDCK1/MDCK1	640	320	320	320	1280	640	1280	1280	640	V152T, V173I
A/Hong Kong/89/2016	6B.2	2016-01-04	MDCK1/MDCK1	640	320	320	320	1280	320	1280	1280	640	V152T, V173I
A/Hong Kong/90/2016	6B.2	2016-01-05	MDCK1/MDCK1	1280	640	640	640	1280	640	2560	1280	1280	V152T, V173I
A/Baleares/35/2016	6B.1	2016-01-05	SIAT1/MDCK1	640	320	320	320	1280	320	1280	1280	640	S162N, I216T
A/Georgia/20/2016	6B.1	2016-01-06	MDCK1	1280	640	640	640	1280	640	2560	2560	1280	S162N, I216T
A/Iceland/1/2016	6B.1	2016-01-08	MDCK0/MDCK1	640	640	320	640	1280	640	1280	1280	1280	S162N, I216T
A/Iceland/2/2016		2016-01-08	MDCK0/MDCK1	1280	640	320	640	1280	320	1280	1280	1280	
A/Iceland/4/2016	6B.1	2016-01-12	MDCK0/MDCK1	640	640	320	640	1280	640	2560	1280	1280	S162N, I216T
A/Iceland/5/2016	6B.1	2016-01-13	MDCK0/MDCK1	1280	640	640	1280	1280	640	2560	2560	1280	S162N, I216T
A/Iceland/6/2016		2016-01-13	MDCK0/MDCK1	640	640	320	640	1280	640	1280	1280	1280	

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

¹ < = <40

Vaccine

Sequences in phylogenetic trees

Table 5-12. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2016-02-03)

				Haemagglutination inhibition titre												
Viruses	Other information	Collection date	Passage history	Post-infection ferret antisera												
				A/Cal	A/Bayern	A/Lviv	A/Chch	A/Astrak	A/St. P	A/St. P	A/HK	A/Sth Afr	A/Slov			
				7/09	69/09	N6/09	16/10	1/11	27/11	100/11	5659/12	3626/13	2903/15			
				Egg	MDCK	MDCK	Egg	MDCK	Egg	Egg	MDCK	Egg	Egg	Egg		
				NIB F05/14	F09/15 ¹	F14/13 ¹	F15/14 ¹	F22/13 ¹	F26/14 ¹	F24/11 ¹	F30/12 ¹	F3/14 ¹	F02/16 ¹			
				1	1	1	4	5	6	7	6A	6B	6B.1			
REFERENCE VIRUSES																
A/California/7/2009		1	2009-04-09	E4/E2	640	320	320	320	640	320	1280	1280	1280	1280		
A/Bayern/69/2009		1	2009-07-01	MDCK5/MDCK1	<	320	160	80	40	<	40	40	40	ND		
A/Lviv/N6/2009		1	2009-10-27	MDCK4/SIAT1/MDCK3	80	640	640	160	80	80	80	160	80	ND		
A/Christchurch/16/2010		4	2010-07-12	E1/E3	1280	1280	1280	2560	1280	640	2560	1280	1280	ND		
A/Astrakhan/1/2011		5	2011-02-28	MDCK1/MDCK5	1280	640	320	640	1280	640	1280	2560	1280	ND		
A/St. Petersburg/27/2011		5	2011-02-14	E1/E3	1280	640	640	640	1280	640	1280	1280	1280	ND		
A/St. Petersburg/100/2011		7	2011-03-14	E1/E3	1280	640	640	640	1280	640	2560	1280	1280	ND		
A/Hong Kong/5659/2012		6A	2012-05-21	MDCK4/MDCK1	320	160	80	160	320	160	640	640	320	ND		
A/South Africa/3626/2013		6B	2013-06-06	E1/E3	640	640	640	320	640	640	1280	1280	1280	1280		
A/Slovenia/2903/2015	clone 37	6B.1	2015-10-26	E4	2560	1280	640	1280	2560	1280	5120	2560	2560	5120	S162N, I216T	
TEST VIRUSES																
A/Antananarivo/4551/2015		6B	2015-10-27	MDCK1	1280	1280	640	1280	1280	1280	2560	2560	2560	2560		
A/Norway/2647/2015		6B	2015-11-03	MDCK1	1280	320	320	640	1280	640	2560	1280	1280	2560		
A/Stockholm/55/2015		6B.1	2015-11-10	MDCK0/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	S162N, I216T	
A/Stockholm/56/2015			2015-11-18	MDCK0/MDCK1	1280	640	320	640	1280	640	2560	2560	1280	2560		
A/Stockholm/57/2015			2015-11-18	MDCK0/MDCK1	1280	320	320	640	1280	320	1280	1280	1280	2560		
A/Stockholm/59/2015			2015-11-19	MDCK0/MDCK1	2560	1280	640	640	2560	1280	2560	2560	2560	5120		
A/Ukraine/517/2015	ngs?		2015-11-20	SIAT2/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560		
A/Stockholm/67/2015		6B.1	2015-11-21	MDCK0/MDCK1	640	320	160	320	640	320	1280	640	640	1280	S162N, I216T	
A/Stockholm/66/2015			2015-11-23	MDCK2/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560		
A/Stockholm/60/2015		6B.1	2015-11-25	MDCK0/MDCK1	1280	640	320	640	1280	640	2560	2560	1280	2560	S162N, I216T	
A/Portugal/SU66/2015			2015-11-26	MDCK2	1280	640	320	640	1280	640	2560	2560	1280	2560		
A/Portugal/SU63/2015		6B	2015-11-27	MDCK1	1280	640	320	640	1280	640	2560	2560	1280	2560		
A/Norway/2774/2015		6B.2	2015-11-30	MDCK2	1280	640	640	640	1280	640	2560	2560	2560	2560	V152T, V173I	
A/Kazakhstan/4166/2015			2015-11-30	SIAT1	2560	1280	1280	1280	2560	1280	5120	5120	5120	5120		
A/Georgia/1305/2015		6B	2015-11-30	MDCK1	1280	640	640	640	1280	640	2560	2560	1280	2560		
A/Czech Republic/96/2015			2015-12-01	SIAT1/MDCK1	1280	640	320	640	1280	640	2560	2560	1280	2560		
A/Czech Republic/95/2015		6B.1	2015-12-01	SIAT1/MDCK1	1280	640	640	1280	1280	2560	2560	2560	1280	2560	S162N, I216T	
A/Portugal/SU81/2015		6B	2015-12-02	MDCK2	1280	640	640	640	1280	640	2560	2560	1280	2560		
A/Navarra/1801/2015		6B.1	2015-12-02	SIAT1/MDCK1	1280	640	160	640	1280	640	2560	1280	1280	2560	S162N, I216T	
A/Norway/2844/2015			2015-12-04	MDCK1	640	640	320	320	640	1280	1280	1280	640	2560		
A/Norway/2879/2015		6B.1	2015-12-05	MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	S162N, I216T	
A/Norway/2846/2015		6B.1	2015-12-07	MDCK1	1280	640	640	640	1280	640	2560	2560	1280	2560	S162N, I216T	
A/Denmark/48/2015		6B.2	2015-12-09	SIAT2/MDCK2/MDCK2	640	320	160	320	640	320	1280	640	640	1280	V152T, V173I	
A/Ghana/DILI-15-1296/2015		6B	2015-12-14	MDCK1	1280	1280	640	1280	2560	1280	2560	2560	2560	2560		
A/Finland/559/2015			2015-12-14	MDCK1	1280	640	320	640	640	640	2560	1280	1280	2560		
A/Moscow/160/2015		6B.1	2015-12-14	MDCK1/MDCK1	1280	1280	640	1280	2560	1280	2560	2560	1280	2560	S162N, I216T	
A/Georgia/1408/2015		6B	2015-12-15	MDCK1	1280	1280	640	1280	2560	1280	2560	2560	2560	5120		
A/Finland/555/2015			2015-12-15	MDCK2	1280	640	320	640	1280	640	2560	1280	1280	2560		
A/Finland/557/2015		6B.1	2015-12-15	MDCK1	640	320	160	320	640	320	1280	1280	640	1280	S162N, I216T	
A/Moscow/163/2015			2015-12-15	MDCK1/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560		
A/Moscow/168/2015		6B.1	2015-12-15	MDCK1/MDCK1	1280	1280	640	640	1280	640	2560	2560	1280	2560	S162N, I216T	
A/Stavropol/3/2015			2015-12-15	MDCK1/MDCK1	640	640	320	320	640	320	1280	1280	640	1280		
A/Urengoy/9/2015			2015-12-15	MDCK1/MDCK1	640	640	320	320	640	320	1280	1280	640	1280		
A/Finland/556/2015		6B.1	2015-12-16	MDCK1	1280	320	320	640	1280	320	1280	1280	1280	1280	S162N, I216T	
A/Ukraine/6907/2015		6B.2	2015-12-16	C2/MDCK1	160	320	320	160	320	160	320	640	320	640	V152T, V173I	
A/Moscow/171/2015		6B.1	2015-12-16	MDCK1/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	S162N, I216T	
A/Stavropol/4/2015		6B.1	2015-12-16	MDCK1/MDCK1	640	320	320	320	640	320	1280	1280	640	1280	S162N, I216T	
A/Novosibirsk/49/2015		6B.1	2015-12-16	MDCK1/MDCK1	640	320	320	320	640	320	1280	640	640	1280	S162N, I216T	
A/Bretagne/2511/2015		6B.1	2015-12-16	MDCK1/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	S162N, I216T	
A/Moscow/172/2015		6B.1	2015-12-17	MDCK1/MDCK1	1280	1280	640	1280	1280	640	2560	2560	1280	2560	S162N, I216T	
A/Moscow/173/2015			2015-12-17	MDCK1/MDCK1	1280	640	320	640	1280	640	2560	2560	1280	2560		
A/Paris/2535/2015		6B.1	2015-12-17	MDCK1/MDCK1	640	320	320	320	640	320	1280	1280	640	1280	S162N, I216T	
A/Urengoy/15/2015			2015-12-19	MDCK1/MDCK1	640	320	320	320	640	320	1280	1280	640	1280		
A/Urengoy/17/2015		6B.1	2015-12-20	MDCK1/MDCK1	640	320	320	320	640	320	1280	1280	640	1280	S162N, I216T	
A/Bayern/20150/2015			2015-12-21	C2/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560		
A/Spain/111793/2015		6B.1	2015-12-23	MDCK1	2560	1280	640	1280	2560	1280	2560	2560	2560	5120	S162N, I216T	
A/Cameroon/15v-9415/2015		6B.1	2015-12-23	MDCK1	1280	640	320	640	1280	640	2560	2560	1280	2560	S162N, I216T	
A/Cameroon/15v-9393/2015		6B.1	2015-12-23	MDCK2	1280	640	320	640	1280	640	2560	2560	1280	2560	S162N, I216T	
A/Khmeltsky/89/2016		6B.1	2015-12-23	SIAT2/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	S162N, I216T	
A/Ukraine/27/2016		6B.1	2015-12-24	SIAT1/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560	S162N, I216T	
A/Ukraine/28/2016			2015-12-25	SIAT1/MDCK1	2560	1280	640	1280	2560	1280	5120	2560	2560	5120	S162N, I216T	
A/Khmeltsky/87/2016		6B.1	2015-12-25	SIAT2/MDCK1	2560	640	640	1280	2560	1280	5120	2560	2560	5120	S162N, I216T	
A/Parma/295/2015		6B.1	2015-12-25	MDCK1/MDCK1	1280	640	320	640	1280	640	2560	2560	1280	2560	S162N, I216T	
A/Bayern/20151/2015			2015-12-26	C2/MDCK1	640	640	320	640	640	320	1280					

Table 5-13. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2016-02-09)

Haemagglutination inhibition titre													
Viruses	Other information	Collection date	Passage history	Post-infection ferret antisera									
				A/Cal	A/Bayern	A/Lviv	A/Chch	A/Astrak	A/SL P	A/SL P	A/HK	A/SH Afr	A/Slov
				7/09	69/09	N6/09	16/10	1/11	27/11	100/11	5659/12	3626/13	2903/2015
				Egg	MDCK	MDCK	Egg	MDCK	Egg	Egg	MDCK	Egg	Egg
				NIB F05/14	F09/15 ¹	F14/13 ¹	F15/14 ¹	F22/13 ¹	F26/14 ¹	F24/11 ¹	F30/12 ¹	F03/14 ¹	F02/16 ²
					1	1	4	5	6	7	6A	6B	6B.1
REFERENCE VIRUSES													
A/California/7/2009	1	2009-04-09	NIBSC E4/E2	640	320	320	320	640	320	1280	1280	640	640
A/Bayern/69/2009	1	2009-07-01	MDCK5/MDCK1	<	320	320	160	80	40	80	80	80	<
A/Lviv/N6/2009	1	2009-10-27	MDCK4/SIAT1/MDCK3	80	1280	640	320	80	80	80	160	80	80
A/Christchurch/16/2010	4	2010-07-12	E1/E3	1280	1280	1280	5120	2560	640	2560	2560	1280	2560
A/Astrakhan/1/2011	5	2011-02-28	MDCK1/MDCK5	1280	640	640	1280	1280	640	2560	2560	1280	2560
A/St. Petersburg/27/2011	6	2011-02-14	E1/E4	1280	640	640	1280	640	2560	1280	1280	1280	1280
A/St. Petersburg/100/2011	7	2011-03-14	E1/E3	1280	1280	640	1280	640	2560	2560	2560	2560	2560
A/Hong Kong/5659/2012	6A	2012-05-21	MDCK4/MDCK2	320	160	160	320	320	320	1280	1280	640	640
A/South Africa/3626/2013	6B	2013-06-06	E1/E3	640	640	640	640	640	640	2560	1280	1280	1280
A/Slovenia/2903/2015	6B.1	2015-10-26	E4	1280	640	640	2560	2560	1280	5120	2560	1280	2560
TEST VIRUSES													
A/Alasi/187166/2015	ngs?	2015-10-13	MDCK1/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560
A/Tomsk/154/2015	ngs?	2015-11-19	MDCK1/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560
A/Denmark/41/2015	ngs?	2015-11-27	SIAT3/MDCK2/MDCK1	640	640	320	640	1280	640	1280	1280	1280	2560
A/Israel/Q-346/2015	ngs?	2015-12-01	C2/MDCK1	320	640	320	640	1280	320	1280	640	640	1280
A/Israel/Q-363/2015	ngs?	2015-12-01	C1/MDCK1	320	640	320	320	640	320	1280	640	640	1280
A/Denmark/43/2015	ngs?	2015-12-03	MDCK3/MDCK1	320	320	160	320	640	320	1280	640	640	1280
A/Moscow/155/2015	ngs?	2015-12-07	MDCK1/MDCK1	640	640	320	640	1280	640	1280	1280	1280	2560
A/Israel/Q-423/2015	ngs?	2015-12-08	C2/MDCK1	640	320	320	640	640	320	1280	640	640	1280
A/Israel/Q-426/2015	ngs?	2015-12-08	C2/MDCK1	640	320	320	640	640	320	1280	640	640	1280
A/Israel/Q-424/2015	ngs?	2015-12-08	C1/MDCK1	2560	1280	640	1280	2560	1280	2560	2560	1280	2560
A/Moscow/157/2015	ngs?	2015-12-11	MDCK2/MDCK1	640	320	320	640	1280	640	2560	1280	1280	2560
A/Moscow/158/2015	ngs?	2015-12-11	MDCK2/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	5120
A/Moscow/159/2015	ngs?	2015-12-11	MDCK2/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560
A/Moscow/167/2015	ngs?	2015-12-12	MDCK1/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560
A/Moscow/165/2015	ngs?	2015-12-13	MDCK1/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560
A/Moscow/166/2015	ngs?	2015-12-13	MDCK1/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560
A/Moscow/169/2015	ngs?	2015-12-13	MDCK1/MDCK1	1280	640	640	640	1280	640	2560	1280	1280	2560
A/Ukraine/6904/2015	ngs?	2015-12-14	C2/MDCK1	320	320	320	320	640	320	640	640	1280	1280
A/Ukraine/6909/2015	ngs?	2015-12-14	C1/MDCK1	320	640	320	320	640	320	640	640	640	1280
A/Moscow/161/2015	ngs?	2015-12-14	MDCK1/MDCK1	1280	640	320	640	640	640	2560	1280	640	1280
A/Moscow/162/2015	ngs?	2015-12-14	MDCK1/MDCK1	1280	640	640	1280	1280	640	2560	2560	2560	5120
A/Moscow/164/2015	ngs?	2015-12-14	MDCK1/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560
A/Moscow/170/2015	ngs?	2015-12-14	MDCK1/MDCK1	640	320	320	640	640	640	1280	1280	1280	2560
A/Ukraine/6905/2015	ngs?	2015-12-15	C2/MDCK1	320	640	320	320	640	320	640	640	640	1280
A/Ukraine/6912/2015	ngs?	2015-12-16	C1/MDCK1	160	640	640	320	640	320	640	640	320	1280
A/Odessa/62/2016	ngs?	2015-12-17	SIAT2/MDCK1	320	320	320	320	640	320	640	640	640	1280
A/Ireland/72156/2015	ngs?	2015-12-17	MDCK1/MDCK1	1280	640	640	1280	1280	640	2560	2560	1280	5120
A/Lisboa/51/2015	ngs?	2015-12-17	SIAT1/MDCK1	1280	640	640	640	1280	640	2560	2560	2560	5120
A/Ukraine/6906/2015	ngs?	2015-12-18	C2/MDCK1	320	320	160	320	640	320	640	640	640	640
A/Ukraine/6908/2015	ngs?	2015-12-18	C1/MDCK1	160	640	640	320	640	320	640	640	640	1280
A/Lisboa/49/2015	ngs?	2015-12-18	SIAT3/MDCK1	2560	1280	640	1280	2560	1280	2560	2560	2560	5120
A/Croatia/2737/2015	ngs?	2015-12-18	MDCKx/MDCK1	1280	640	640	640	1280	640	2560	2560	1280	2560
A/Croatia/2738/2015	ngs?	2015-12-18	MDCKx/MDCK1	1280	640	320	640	1280	640	2560	2560	1280	2560
A/Khmelnitsky/81/2016	ngs?	2015-12-20	SIAT2/MDCK1	1280	640	320	640	1280	640	2560	2560	1280	2560
A/Ukraine/25/2016	ngs?	2015-12-21	SIAT1/MDCK1	1280	640	640	640	1280	640	2560	2560	1280	2560
A/Khmelnitsky/78/2016	ngs?	2015-12-21	SIAT2/MDCK1	1280	640	320	640	640	640	1280	1280	1280	2560
A/Khmelnitsky/85/2016	ngs?	2015-12-21	SIAT2/MDCK1	1280	640	640	640	1280	640	2560	2560	1280	2560
A/Lisboa/65/2015	ngs?	2015-12-21	SIAT2/MDCK1	1280	640	640	640	1280	640	2560	2560	1280	5120
A/Croatia/2743/2015	ngs?	2015-12-21	MDCKx/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560
A/Lisboa/53/2015	ngs?	2015-12-22	SIAT1/MDCK1	1280	1280	640	1280	1280	640	2560	2560	1280	2560
A/Moldova/682-52/2015	ngs?	2015-12-22	MDCK3/MDCK1	1280	640	640	640	1280	1280	2560	2560	2560	2560
A/Ukraine/6915/2015	ngs?	2015-12-24	MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560
A/Algeria/VR40-15/2015	ngs?	2015-12-27	MDCK1	1280	320	320	640	1280	640	2560	1280	1280	2560
A/Ukraine/6910/2015	ngs?	2015-12-28	C1/MDCK1	320	320	160	160	320	320	640	640	640	1280
A/Ireland/169/2015	ngs?	2015-12-29	MDCK1/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560
A/Lisboa/66/2015	ngs?	2015-12-29	SIAT1/MDCK1	2560	1280	640	640	1280	640	2560	2560	1280	5120
A/Lisboa/68/2016	ngs?	2015-12-29	SIAT1/MDCK1	2560	1280	640	1280	2560	1280	2560	2560	2560	5120
A/Istanbul/194/2015	ngs?	2015-12-29	MDCK1/MDCK1	1280	640	320	640	1280	640	2560	2560	1280	5120
A/Ukraine/6884/2015	ngs?	2015-12-30	C1/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560
A/Algeria/G0282/16/2015	ngs?	2015-12-31	MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560
A/Algeria/G0283/16/2015	ngs?	2015-12-31	MDCK1	640	320	320	640	1280	640	2560	1280	1280	2560
A/Algeria/G0281/16/2016	ngs?	2016-01-01	MDCK1	640	640	320	640	1280	640	2560	1280	1280	2560
A/Lisboa/2/2016	ngs?	2016-01-02	SIAT2/MDCK2	1280	640	320	640	1280	640	1280	1280	1280	2560
A/Lisboa/5/2016	ngs?	2016-01-03	SIAT1/MDCK1	1280	640	640	1280	1280	1280	2560	2560	1280	2560
A/Ireland/767/2016	ngs?	2016-01-04	MDCK1/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560
A/Lisboa/1/2016	ngs?	2016-01-04	SIAT1/MDCK1	1280	1280	640	1280	1280	1280	2560	1280	1280	2560
A/Lisboa/7/2016	ngs?	2016-01-04	SIAT1/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	2560
A/Ireland/768/2016	ngs?	2016-01-05	MDCK1/MDCK1	1280	640	640	1280	1280	640	2560	2560	1280	2560
A/Ireland/2133/2016	ngs?	2016-01-06	MDCK1/MDCK1	1280	640	640	640	1280	640	2560	2560	2560	5120
A/Galat/190006/2016	ngs?	2016-01-06	MDCK1/MDCK1	1280	1280	1280	1280	2560	1280	5120	2560	1280	5120
A/Lisboa/4/2016	ngs?	2016-01-06	SIAT1/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	5120
A/Ireland/179/2016	ngs?	2016-01-07	MDCK1/MDCK1	1280	640	320	640	1280	640	2560	1280	1280	

Table 5-14. Summary of HI reactivity of influenza A(H1N1)pdm09 Test viruses with collection dates after 2015-08-31

HI Table	Fold reduction compared to homologous titre	Antiserum to Virus Host Genetic group	No of test viruses	Number of test viruses reacting within 2- or 4-fold of the antiserum homologous titre									
				A/Cal 7/09	A/Bayern 69/09	A/Lviv N6/09	A/Chch 16/10	A/Astrak 1/11	A/St P 27/11	A/St P 100/11	A/HK 5659/12	A/Sth Afr 3626/13	A/Slov 2903/15
				Egg	Cell	Cell	Egg	Cell	Egg	Egg	Cell	Egg	Egg
				1	1	1	4	5	6	7	6A	6B	6B.1
5-4	2-fold		6	5	6	1	0	6	6	6	6	6	
	4-fold			6	6	6	1	6	6	6	6	6	
5-5	2-fold		9	6	9	8	1	9	9	9	9	9	
	4-fold			9	9	9	6	9	9	9	9	9	
5-6	2-fold		9	8	9	9	0	9	9	9	9	9	
	4-fold			9	9	9	6	9	9	9	9	9	
5-7	2-fold		9	0	9	9	0	9	9	9	9	9	
	4-fold			8	9	9	5	9	9	9	9	9	
5-8	2-fold		4	4	4	4	0	0	4	4	4	4	
	4-fold			4	4	4	3	4	4	4	4	4	
5-9	2-fold		35	34	35	31	0	27	35	35	35	35	
	4-fold			35	35	35	1	35	35	35	35	35	
5-10	2-fold		33	32	32	4	1	31	32	32	32	32	
	4-fold			33	33	29	19	33	33	33	33	33	
5-11	2-fold		42	41	41	34	3	38	39	35	42	42	
	4-fold			42	42	40	30	41	41	41	42	42	
5-12	2-fold		89	88	89	82	13	88	88	88	89	88	89
	4-fold			89	89	89	73	89	89	88	89	89	89
5-13	2-fold		77	75	77	73	0	76	77	69	77	76	76
	4-fold			77	77	77	15	77	77	77	77	77	77
Total No			313										
	Number reacting within 2-fold			293	311	255	18	297	308	296	312	310	165
	Number reacting within 4-fold			312	313	307	159	312	312	311	313	313	166
	Percentage reacting within 2-fold			93.61	99.36	81.47	5.75	94.89	98.40	94.57	99.68	99.04	99.40
	Percentage reacting within 4-fold			99.68	100.00	98.08	50.80	99.68	99.68	99.36	100.00	100.00	100.00

Vaccine

Table 6-1. Antigenic analysis of influenza A(H1N1)pdm09 viruses - Plaque Reduction Neutralisation (MDCK cell) 2016-01-28

Viruses	Collection Date	Passage History	Neutralisation titre ¹												HA1 substitutions compared to A/California/7/2009: Egg adaptation HA substitutions compared to the corresponding cell isolate
			Post-infection ferret antisera												
			A/California 7/09		A/Bayern 69/09		A/St. Petersburg 27/11		A/South Africa 3626/13		A/Slovenia 2903/2015				
			Egg CDC F72/14		MDCK F09/15		Egg F26/14		Egg F03/14		MDCK F01/16				
			1		1		6		6B		6B.1				
			2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read			
REFERENCE VIRUSES															
A/California/7/2009	1	2009-04-09	E3/E1	1280	1516	640	737	1280	1117	2560	2405	5120	5632	D222D/G Q223Q/R	
A/Bayern/69/2009	1	2009-07-01	MDCK5/MDCK1	40	40	160	226	40	20	40	42	40	40	G155E	
A/St. Petersburg/27/2011	6	2011-02-14	E3	640	545	320	281	320	473	1280	987	2560	2703	Q223R K154K/E (K>E)	
A/South Africa/3626/2013	6B	2013-06-06	E1/E2	320	369	640	953	320	335	1280	1124	640	853	D127X	
A/Slovenia/2903/2015	6B.1	2015-10-26	E2	1280	1243	320	466	640	616	1280	1353	2560	2466	HA S84N, S162N, K163Q,I216T	
TEST VIRUSES															
Additional information															
A/Saudi Arabia/4001/2015	6B.1	2015-09-21	SIAT2	2560	3511	1280	1573	1280	1819	2560	3698	>5120	>5120	HA S84N, S162N, K163Q,I216T	
A/Saudi Arabia/4009/2015	6B.1	2015-09-21	MDCK1	5120	3840	2560	1920	2560	2560	5120	7314	>5120	>5120	HA S84N, S162N, K163Q,I216T	
A/Saudi Arabia/78093/2015	6B.1	2015-09-22	SIAT2	2560	3733	1280	1629	2560	2269	5120	4096	>5120	>5120	HA S84N, S162N, K163Q,I216T	
A/Saudi Arabia/78169/2015	6B.1	2015-09-22	SIAT2	2560	3339	1280	1856	2560	2265	5120	3881	>5120	>5120	HA S84N, S162N, K163Q,I216T	
A/Saudi Arabia/78649/2015	6B.1	2015-09-24	MDCK1	2560	2496	1280	1674	2560	1958	5120	3840	>5120	>5120	HA S84N, S162N, K163Q,I216T	
A/Saudi Arabia/78524/2015	6B.1	2015-09-24	MDCK1	5120	4716	2560	2737	2560	2720	5120	4736	>5120	>5120	HA S84N, S162N, K163Q,I216T	
A/Saudi Arabia/78558/2015	6B	2015-09-24	MDCK1	2560	3673	1280	1128	1280	1707	2560	3313	>5120	>5120	HA K163Q	
A/Saudi Arabia/78741/2015	6B.1	2015-09-25	MDCK2	5120	3947	1280	1886	2560	2633	2560	3413	>5120	>5120	HA S84N, S162N, K163Q,I216T	
A/Norway/2633/2015	6B	2015-10-27	MDCK1/MDCK1	2560	3520	1280	1625	1280	1455	2560	3673	>5120	>5120	HA S84N, K163Q	
A/Norway/2646/2015	6B	2015-11-03	MDCK1/MDCK1	5120	5120	2560	2377	1280	1732	5120	5120	>5120	>5120	HA S84N, K163Q	
A/Norway/2650/2015	6B.1	2015-11-03	MDCK1/MDCK1	2560	2337	1280	1536	1280	1667	2560	3072	5120	5120	HA S84N, S162N, K163Q,I216T	
A/Norway/2674/2015	6B	2015-11-04	MDCK2	2560	3129	1280	1422	1280	1338	2560	3214	5120	5120	HA S84N, K163Q	
A/Norway/2685/2015	6B.1	2015-11-13	MDCK1	5120	5120	2560	2160	2560	2560	5120	4969	>5120	>5120	HA S84N, S162N, K163Q,I216T	
				Vaccine											

¹ Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 80% reduction in plaque formation

Table 6-2. Antigenic analysis of influenza A(H1N1)pdm09 viruses - Plaque Reduction Neutralisation (MDCK cell) 2016-02-03

Neutralisation titre ¹														HA1 substitutions compared to A/California/7/2009: Egg adaptation HA substitutions compared to the corresponding cell isolate
Viruses	Collection Date	Passage History	Post-infection ferret antisera											
			A/California 7/09		A/Bayern 69/09		A/St. Petersburg 27/11		A/South Africa 3626/13		A/Slovenia 2903/2015			
			Egg CDC F72/14		MDCK F09/15		Egg F26/14		Egg F03/14		MDCK F01/16			
			1		1		6		6B		6B.1			
			2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read		
REFERENCE VIRUSES														D222D/G Q223Q/R G155E Q223R K154K/E (K>E) D127X HA S84N, S162N, K163Q,I216T
A/California/7/2009	1	2009-04-09	E3/E1	1280	1516	640	737	1280	1117	2560	2405	5120	5632	
A/Bayern/69/2009	1	2009-07-01	MDCK5/MDCK1	40	40	160	226	40	20	40	42	40	40	
A/St. Petersburg/27/2011	6	2011-02-14	E3	640	545	320	281	320	473	1280	987	2560	2703	
A/South Africa/3626/2013	6B	2013-06-06	E1/E2	320	369	640	953	320	335	1280	1124	640	853	
A/Slovenia/2903/2015	6B.1	2015-10-26	MDCKx/MDCK1	5120	3956	2560	2710	2560	1984	5120	5421	10240	10240	
TEST VIRUSES														Additional information
A/Iran/112039/2015	6B.1	2015-10-05	SIAT1/MDCK1	10240	11378	2560	2880	5120	5535	5120	7111	>10240	>10240	HA S84N, S162N, K163Q,I216T
A/England/350/2015	6B.1	2015-10-06	SIAT1/MDCK1	5120	8271	2560	2133	5120	4053	5120	6777	>10240	>10240	HA S84N, S162N, K163Q,I216T
A/Berlin/166/2015	6B	2015-10-21	C2/MDCK1	>10240	>10240	1280	1629	5120	3956	10240	8014	10240	8777	HA K163Q
A/Bremen/25/2015	6B	2015-11-09	C2/MDCK1	10240	11378	2560	2062	5120	3840	10240	8014	>10240	>10240	HA S84N, K163Q,
A/Berlin/169/2015	6B	2015-11-09	C2/MDCK1	10240	11378	5120	6174	2560	2560	5120	6982	10240	9309	HA S84N, K163Q
A/Finland/541/2015	6B.1	2015-11-09	MDCK1/MDCK1	5120	4446	2560	3291	5120	3988	10240	10240	>10240	>10240	HA S84N, S162N, K163Q,I216T
A/Berlin/168/2015	6B.1	2015-11-10	C2/MDCK1	10240	8046	2560	2560	2560	3129	5120	7411	10240	9638	HA S84N, S162N, K163Q,I216T
A/Netherlands/2915/2015	6B.1	2015-11-10	MDCK1/MDCK1	5120	7168	2560	3638	5120	4480	10240	10240	>10240	>10240	HA S84N, S162N, K163Q,I216T
A/Berlin/167/2015	6B.1	2015-11-13	C2/MDCK1	5120	6328	2560	2560	2560	2291	5120	6720	10240	10779	HA S84N, S162N, K163Q,I216T
A/Sachsen//102/2015	6B.1	2015-11-13	C2/MDCK1	5120	6758	5120	5705	5120	5486	5120	6981	10240	14628	HA S84N, S162N, K163Q,I216T
A/Nordrhein-Westfalen/103/2015	6B.1	2015-11-13	C2/MDCK1	5120	3840	1280	1778	2560	2133	5120	5255	10240	9056	HA S84N, S162N, K163Q,I216T
A/England/365/2015	6B.1	2015-11-17	SIAT1/MDCK1	5120	6051	2560	2027	1280	1792	5120	6737	>10240	>10240	HA S84N, S162N, K163Q,I216T
A/Bayern/147/2015	6B.1	2015-11-19	C2/MDCK1	10240	12047	5120	4422	5120	5383	5120	7680	>10240	>10240	HA S84N, S162N, K163Q,I216T
A/Nordrhein-Westfalen/102/2015	6B.1	2015-11-23	C2/MDCK1	5120	6179	2560	3840	5120	4224	5120	7089	>10240	>10240	HA S84N, S162N, K163Q,I216T
A/Iran/116675/2015	6B.1	2015-11-28	E1/E1	1280	1529	1280	1280	640	985	2560	2348	5120	5242	HA S84N, S162N, K163Q,I216T
A/Iran/1337/2015	6B.1	2015-11-28	SIAT1/MDCK1	5120	4422	2560	2377	5120	4608	5120	6583	>10240	>10240	HA S84N, S162N, K163Q,I216T
A/Norway/2774/2015	6B.2	2015-11-30	MDCK2	640	825	320	420	640	763	2560	3040	2560	2070	HA R113K,D122E,V152T, K163Q,V173I
A/Finland/553/2015	6B.1	2015-12-06	MDCK1/MDCK1	5120	7124	2560	3520	5120	4480	5120	7040	>10240	>10240	HA S84N, S162N, K163Q,I216T
A/Ghana/DILI-15-1281/2015	6B	2015-12-08	Cx/MDCK1	5120	4189	1280	1152	1280	1182	1280	1827	10240	8960	HA K163Q
A/Latvia/12-037196/2015	6B.1	2015-12-15	C2/MDCK1	5120	4023	2560	2560	2204	2560	5120	5514	10240	8704	HA S84N, S162N, K163Q,I216T
A/Latvia/12-042353/2015	6B.1	2015-12-16	C1/MDCK1	5120	6485	2560	2240	5120	3840	10240	12047	>10240	>10240	HA S84N, S162N, K163Q,I216T
A/Latvia/12-063862/2015	6B.1	2015-12-29	C1/MDCK1	2560	2560	1280	1463	1280	2112	2560	3145	10240	10779	HA S84N, S162N, K163Q,I216T

Vaccine

¹ Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 80% reduction in plaque formation

Figure 1. Phylogenetic comparison of influenza A(H1N1)pdm09 HA genes

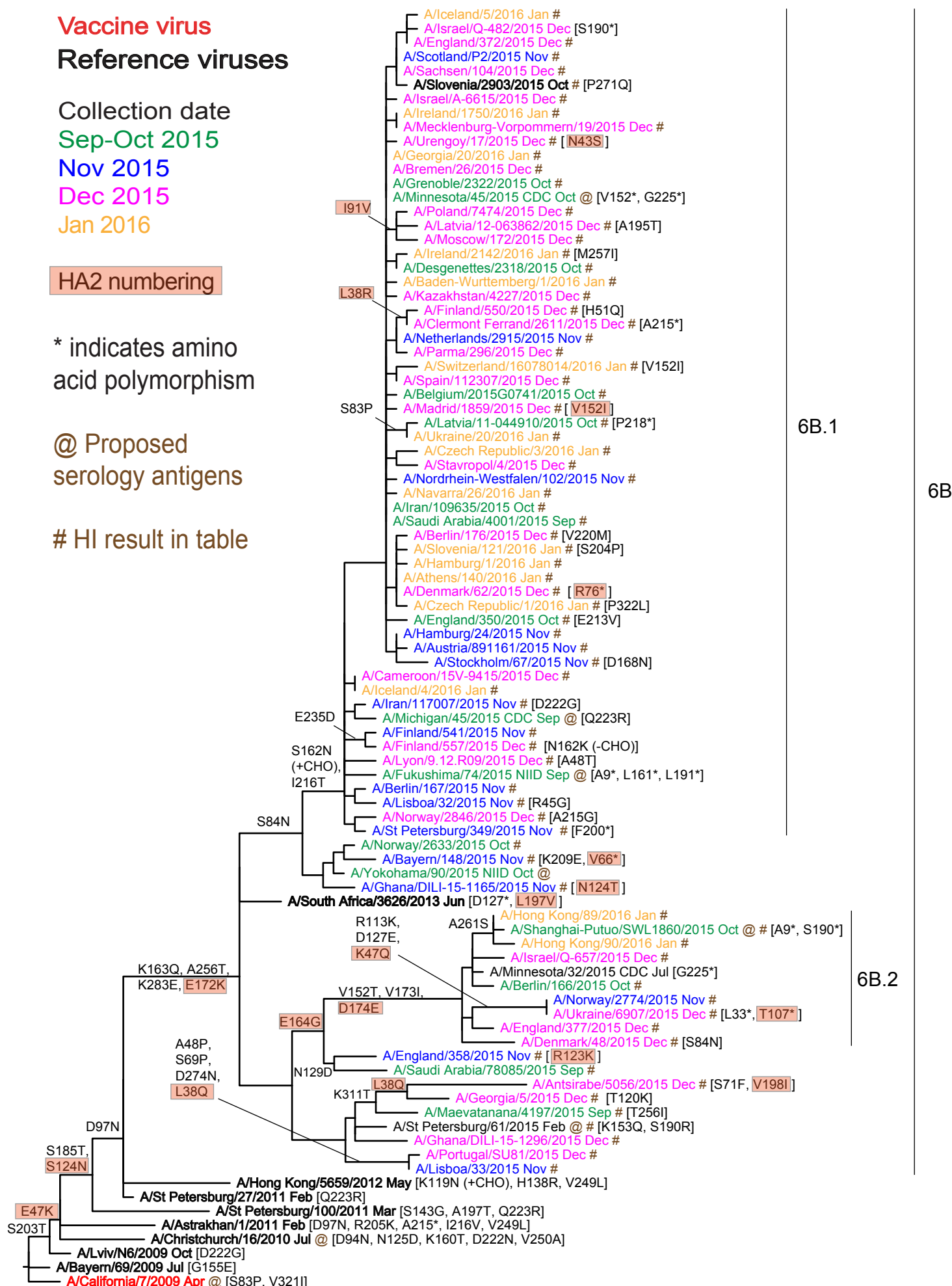
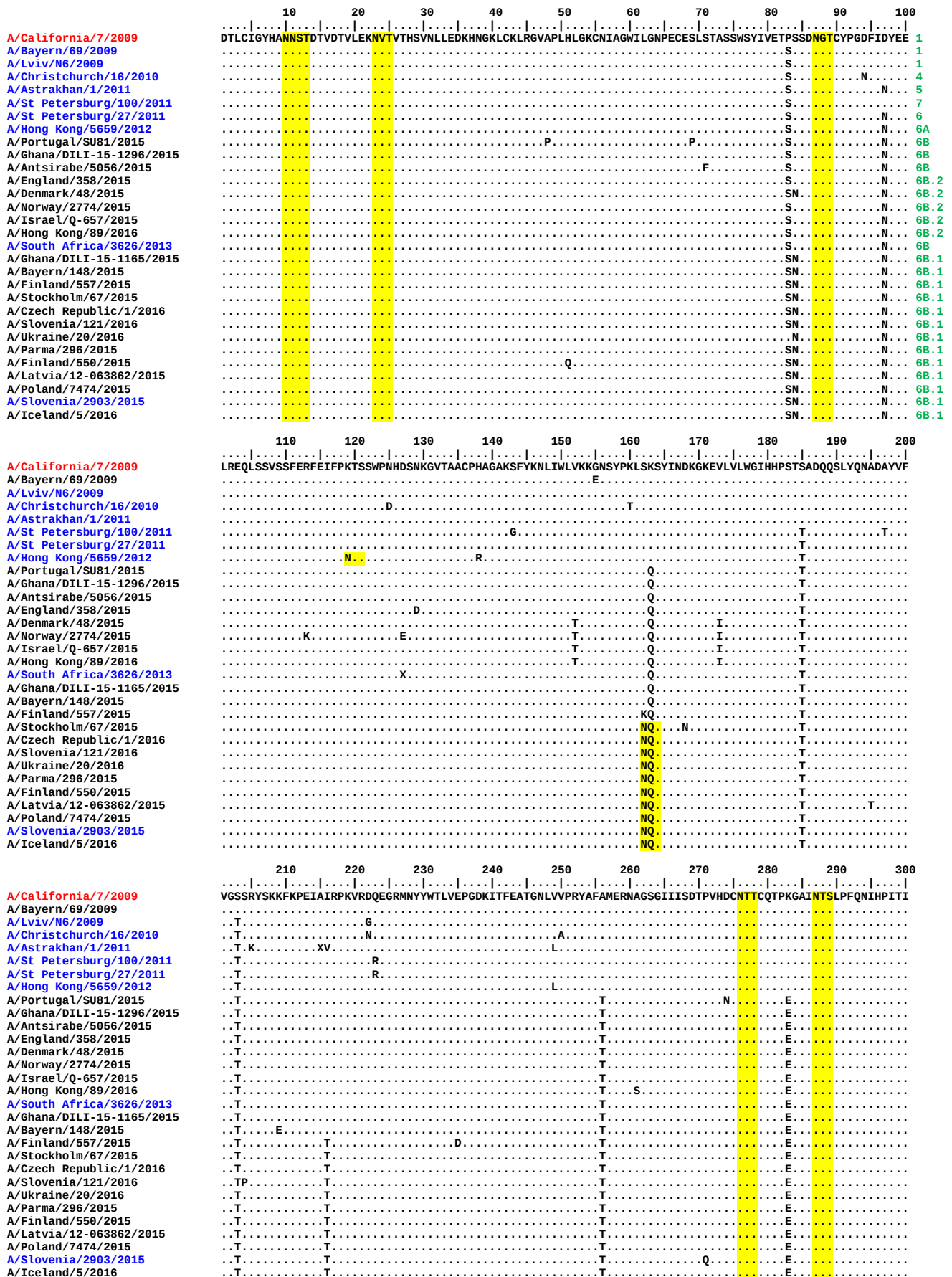


Figure 2. HA protein alignment for A(H1N1)pdm09 viruses

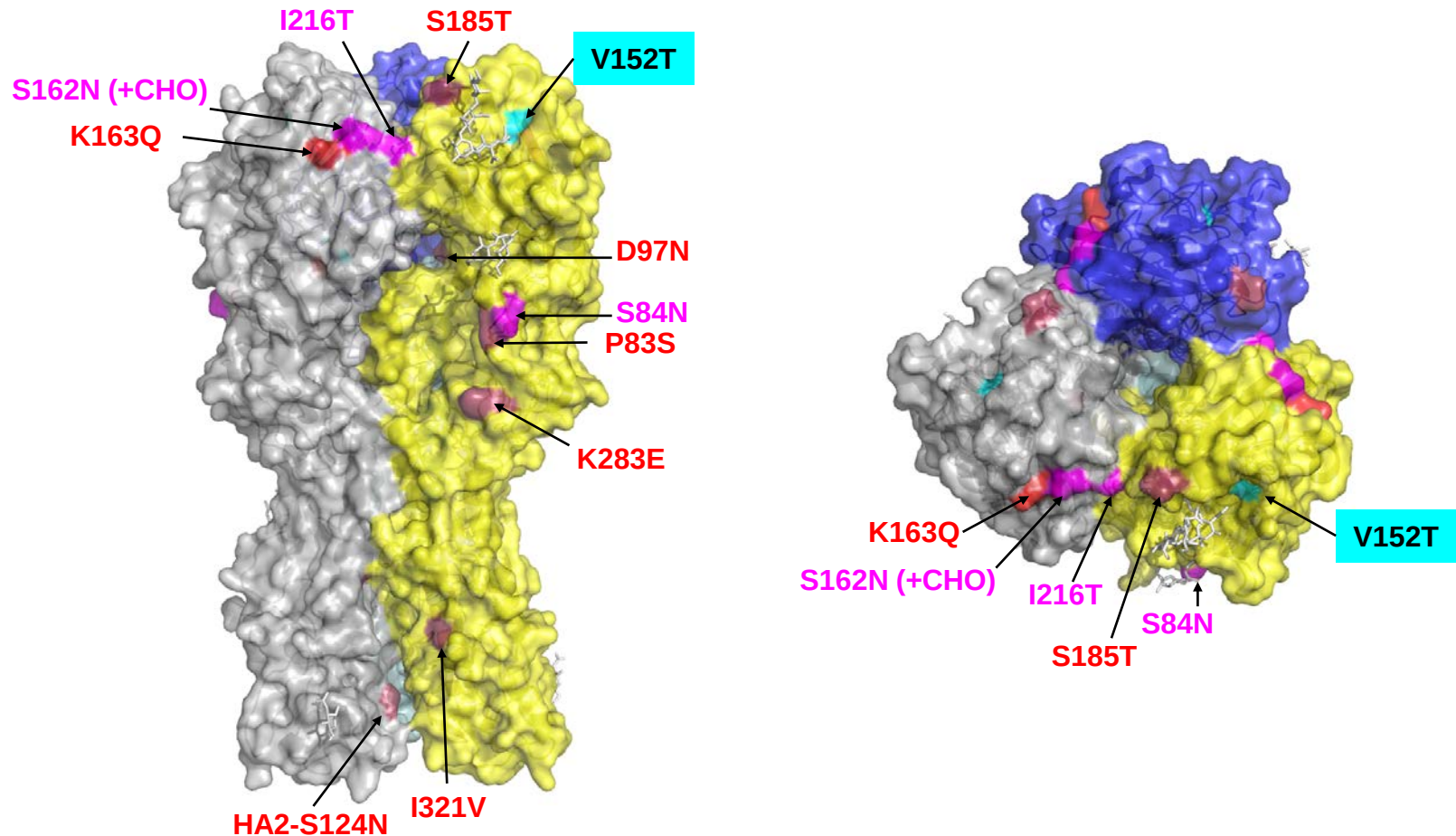
Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site



	310	320	330	340	350	360	370	380	390	400
A/California/7/2009	GKCPKYVKSTKLRLATGLRNIPSIQSR	GLFGA	IAGFIEGGWTGMVDGWYGYHHQNEQGS	GYAADLKSTQNA	IDEITNKVNSVIEKMNTQFTAVGKEFNHL					
A/Bayern/69/2009		V								
A/Lviv/N6/2009		V								
A/Christchurch/16/2010		V							K	
A/Astrakhan/1/2011		V							K	
A/St Petersburg/100/2011		V							K	
A/St Petersburg/27/2011		V							K	
A/Hong Kong/5659/2012		V							K	
A/Portugal/SU81/2015		V					Q		K	
A/Ghana/DILI-15-1296/2015		V							K	
A/Antsirabe/5056/2015		T	V				Q		K	
A/England/358/2015		V							K	
A/Denmark/48/2015		V							K	
A/Norway/2774/2015		V						Q		
A/Israel/Q-657/2015		V							K	
A/Hong Kong/89/2016		V							K	
A/South Africa/3626/2013		V							K	
A/Ghana/DILI-15-1165/2015		V							K	
A/Bayern/148/2015		V							K	X
A/Finland/557/2015		V							K	
A/Stockholm/67/2015		V							K	
A/Czech Republic/1/2016		VL							K	
A/Slovenia/121/2016		V							K	
A/Ukraine/20/2016		V							K	
A/Parma/296/2015		V							K	
A/Finland/550/2015		V					R		K	
A/Latvia/12-063862/2015		V							K	
A/Poland/7474/2015		V							K	
A/Slovenia/2903/2015		V							K	
A/Iceland/5/2016		V							K	
	410	420	430	440	450	460	470	480	490	500
A/California/7/2009	EKRLENLNKKVDDGFLDIWTYNAELLVLL	ENERTLDYHDSNVKNLYEKVRSQ	LKNNAKEIGNGCFEFYHKDNTCMESV	KNGTYDYPKYSEEA	KLNR	E				
A/Bayern/69/2009										
A/Lviv/N6/2009										
A/Christchurch/16/2010										
A/Astrakhan/1/2011										
A/St Petersburg/100/2011					N					
A/St Petersburg/27/2011					N					
A/Hong Kong/5659/2012					N					
A/Portugal/SU81/2015					N					K
A/Ghana/DILI-15-1296/2015					N					K
A/Antsirabe/5056/2015					N					K
A/England/358/2015					KN				G	K
A/Denmark/48/2015					N				G	K
A/Norway/2774/2015					N				G	K
A/Israel/Q-657/2015					N				G	K
A/Hong Kong/89/2016					N				G	K
A/South Africa/3626/2013					N					K
A/Ghana/DILI-15-1165/2015					T					K
A/Bayern/148/2015					N					K
A/Finland/557/2015					N					K
A/Stockholm/67/2015					N					K
A/Czech Republic/1/2016					N					K
A/Slovenia/121/2016					N					K
A/Ukraine/20/2016					N					K
A/Parma/296/2015					N					K
A/Finland/550/2015					N					K
A/Latvia/12-063862/2015			V		N					K
A/Poland/7474/2015			V		N					K
A/Slovenia/2903/2015					N					K
A/Iceland/5/2016					N					K
	510	520	530	540						
A/California/7/2009	DGVKLESTRIYQILAIYSTVASSLVLVSL	GAISFWMCSNGSLQCRICI								
A/Bayern/69/2009										
A/Lviv/N6/2009										
A/Christchurch/16/2010										
A/Astrakhan/1/2011										
A/St Petersburg/100/2011										
A/St Petersburg/27/2011										
A/Hong Kong/5659/2012										
A/Portugal/SU81/2015										
A/Ghana/DILI-15-1296/2015										
A/Antsirabe/5056/2015					I					
A/England/358/2015										
A/Denmark/48/2015	E									
A/Norway/2774/2015	E									
A/Israel/Q-657/2015	E									
A/Hong Kong/89/2016	E									
A/South Africa/3626/2013			V							
A/Ghana/DILI-15-1165/2015										
A/Bayern/148/2015										
A/Finland/557/2015										
A/Stockholm/67/2015										
A/Czech Republic/1/2016										
A/Slovenia/121/2016										
A/Ukraine/20/2016										
A/Parma/296/2015										
A/Finland/550/2015										
A/Latvia/12-063862/2015										
A/Poland/7474/2015										
A/Slovenia/2903/2015										
A/Iceland/5/2016										

Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

Figure 3. H1 HA locations of amino acids substitutions in genetic subgroups 6B.1 and 6B.2



Amino acid substitutions compared with A/California/7/2009 (see Figure 1). Those that define genetic group 6B are in **red** (S203T, A256T with E47K, E172K in HA2 not visible on this structure), those that define subgroup 6B.1 in **magenta**, those that define subgroup 6B.2 in **cyan** (V173I with E164G, D174E in HA2 not visible on this structure). A number of viruses in subgroup 6B.2 carry additional substitutions (e.g. R113K, D127E with K47Q in HA2).

Figure 4. Phylogenetic comparison of influenza A(H1N1)pdm09 NA genes

Vaccine virus

Reference viruses

Collection date

Sep/Oct 2015

Nov 2015

Dec 2015

Jan 2016

* indicates amino acid polymorphism

@ Proposed serology antigens

HI result in table

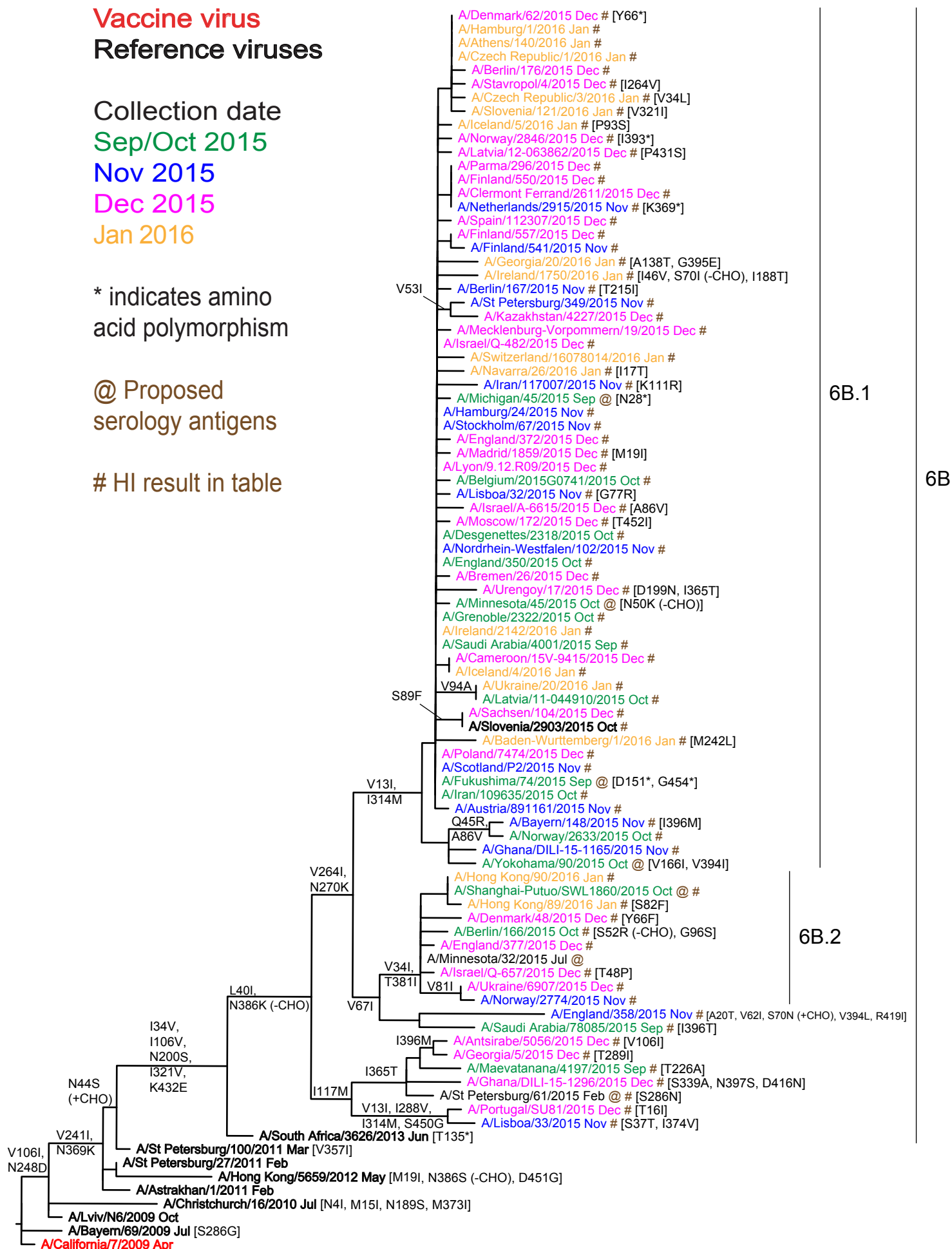
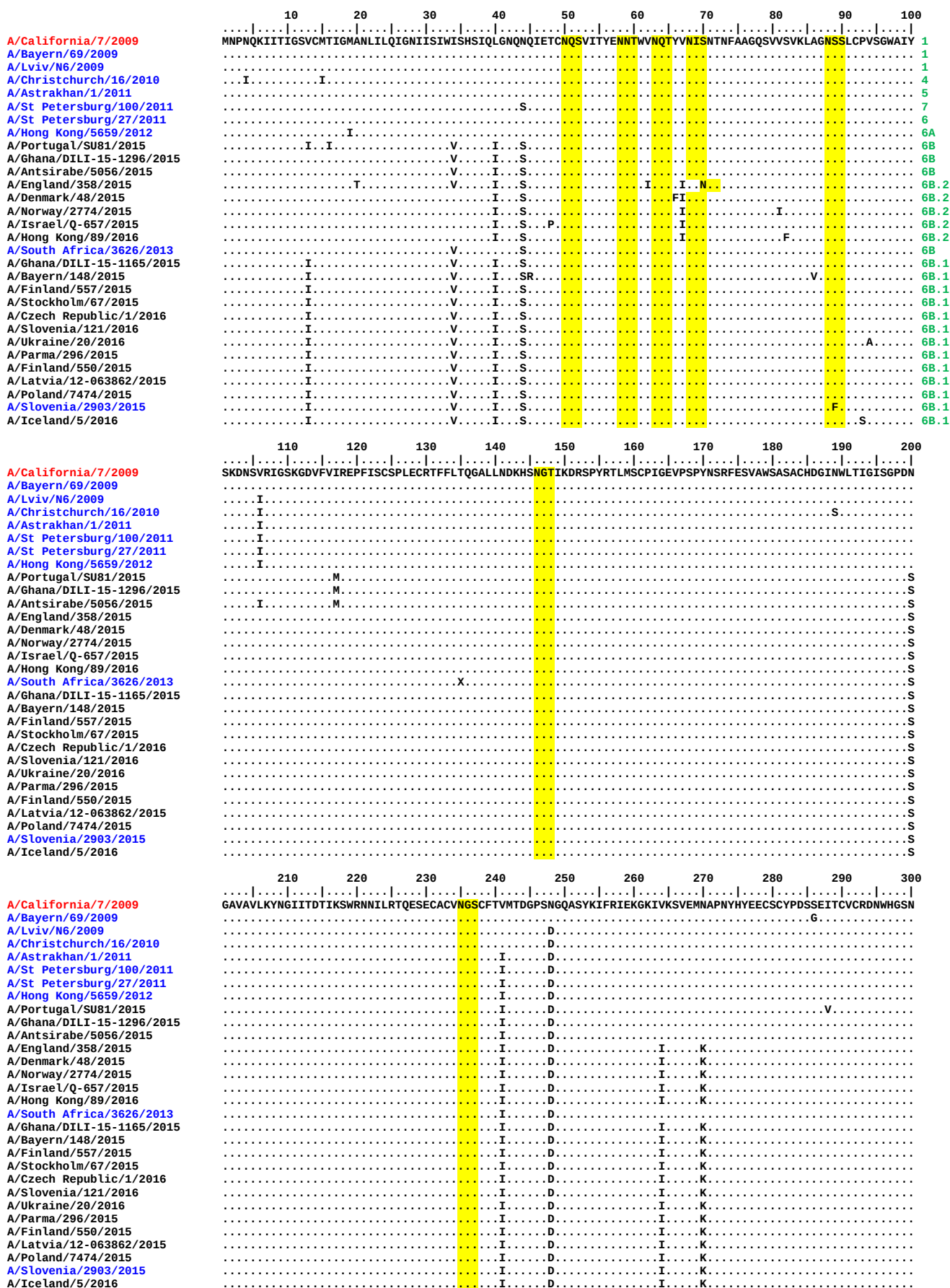


Figure 5. NA protein alignment for A(H1N1)pdm09 viruses

Vaccine virus: Reference virus: Genetic group: Potential N-linked glycosylation site



	310	320	330	340	350	360	370	380	390	400
A/California/7/2009	RPWVSFNQ	LEYQIGYIC	SGIFGDN	PRPNDKT	GSCGPVSS	NGANGVK	GFSGFYGN	GVWIGRT	KSISSRNG	FEMIWD
A/Bayern/69/2009										
A/Lviv/N6/2009										
A/Christchurch/16/2010								I		
A/Astrakhan/1/2011							K			
A/St Petersburg/100/2011						I	K			
A/St Petersburg/27/2011							K			
A/Hong Kong/5659/2012							K		S	
A/Portugal/SU81/2015	M	V					K		K	
A/Ghana/DILI-15-1296/2015		V		A			T	K		S
A/Antsirabe/5056/2015		V					T	K		M
A/England/358/2015		V					K		K	L
A/Denmark/48/2015		V					K		I	K
A/Norway/2774/2015		V					K		I	K
A/Israel/Q-657/2015		V					K		I	K
A/Hong Kong/89/2016		V					K		I	K
A/South Africa/3626/2013		V					K			
A/Ghana/DILI-15-1165/2015	M	V					K		K	
A/Bayern/148/2015	M	V					K		K	M
A/Finland/557/2015	M	V					K		K	
A/Stockholm/67/2015	M	V					K		K	
A/Czech Republic/1/2016	M	V					K		K	
A/Slovenia/121/2016	M						K		K	
A/Ukraine/20/2016	M	V					K		K	
A/Parma/296/2015	M	V					K		K	
A/Finland/550/2015	M	V					K		K	
A/Latvia/12-063862/2015	M	V					K		K	
A/Poland/7474/2015	M	V					K		K	
A/Slovenia/2903/2015	M	V					K		K	
A/Iceland/5/2016	M	V					K		K	
	410	420	430	440	450	460				
A/California/7/2009	GYSGSFVQ	HPELTGL	DCIRPCF	WVELIRG	RPKENTI	IWTSGSSIS	FCGVNSD	TVGWSW	PDGAELP	FTIDK
A/Bayern/69/2009										
A/Lviv/N6/2009										
A/Christchurch/16/2010										
A/Astrakhan/1/2011										
A/St Petersburg/100/2011										
A/St Petersburg/27/2011										
A/Hong Kong/5659/2012							G			
A/Portugal/SU81/2015						E				
A/Ghana/DILI-15-1296/2015		N				E				
A/Antsirabe/5056/2015						E				
A/England/358/2015		I				E				
A/Denmark/48/2015						E				
A/Norway/2774/2015						E				
A/Israel/Q-657/2015						E				
A/Hong Kong/89/2016						E				
A/South Africa/3626/2013						E				
A/Ghana/DILI-15-1165/2015						E				
A/Bayern/148/2015						E				
A/Finland/557/2015						E				
A/Stockholm/67/2015						E				
A/Czech Republic/1/2016						E				
A/Slovenia/121/2016						E				
A/Ukraine/20/2016						E				
A/Parma/296/2015						E				
A/Finland/550/2015						E				
A/Latvia/12-063862/2015					SE					
A/Poland/7474/2015						E				
A/Slovenia/2903/2015						E				
A/Iceland/5/2016						E				

Vaccine virus: Reference virus: Genetic group: Potential N-linked glycosylation site

Influenza A(H3N2) virus analyses.

Influenza A(H3N2) viruses collected since 2015-08-31 have been received from NICs in 32 countries or administrative regions in four WHO Regions: African (4: Algeria, Cameroon, Ghana and Zambia), Eastern Mediterranean (5: Iran, Jordan, Oman, Saudi Arabia and Tunisia), European (22 countries) and Western Pacific (1: Hong Kong SAR). Influenza A(H3N2) viruses have become increasingly difficult to characterise antigenically by HI assay due to either the variable agglutination of red blood cells (RBCs) from guinea pig, turkey and humans or the loss of the ability of viruses to agglutinate any of these RBCs. Of the successfully propagated viruses collected since 2015-0-31, just over 44% had sufficient HA titre to be analysed by HI assays conducted using guinea pig RBCs in the presence of 20nM oseltamivir, added to circumvent NA-mediated binding to the RBCs (**Tables 2 & 7, Figure 6**). The remainder were unable to agglutinate guinea pig RBCs at all (45% of viruses collected since 2015-08-31: **Table 7**) or were unable to agglutinate RBCs in the presence of 20nM oseltamivir (11% of viruses). Of the 36 viruses in these last two categories that were sequenced, all belonged to genetic subgroup 3C.2a. Nine (75%) of the 12 viruses in genetic subgroup 3C.2a with collection dates after 2015-08-31, that were able to bind guinea pig RBCs in the presence of oseltamivir, showed either loss of or polymorphism at the 158/160 glycosylation motif in HA1 acquired by viruses in this genetic subgroup (**Tables 9-3 to 9-6**). Of the viruses able to agglutinate turkey RBCs, for which the addition of oseltamivir reduced their respective titres on guinea pig RBCs by two-fold or less and for which the respective HA sequences were known, ~80% (19 of 24) had HA sequences that fell into genetic clade 3C.3a (**Figure 6**).

HI analyses.

A total of 46 viruses with collection dates after 2015-08-31 could be analysed by HI assay and summaries of the HI results is shown in **Tables 3 and 8**. Individual HI results are shown in **Tables 9-1 to 9-7**. Test viruses in each table are sorted by date of collection and those below the red horizontal lines in the tables are viruses with collection dates after 2015-08-31. The HA genetic group is indicated for those viruses that have been sequenced and those included in phylogenetic trees are highlighted. Notable substitutions in the HA glycoprotein are also shown. Two sets of HI results for egg-propagated cultivars of viruses from China and Australia are shown in **Tables 9-1 and 9-3**.

A summary of the HI results for all H3N2 viruses that could be tested by HI since our report for the September 2015 consultation, sorted based on gene sequencing to be in one of four genetic clades (3C.2a, 3C.3, 3C.3a or 3C.3b), is shown in **Table 9-8**.

A summary of the HI results for viruses collected since 2015-08-31 with antisera raised against the cell culture-propagated cultivar of A/Hong Kong/5738/2014 and

egg-propagated A/Hong Kong/4801/2014 is shown in **Table 8**. The antiserum raised against A/Hong Kong/5738/2014 recognised 97% of viruses at titres within 4-fold of the titre of the antiserum with the homologous virus, 59% within 2-fold of the homologous titre. However, the antiserum raised against egg-propagated A/Hong Kong/4801/2014 recognised only 53% of test viruses with collection dates after 2015-08-31 at titres within 4-fold of the titre of the antiserum with the homologous virus and only 35% at a titre within 2-fold of the titre with the homologous virus. In **Tables 9-4 to 9-7** an antiserum raised against a cell culture-propagated cultivar of A/Hong Kong/4801/2014 was also used. For test viruses characterised genetically and antigenically by HI since the September 2015 report, this antiserum recognised 64% of those viruses in genetic subgroup 3C.2a, 79% of those in genetic subgroup 3C.3a and all those in genetic subgroup 3C.3b, at titres within 4-fold of the titre with the homologous virus; but none of those in genetic group 3C were recognised at titres within 4-fold of the titre for the homologous virus (**Table 9-8**).

A/Georgia/532/2015 is a cell culture-propagated reference virus that carries the amino acid substitutions R142K and Q197R in HA1 and shows polymorphism at the 158-160 glycosylation motif. Antiserum raised against this virus recognised only 35% of test viruses collected since 2015-08-31 at titres within 4-fold of that for the homologous virus; of those analysed genetically, it recognised 30% of the genetic subgroup 3C.2a viruses, 36% of those in genetic subgroup 3C.3a and 42% of those in genetic group 3C.3 but only one of the 13 viruses in genetic subgroup 3C.3b, at titres within 4-fold of the titre for the homologous virus.

Antiserum raised against the previously recommended vaccine virus, egg-propagated A/Switzerland/9715293/2013, recognised only six of the test viruses (12% of the total) collected since 2015-08-31 at titres within 4-fold of the titre for the homologous virus. Of those analysed genetically and analysed by HI since the report for the September 2015 consultation a total of 13 viruses (15% of the total) were recognised at a titre within 4-fold of the homologous titre (**Table 9-8**). This antiserum recognised 54% of viruses in genetic group 3C.3, 14.3%, 7.7% and 2.9% of those in genetic subgroups 3C.3a, 3C.3b and 3C.2a respectively, at titres within 4-fold of the titre with the homologous virus. The antiserum raised against cell culture-propagated A/Switzerland/9715293/2013 had a low homologous titre of 80, so it was not possible to estimate the number of viruses that were recognised by this antiserum at a titre within 4-fold of the homologous titre. However, 39 viruses (44% of the total) were recognised at a titre >2-fold reduced compared to the low homologous titre.

Compared to antiserum raised against the egg-propagated cultivar of A/Switzerland/9715293/2013, antiserum raised against egg-propagated A/Stockholm/6/2014 recognised substantially more (74%) of the test viruses collected since 2015-08-31 at titres within 4-fold of the titre with the homologous virus. For viruses analysed since our report for the September 2015 consultation that were analysed genetically: 66% of the 3C.2a viruses, all of the 3C.3a, 92% of the

3C.3b and all of the 3C.3 viruses were recognised by the antiserum raised against egg-propagated A/Stockholm/6/2014 at titres within 4-fold of the titre with the homologous virus (**Table 9-8**). Antiserum raised against the cell-propagated cultivar of A/Stockholm/6/2014 recognised well the test viruses that had been analysed genetically, with 74% of the 3C.2a viruses, 64% of 3C.3a viruses and all of the 3C.3a and 3C.3b viruses being recognised at titres within 4-fold of the titre for the homologous virus (**Table 9-8**).

Post-infection ferret antiserum raised against egg-propagated A/Texas/50/2012, a former vaccine virus, recognised the vast majority of cell culture-propagated test viruses poorly in HI assays showing ≥ 8 -fold decreased titres compared to the titre of the antiserum with the homologous virus. Only one virus collected since 2015-08-31 was recognised at a titre within 4-fold of the titre of the antiserum with the homologous virus. Of those viruses tested since the report for September 2015 for which gene sequencing was completed, 6 viruses (7% of the total) were recognised by this antiserum at titres within 4-fold of the homologous titre and all had HA genes in genetic group 3C.3 or subgroup 3C.3b. This represented 67% and 54%, respectively, of the viruses from these two genetic clades.

An antiserum raised against a cell culture-propagated reference virus in genetic subgroup 3C.3b, A/Netherlands/525/2014, was used in the HI tests. It recognised none of the test viruses in genetic subgroups 3C.2a or 3C.3a at titres within 4-fold of the titre for the homologous virus. However, the antiserum recognised eight of the 13 (61%) viruses in genetic group 3C.3, and eight of the 13 viruses assigned to genetic subgroup 3C.3b, at titres within 4-fold of the titre for the homologous virus (**Table 9-8**).

Virus Neutralisation.

Plaque reduction neutralisation assays (PRNA) were used to complement HI tests and allowed antigenic characterisation of viruses that could not be assayed by HI.

The results of PRNA performed with MDCK-SIAT1 cells are shown in **Tables 10-1 to 10-3**. The genetic groups and noteworthy sequence features in the HA are marked. A total of 47 test viruses were analysed, of which 70% were from genetic subgroup 3C.2a, 28% from genetic subgroup 3C.3a, while a single virus was of the 3C.3b subgroup.

An antiserum raised against the egg-propagated vaccine virus A/Hong Kong/4801/2014 recognised 39% of the test viruses belonging to genetic subgroup 3C.2a, but none of the 13 viruses belonging to genetic subgroup 3C.3a nor the subgroup 3C.3b virus, at titres within 4-fold of the titre for the homologous.

However, an antiserum raised against cell culture-propagated A/Hong Kong/7295/2014 recognised all of the test viruses belonging to genetic subgroup 3C.2a at a titre within 4-fold of the titre for the homologous virus. It also recognised

38% (5/13) of viruses in genetic subgroup 3C.3a, and the subgroup 3C.3b virus, at titres within 4-fold of the titre with the homologous virus. It should be noted though that the viruses in genetic subgroups 3C.3a and 3C.3b were recognised at a lower titre relative to the homologous titre than those in genetic subgroup 3C.2a.

A/South Africa/R2651/2015 is a cell culture-propagated reference virus in genetic subgroup 3C.2a that carries the amino acid substitutions Q197R and I214T in HA1. The antiserum raised against this virus recognised 70% of test viruses in genetic subgroup 3C.2a at titres within 4-fold of the antiserum with the homologous virus. None of the test viruses in genetic subgroups 3C.3a and 3C.3b were recognised at titres within 4-fold of the homologous titre for this antiserum.

A/South Africa/R2665/2015 is a cell culture-propagated reference virus in genetic subgroup 3C.2a that carries the amino acid substitutions R142K and Q197R in HA1. The antiserum raised against this virus has a high homologous titre and recognised only 21% of the test viruses in genetic subgroup 3C.2a at titres within 4-fold of the homologous titre. Again, none of the test viruses in genetic subgroups 3C.3a and 3C.3b were recognised at titres within 4-fold of the homologous titre.

The antiserum raised against cell-culture propagated A/Switzerland/9715293/2013, had a low homologous titre of 137 (160). This antiserum recognised 90% (30/33) of 3C.2a viruses, all 13 3C.3a viruses and the 3C.3b test virus at titres within 4-fold of the titre with the homologous virus.

The antiserum raised against A/Netherlands/525/2014, a genetic subgroup 3C.3b virus, recognised the subgroup 3.3b test virus very well at a titre within 2-fold of the titre of the antiserum with the homologous virus. All test viruses from genetic subgroups 3C.2a and 3C.3a were recognised poorly by this antiserum.

Genetic analyses.

Phylogenetic analysis of the HA genes of representative recently circulating H3N2 viruses is shown in **Figure 7**. The HA genes fell within genetic group 3C. This group has three subdivisions: 3C.1, 3C.2 and 3C.3; further subdivisions have emerged in genetic group subdivisions 3C.2 and 3C.3 yielding subgroups designated as 3C.2a, 3C.3a and 3C.3b respectively (**Figure 7**).

The previously used vaccine virus A/Texas/50/2012 belongs to genetic subdivision 3C.1. Viruses received with collection dates since 2015-08-31 that have been sequenced fell into subgroups 3C.2a (48: 68.6%), 3C.3b (1: 1.4%), 3C.3a (20: 28.6%) and genetic group subdivision 3C.3 (1: 1.4%). Amino acid substitutions that define these new subgroups compared with A/Texas/50/2012 are:

(3C.2a) L3I, N128T (resulting in the gain of a potential glycosylation site), **N144S** (resulting in the loss of a potential glycosylation site), **N145S, F159Y, K160T** (resulting in the gain of a potential glycosylation site), **V186G[@], P198S, F219S, N225D** and **Q311H** in **HA1**, e.g. A/Hong Kong/5738/2014. Some

viruses in this subgroup also carry the substitution **Q197R** in **HA1**, some combined with **R142K**. Another cluster within the subgroup has substitutions **N171K** in **HA1** and **I77V** and **G155E** in **HA2**.

(3C.3a) T128A (resulting in the loss of a potential glycosylation site), **A138S**, **R142G**, **N145S**, **F159S**, **V186G**[@], **P198S**, **F219S** and **N225D** in **HA1**, e.g. A/Switzerland/9715293/2013. Many recent viruses also carry **Y94H** in **HA1**.

(3C.3b) E62K, **K83R**, **N122D** (resulting in the loss of a potential glycosylation site), **T128A** (resulting in the loss of a potential glycosylation site), **R142G**, **N145S**, **L157S**, **V186G**[@], **P198S**, **F219S** and **R261Q** in **HA1** with **V18K** in **HA2**, e.g. A/Norway/2605/2015. Some also carry the substitution **Q917H** in **HA1**, e.g. A/Netherlands/525/2014.

[@]Note: the **G186V** substitution in HA1 occurred during adaptation of A/Texas/50/2012 to propagation in hens' eggs.

Figure 8 shows an HA amino acid sequence alignment for a representative selection of the viruses used to generate the phylogeny (**Figure 7**). The vaccine viruses A/Switzerland/9715293/2013 and A/Hong Kong/4801/2014 are shown in red, glycosylation motifs are highlighted, the reference viruses are coloured blue and the HA1/HA2 proteolytic cleavage site is highlighted. The genetic group/subgroup to which each of the sequences belongs is also shown.

Figure 9 shows the location on a 2005 H3 HA structure of amino acid substitutions that define the main genetic subgroups 3C.2a (**Figure 9A**) and 3C.3a (**Figure 9B**) each compared with the former vaccine virus A/Texas/50/2012.

Figure 10 shows a NA phylogenetic tree for representative viruses with their HA genetic groups indicated. **Figure 11** shows a NA amino acid sequence alignment for a representative selection of these viruses. As above, vaccine viruses are shown in red, glycosylation motifs are highlighted, the reference viruses are coloured blue and the genetic group/subgroup to which the corresponding HA gene sequences belong is also shown.

The NA gene sequences of recent viruses with HA genes in genetic group 3C cluster similarly to the HA sequences. However, the NA genes of most recently collected viruses in subgroup 3C.3a have acquired several substitutions in their NA glycoproteins (**K128R**, **V143M**, **S315R**, **I397L** and **R400K**). Those in genetic subgroup 3C.2a carry **T267K** and **I380V** with a subcluster carrying **S245N** (resulting in the acquisition of a glycosylation motif and **S247T**). Other subclusters are apparent in the NA genes of viruses in genetic group 3C.2a: additional to the substitutions **S245N** and **T267T** others also have (i) **V149A** substitution or (ii) **P468H** with significant number of viruses in this subcluster also carrying **D339N** substitution. **Figure 12** indicates the positions of the amino acid substitutions on the structure of an N2 NA for four reference viruses in the three main genetic clusters.

Antiviral resistance

Phenotypic testing to assess susceptibility to oseltamivir and zanamivir was performed on 34 viruses recovered from samples with collection dates after 2015-08-31. All showed normal inhibition (NI) by both neuraminidase inhibitors (**Table 15**).

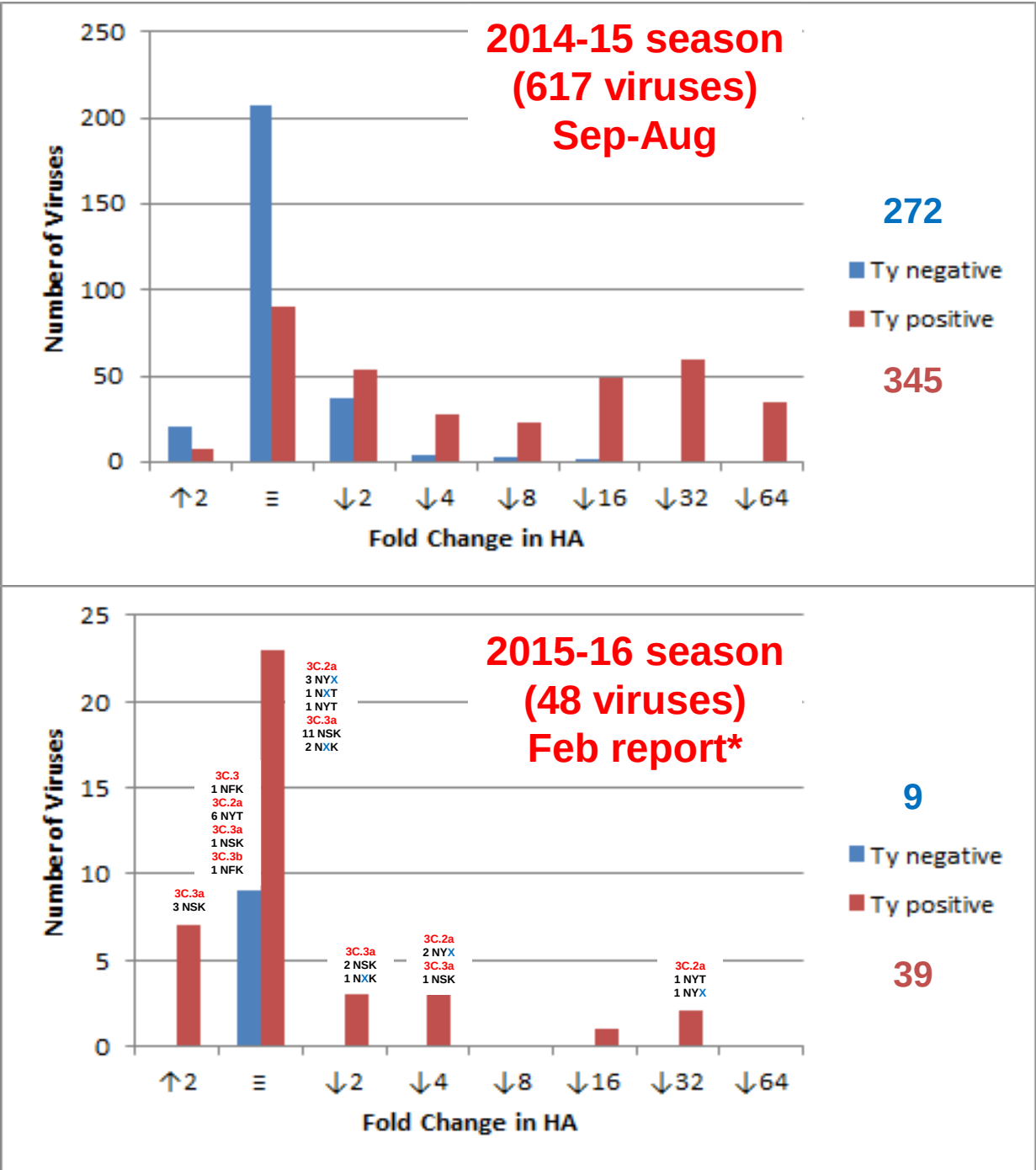
Conclusion.

Viruses in genetic subgroup 3C.2a were the most common but many viruses in genetic subgroup 3C.2a did not agglutinate RBCs and could not therefore be characterised by HI assay. Of the viruses with collection dates after 2015-08-31 in genetic subgroup 3C.2a, that were able to agglutinate RBCs in the presence of oseltamivir, 75% showed selection of amino acid substitutions in the HA1 158 to 160 glycosylation motif during culture resulting in the loss or partial loss of the glycosylation site.

In HI assays the antiserum raised against a cell culture-propagated 3C.2a virus (polymorphic for the glycosylation motif at residues 158 to 160 in HA1) recognised viruses in genetic subgroups 3C.2a, 3C.3a and 3C.3b similarly well. The antiserum raised against the egg-propagated vaccine virus A/Hong Kong/4801/2014 recognised the majority of viruses at titres within 4-fold of the homologous titre as did the antiserum raised against the egg-propagated virus A/Stockholm/6/2014, a virus from genetic subgroup 3C.3a. The antiserum raised against egg-propagated A/Switzerland/9715293/2013, also a genetic subgroup 3C.3a virus, recognised only a minority of the test viruses. The antiserum raised against a genetic subgroup 3C.3b virus recognised only test viruses in this genetic subgroup.

In PRNA, genetic subgroup 3C.2a viruses were recognised best by antisera raised against two cell culture-propagated viruses in genetic group 3C.2a but not by an antiserum raised against a virus with a **R142K** substitution in **HA1**. However, antiserum raised against the egg-propagated vaccine virus A/Hong Kong/4801/2014 (3C.2a) recognised a lower proportion of viruses in genetic subgroup 3C.2a.

Figure 6. Analysis of H3N2 Influenza Virus Agglutination of Guinea Pig RBCs and the Effect of 20nM Osetamivir



* Clade and sequences for HA1 positions 158-160 are shown for viruses characterised genetically

Fold shift in HA activity in the presence of 20nM osetamivir:
[Enhancement (↑2) Equivalence (≡) Reduction (↓2-↓64)]

Table 7. A(H3N2) viruses recovered from specimens collected after 2015-08-31, as assessed by NA activity (MUNANA-based assay)

RBC binding phenotype ¹			HI assay possible	Number of isolates	Number sequenced ²	Number in clade 3C.3b	Number in clade 3C.3a	Number in clade 3C.2a	Number in clade 3.3
Turkey	Guinea pig	Guinea pig + 20nM oseltamivir							
+	+	+	Yes	37	25 (68%)	0	17	8	0
-	+	+	Yes	9	9 (100%)	1	3	4	1
-	-	+	Yes	0	0	0			
+	-	+	Yes	0	0	0			
			Totals	46	34 (74%)	1	20	12	1
+	+	-	No	3	2 (67%)	0	0	2	0
+	-	-	No	0	0	0	0	0	0
-	+	-	No	9	5 (56%)	0	0	5	0
-	-	-	No	47	29 (62%)	0	0	29	0
			Totals	59	36 (61%)	0	0	36	0
			Totals	105	70 (67%)	1	20	48	1

1. Negative (-) corresponds to HA titres of <4, while positive (+) corresponds to HA titres of ≥4

2. As of 2016-02-12

* These viruses showed either **loss** [NYK] or **polymorphism** [K/NYT or NYK/T] at the 158/160 glycosylation motif [NYT] in HA1

Table 8. Summary of influenza A(H3N2) viruses analysed, collected after 2015-08-31

MONTH	H3N2							
Country	Number received	Number propagated¹	≤2 fold²	4 fold²	≥8 fold²	≤2 fold³	4 fold³	≥8 fold³
SEPTEMBER								
Iran	1	1	1			1		
Saudi Arabia	9	2	1	1				2
United Kingdom	3	3	1	2				3
OCTOBER								
China	1	1	1				1	
Norway	2	2		2				2
Zambia	2	2			2			2
NOVEMBER								
Austria	4	4	3	1		2	1	1
China	3	3	2	1		1	1	1
Finland	2	1	1			1		
France	1	1		1				1
Norway	1	1	1			1		
Sweden	4	4	4			3	1	
Zambia	5	in process			2			2
DECEMBER								
Austria	3	2		2				2
Finland	1	1		1				1
France	2	2	2			1	1	
Germany	3	3	2	1		2		1
Italy	5	2	1		1			2
Norway	2	1	1				1	
Portugal	3	2	1	1			1	1
Slovenia	5	1		1			1	
Turkey	13	3	3			1	2	
JANUARY								
Switzerland	1	1	1			1		
Turkey	11	in process	1			1		
	87	43	27	14	5	15	10	21
Countries - 16			58.7%	30.4%	10.9%	32.6%	21.7%	45.7%

1. Propagated to sufficient titre to perform HI assay in presence of 20nM oseltamivir

2. Compared to homologous titres for ferret antisera raised against tissue culture-propagated virus A/Hong Kong/5738/2014 (3C.2a)

3. Compared to homologous titres for ferret antisera raised against egg-propagated virus A/Hong Kong/4801/2014 (3C.2a)

Table 9-1. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2015-09-08

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre											
				Post-infection ferret antisera											
				A/Perth 16/09 Egg	A/Texas 50/12 Egg	A/Samara 73/13 SIAT	A/Stock 6/14 SIAT	A/Stock 6/14 Egg	A/Switz 9715293/13 SIAT	A/Switz 9715293/13 Egg	A/Neth 525/14 SIAT	A/HK 146/13 Egg	A/HK 5738/14 MDCK	A/HK 4801/14 Egg	A/Georgia 532/15 SIAT
				F18/11 ⁻¹	F36/12 ⁻¹	F24/13 ⁻¹	F14/14 ⁻¹	F20/14 ⁻¹	NIB F13/14 ⁻¹	F32/14 ⁻¹	F23/15 ⁻¹	F10/15 ⁻¹	F30/14 ⁻¹	F12/15 ⁻¹	F33/15 ⁻¹
	Passage history														
	Ferret number														
	Genetic group														
REFERENCE VIRUSES															
A/Perth/16/2009		2009-07-04	E3/E3	640	320	160	40	160	<	80	80	80	<	40	<
A/Texas/50/2012	3C.1	2012-04-15	E5/E2	320	5120	640	160	640	80	640	320	320	160	80	160
A/Samara/73/2013	3C.3	2013-03-12	C1/SIAT3	160	1280	640	320	640	160	320	320	640	320	160	320
A/Stockholm/6/2014	3C.3a	2014-02-06	SIAT1/SIAT2	<	160	320	640	320	320	160	80	160	160	160	160
A/Stockholm/6/2014	3C.3a	2014-02-06	E4/E1	80	640	80	160	320	160	640	80	80	160	40	80
A/Switzerland/9715293/2013	3C.3a	2013-12-06	SIAT1/SIAT2	<	80	160	640	160	80	160	40	40	160	80	80
A/Switzerland/9715293/2013	clone 123	2013-12-06	E4/E1	40	640	160	160	320	80	640	40	80	160	80	160
A/Netherlands/525/2014	3C.3b	2014-12-17	SIAT2/SIAT3	160	640	320	320	160	80	160	1280	160	80	80	160
A/Hong Kong/146/2013	3C.2	2013-01-11	E3/E3	640	2560	640	160	640	80	640	320	1280	320	160	320
A/Hong Kong/5738/2014	3C.2a	2014-04-04	MDCK1/MDCK3	<	160	160	320	160	80	80	40	80	160	160	320
A/Hong Kong/4801/2014	isolate 1	2014-02-26	E6/E1	40	80	160	160	160	40	40	80	40	320	320	320
A/Georgia/532/2015	plaque 20	2015-03-09	SIAT2	<	160	160	160	160	<	40	40	40	160	160	640
TEST VIRUSES															
A/Guyane/1742/2015	3C.2a	2015-01-23	MDCK2/SIAT1	<	80	160	320	160	<	80	<	40	160	160	80
A/Dijon/1484/2015	3C.3b	2015-03-09	MDCK1/SIAT1	80	320	320	320	160	40	80	640	80	80	80	80
A/South Australia/211/2015	3C.2a	2015-03-13	E4/E1	<	40	80	160	80	40	40	40	40	160	320	160
A/Bretagne/1564/2015	3C.3	2015-03-19	MDCK2/SIAT1	80	320	320	160	80	40	80	640	80	80	80	80
A/Guyane/1749/2015	3C.3	2015-03-20	MDCK2/SIAT1	40	320	80	160	80	<	40	640	40	40	80	<
A/Alsace/1571/2015	3C.3b	2015-03-21	MDCK2/SIAT1	80	640	320	320	320	40	160	1280	80	80	160	80
A/Guyane/1750/2015	3C.2a	2015-03-23	MDCK2/SIAT1	<	80	80	160	80	<	40	<	<	80	160	80
A/Lorraine/1641/2015	3C.3b	2015-03-30	MDCK1/SIAT1	160	320	160	320	160	40	80	640	80	80	80	80
A/Bretagne/1661/2015	3C.3	2015-04-01	MDCK2/SIAT1	80	320	160	320	160	40	80	640	80	80	80	80
A/Brisbane/82/2015	3C.2a	2015-04-22	E3/E1	<	80	80	80	80	40	<	40	<	160	160	1280
A/Norway/2326/2015	3C.3b	2015-05-10	SIAT1/SIAT1	40	320	160	320	160	40	40	320	80	80	40	40
A/Victoria/511/2015	3C.3b	2015-05-16	SIAT2/SIAT1	80	320	160	320	80	40	80	640	80	80	80	80
A/Victoria/511/2015	3C.3b	2015-05-16	E3/E1	160	5120	320	160	640	80	320	1280	160	160	40	160
A/Norway/2506/2015	3C.2a	2015-06-05	SIAT1/SIAT2	<	80	80	160	80	<	40	<	40	80	80	40
A/Norway/2504/2015	3C.3b	2015-06-07	SIAT1/SIAT2	40	320	160	320	160	40	80	640	80	80	80	80
A/Norway/2475/2015	3C.2a	2015-06-09	SIAT1/SIAT1	<	80	80	160	80	<	80	<	<	80	80	80

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

¹ < = <40

Vaccine
NH 2015-16

Vaccine
SH 2016

Sequences in phylogenetic trees

Table 9-2. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2015-09-15 to 2015-09-25

Haemagglutination inhibition titre															
Viruses	Other information	Collection date	Passage history	Post-infection ferret antisera											
				A/Texas	A/Samara	A/Stock	A/Stock	A/Switz	A/Switz	A/Neth	A/HK	A/HK	A/HK	A/Georgia	
				50/12	73/13	6/14	6/14	9715293/13	9715293/13	525/14	146/13	5738/14	4801/14	532/15	
				Egg	SIAT	SIAT	Egg	SIAT	Egg	SIAT	Egg	MDCK	Egg	SIAT	
				F36/12 ⁻¹	F24/13 ⁻¹	F14/14 ⁻¹	F20/14 ⁻¹	NIB F13/14 ⁻¹	F32/14 ⁻¹	F23/15 ⁻¹	F10/15 ⁻¹	F30/14 ⁻¹	F12/15 ⁻¹	F33/15 ⁻¹	
				3C.1	3C.3	3C.3a	3C.3a	3C.3a	3C.3a	3c.3b	3C.2	3C.2a	3C.2a	3C.2a	
REFERENCE VIRUSES															
A/Texas/50/2012		3C.1	2012-04-15	E5/E2	5120	640	160	640	40	640	320	320	160	80	160
A/Samara/73/2013		3C.3	2013-03-12	C1/SIAT3	1280	640	320	320	80	320	320	320	320	160	320
A/Stockholm/6/2014	isolate 2	3C.3a	2014-02-06	SIAT1/SIAT2	160	320	320	160	160	160	40	80	80	80	80
A/Stockholm/6/2014		3C.3a	2014-02-06	E4/E1	640	80	160	320	80	640	40	80	160	40	80
A/Switzerland/9715293/2013		3C.3a	2013-12-06	SIAT1/SIAT2	40	80	320	160	80	80	<	<	80	40	40
A/Switzerland/9715293/2013		clone 123	3C.3a	2013-12-06	E4/E1	640	160	320	320	80	640	40	80	160	80
A/Netherlands/525/2014		3C.3b	2014-12-17	SIAT2/SIAT3	640	320	320	160	80	160	1280	160	80	160	160
A/Hong Kong/146/2013		3C.2	2013-01-11	E3/E3	2560	640	80	640	40	640	320	640	320	160	320
A/Hong Kong/5738/2014		3C.2a	2014-04-30	MDCK1/MDCK3	80	160	320	160	40	80	40	80	160	160	160
A/Hong Kong/4801/2014	isolate 1	3C.2a	2014-02-26	E6/E1	80	160	160	160	40	40	80	40	320	320	320
A/Georgia/532/2015	plaque 20	3C.2a	2015-03-09	SIAT2	160	320	320	160	40	80	80	160	160	160	640
TEST VIRUSES															
A/Gifu/46/2014		3C.2a	2014-09-09	MCDK3/SIAT2/SIAT1	80	160	320	80	<	40	<	40	160	80	160
A/Chiba/61/2014		3C.3a	2014-11-13	MDCK2/SIAT1/SIAT1	40	40	320	80	40	40	40	<	40	40	<
A/Hyogo/4001/2014		3C.2a	2014-12-08	MCDK1/SIAT1/SIAT1	40	80	80	40	<	<	<	<	80	40	40
A/Pennsylvania/09/2015		3C.3b	2015-01-07	SIAT2/SIAT1	320	160	160	80	<	80	640	80	40	40	40
A/Zambia/04-00021/2015		3C.3	2015-01-26	MDCK2/SIAT1	640	320	320	160	40	160	160	160	160	80	80
A/Mozambique/IR424/2015		3C.2a	2015-01-26	SIAT1	160	160	160	160	40	80	40	80	160	160	160
A/Mozambique/IR436/2015		3C.2a	2015-01-30	SIAT1	80	80	160	160	<	40	<	40	160	160	80
A/Iceland/32/2015		3C.3	2015-02-06	MDCK1/SIAT1	1280	320	160	160	40	160	1280	160	160	80	160
A/Mozambique/IR451/2015		3C.2a	2015-02-09	SIAT1	80	80	160	80	<	40	<	40	160	80	80
A/Hungary/423/2015		3C.2a	2015-02-09	MDCK3/SIAT2	80	160	160	80	<	40	<	40	160	80	160
A/Hungary/438/2015		3C.3b	2015-02-09	MDCK3/SIAT2	160	160	160	80	<	40	320	40	40	40	40
A/Hungary/436/2015		3C.3b	2015-02-09	MDCK3/SIAT3	160	80	160	40	<	40	640	<	40	40	<
A/Hokkaido/6/2015		3C.3a	2015-02-09	MDCK4/SIAT1/SIAT1	80	80	320	80	40	80	40	<	40	40	40
A/Iceland/34/2015		3C.3	2015-02-09	MDCKx/SIAT1	1280	320	160	160	40	320	1280	160	160	80	160
A/Iceland/35/2015		3C.2a	2015-02-10	MDCKx/SIAT1	40	80	160	40	<	40	<	40	160	80	320
A/Mozambique/IR493/2015		3C.2a	2015-02-19	SIAT1	80	80	160	80	<	40	<	40	80	80	40
A/Iceland/37/2015		3C.3	2015-02-20	MDCKx/SIAT1	1280	640	320	160	40	320	1280	160	160	80	160
A/Iceland/38/2015		3C.3b	2015-02-24	MDCK1/SIAT1	320	160	160	160	<	80	640	80	40	40	40
A/Wisconsin/20/2015		3C.2a	2015-02-24	SIAT3/SIAT1	80	160	160	80	<	40	<	40	160	80	320
A/Iceland/41/2015		3C.3b	2015-03-02	MDCKx/SIAT1	640	320	320	160	40	80	1280	80	80	80	80
A/Fukui/78/2015		3C.3a	2015-03-02	MDCK2/SIAT1/SIAT1	40	80	320	160	40	80	<	40	80	80	40
A/Iceland/43/2015		3C.3	2015-03-06	MDCKx/SIAT1	1280	640	320	160	40	160	2560	320	160	160	160
A/Iceland/45/2015		3C.3	2015-03-13	MDCKx/SIAT1	1280	640	320	320	40	320	1280	320	160	160	160
A/Zambia/02-00106/2015		3C.3	2015-03-25	MDCK2/SIAT1	640	320	320	160	40	160	160	160	160	80	80
A/Jordan/30226/2015		3C.2a	2015-04-14	MDCK1/SIAT1	40	40	160	<	<	<	<	<	80	40	<
A/Fukushima/69/2015		3C.2a	2015-04-24	MDCK1/SIAT1/SIAT1	40	80	160	80	<	40	<	40	80	80	80
A/Mozambique/IR803/2015		3C.2a	2015-05-06	SIAT2	40	80	160	40	<	40	<	40	80	40	40
A/Norway/2319/2015			2015-05-13	SIAT2/SIAT2	320	160	160	80	40	40	640	80	40	80	80
A/Iceland/6879/2015		3C.3	2015-05-13	SIAT1	160	80	80	80	<	40	320	40	<	<	<
A/Iceland/7018/2015		3C.3	2015-05-17	SIAT2	320	320	160	80	40	640	80	80	80	40	80
A/Norway/2501/2015		3C.3b	2015-06-01	SIAT2/SIAT2	320	320	320	80	40	640	80	80	80	80	40

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

⁻¹ < = <40

Vaccine
NH 2015-16

Vaccine
SH 2016

Sequences in phylogenetic trees

Table 9-3. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2015-10-02, 2015-10-06, 2015-10-23, 2015-11-15

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre											Georgia
				Post-infection ferret antisera											
				A/Texas	A/Samara	A/Stock	A/Stock	A/Switz	A/Switz	A/Neth	A/HK	A/HK	A/HK	A/HK	
				50/12	73/13	6/14	6/14	9715293/13	9715293/13	525/14	146/13	5738/14	4801/14	532/14	
	Passage history			Egg	SIAT	SIAT	Egg	SIAT	Egg	SIAT	Egg	MDCK	Egg	SIAT	
	Ferret number			F36/12 ¹	F24/13 ¹	F14/14 ¹	F20/14 ¹	NIB F13/14 ¹	F32/14 ¹	F23/15 ¹	F10/15 ¹	F30/14 ¹	F12/15 ¹	F33/15 ¹	
	Genetic group			3C.1	3C.3	3C.3a	3C.3a	3C.3a	3C.3a	3c.3b	3C.2	3C.2a	3C.2a	3C.2a	
REFERENCE VIRUSES															
A/Texas/50/2012		3C.1	2012-04-15	E5/E2	5120	640	160	640	40	640	320	320	160	80	160
A/Samara/73/2013		3C.3	2013-03-12	C1/SIAT3	2560	1280	640	640	160	640	640	640	640	320	320
A/Stockholm/6/2014	isolate 2	3C.3a	2014-02-06	SIAT1/SIAT3	160	160	640	160	80	160	40	80	160	160	80
A/Stockholm/6/2014		3C.3a	2014-02-06	E4/E1	640	80	160	320	80	640	80	80	160	40	80
A/Switzerland/9715293/2013	clone 123	3C.3a	2013-12-06	SIAT1/SIAT2	80	160	640	160	80	80	40	40	80	80	80
A/Switzerland/9715293/2013		3C.3a	2013-12-06	E4/E1	640	160	320	320	80	1280	40	80	160	80	160
A/Netherlands/525/2014		3C.3b	2014-12-17	SIAT2/SIAT3	640	320	320	160	40	160	640	160	80	80	80
A/Hong Kong/146/2013		3C.2	2013-01-11	E3/E3	2560	640	80	640	40	640	320	640	320	160	320
A/Hong Kong/5738/2014		3C.2a	2014-04-30	MDCK1/MDCK3	160	160	320	160	40	80	40	80	160	160	160
A/Hong Kong/4801/2014	isolate 1	3C.2a	2014-02-26	E6/E1	80	160	160	80	40	40	80	40	320	640	640
A/Georgia/532/2015	plaque 20	3C.2a	2015-03-09	SIAT2	320	320	320	160	40	80	40	160	160	160	1280
TEST VIRUSES															
A/Serbia/361/2015		3C.2a	2015-01-19	C2/SIAT2	160	160	320	80	<	40	<	80	160	80	160
A/Jiangxi-Donghu/1218/2015		3C.2a	2015-02-11	MDCK2/SIAT3/SIAT1	160	320	320	160	80	160	80	160	160	160	320
A/Jilin-Yanji/1179/2015		3C.3a	2015-03-03	MDCK1/SIAT2/SIAT1	80	160	320	160	80	80	40	80	80	80	80
A/Hubei-Dongbao/1205/2015		3C.2a	2015-04-09	MDCK4/SIAT2/SIAT1	80	160	160	80	<	40	<	40	80	80	40
A/Jiangsu-Qingpu/1660/2015		3C.3a	2015-04-29	E4/E1	640	160	320	1280	80	1280	160	160	160	80	80
A/Liaoning-Baita/1293/2015		3C.2a	2015-05-06	MDCK2/SIAT2/SIAT1	80	160	160	80	<	40	<	40	80	80	80
A/Hunan-Wuling/11088/2015		3C.2a	2015-06-15	MDCK2/SIAT2/SIAT1	160	320	320	160	40	80	40	80	160	160	160
A/England/333/2015		3C.2a	2015-09-04	MDCK2/SIAT1	<	40	40	40	<	<	<	<	40	40	<
A/England/334/2015		3C.2a	2015-09-06	MDCK2/SIAT1	40	80	80	80	40	40	<	40	80	80	40
A/England/335/2015		3C.2a	2015-09-29	SIAT1/SIAT1	<	40	80	80	<	<	<	<	40	40	40
															158/160 NYX
															158/160 NYX
															158/160 NXH

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

¹ < = <40

Vaccine
NH 2015-16

Vaccine
SH 2016

Sequences in phylogenetic trees

Table 9-4. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2015-11-20, 2015-12-04, 2015-12-11

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre											
				Post-infection ferret antisera											
				A/Texas	A/Samara	A/Stock	A/Stock	A/Switz	A/Switz	A/Neth	A/HK	A/HK	A/HK	A/HK	A/HK
				50/12	73/13	6/14	6/14	9715293/13	9715293/13	525/14	146/13	5738/14	4801/14	4801/14	4801/14
	Passage history			Egg	SIAT	SIAT	Egg	SIAT	Egg	SIAT	Egg	MDCK	MDCK	Egg	Egg
	Ferret number			F36/12 ¹	F24/13 ¹	F14/14 ¹	F20/14 ¹	F18/15 ¹	F32/14 ¹	F23/15 ¹	F10/15 ¹	F30/14 ¹	F43/15 ¹	F12/15 ¹	F12/15 ¹
	Genetic group			3C.1	3C.3	3C.3a	3C.3a	3C.3a	3C.3a	3c.3b	3C.2	3C.2a	3C.2a	3C.2a	3C.2a
REFERENCE VIRUSES															
A/Texas/50/2012		3C.1	2012-04-15	E5/E2	5120	640	160	640	40	640	320	320	160	320	80
A/Samara/73/2013		3C.3	2013-03-12	C1/SIAT3	2560	1280	640	640	160	640	640	640	640	640	320
A/Stockholm/6/2014		3C.3a	2014-02-06	SIAT1/SIAT3	160	160	640	160	80	160	40	80	160	160	160
A/Stockholm/6/2014	isolate 2	3C.3a	2014-02-06	E4/E1	640	80	160	320	80	640	80	80	160	320	40
A/Switzerland/9715293/2013		3C.3a	2013-12-06	SIAT1/SIAT2	80	160	640	160	80	80	40	40	80	40	80
A/Switzerland/9715293/2013	clone 123	3C.3a	2013-12-06	E4/E1	640	160	320	320	80	1280	40	80	160	160	80
A/Netherlands/525/2014		3C.3b	2014-12-17	SIAT2/SIAT3	1280	320	320	160	40	160	1280	160	80	160	160
A/Hong Kong/146/2013		3C.2	2013-01-11	E3/E3	2560	640	80	640	40	640	320	640	320	640	160
A/Hong Kong/5738/2014		3C.2a	2014-04-30	MDCK3/MDCK3	160	160	320	160	40	80	40	80	160	320	160
A/Hong Kong/4801/2014		3C.2a	2014-02-26	MDCK4/MDCK2	160	160	320	160	80	80	40	80	160	320	160
A/Hong Kong/4801/2014	isolate 1	3C.2a	2014-02-26	E6/E2	80	160	160	80	80	40	80	40	320	640	640
TEST VIRUSES															
A/Novosibirsk/40/2015		3C.2a	unknown	C1/SIAT1	40	80	160	40	<	<	<	40	80	80	40
A/Omsk/26/2015		3C.3a	unknown	C1/SIAT1	80	160	320	160	80	80	40	80	80	160	80
A/St Petersburg/207/2015		3C.3a	unknown	C1/SIAT1	160	160	640	320	80	160	40	80	160	160	160
A/Novosibirsk/13/2015		3C.3a	unknown	C1/SIAT1	160	320	640	320	160	160	40	80	160	160	160
A/St Petersburg/299/2015			unknown	C1/SIAT1	160	160	160	80	40	40	<	80	80	160	80
A/Norway/2587/2015		3C.2a	2015-08-07	MDCK/SIAT1/SIAT1	40	40	80	80	<	40	<	40	40	40	80
A/Saudi Arabia/78572/2015		3C.2a	2015-09-24	SIAT3	80	<	160	80	<	40	<	40	80	40	80
A/Saudi Arabia/78603/2015		3C.2a	2015-09-24	SIAT2	40	<	80	40	<	<	<	40	40	40	40
A/Norway/2605/2015		3C.3	2015-10-02	MDCK/SIAT1/SIAT1	320	80	160	80	<	40	320	40	40	40	40
A/Norway/2629/2015		3C.2a	2015-10-22	MDCK/SIAT1/SIAT3	40	<	80	40	40	40	<	40	40	40	40

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

¹ < = <40

Vaccine
NH 2015-16

Vaccine
SH 2016

Sequences in phylogenetic trees

Table 9-5. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2016-01-08, 2016-01-13, 2016-01-15, 2016-01-20, 2016-01-22

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre											
				Post-infection ferret antisera											
				A/Texas	A/Samara	A/Stock	A/Stock	A/Switz	A/Switz	A/Neth	A/HK	A/HK	A/HK	A/HK	A/Georgia
				50/12	73/13	6/14	6/14	9715293/13	9715293/13	525/14	146/13	5738/14	4801/14	4801/14	532/15
	Passage history			Egg	SIAT	SIAT	Egg	SIAT	Egg	SIAT	Egg	MDCK	MDCK	Egg	SIAT
	Ferret number			F36/12 ¹	F35/15 ¹	F14/14 ¹	F20/14 ¹	F18/15 ¹	F32/14 ¹	F23/15 ¹	F10/15 ¹	F30/14 ¹	F43/15 ¹	F12/15 ¹	F33/15 ¹
	Genetic group			3C.1	3C.3	3C.3a	3C.3a	3C.3a	3C.3a	3C.3b	3C.2	3C.2a	3C.2a	3C.2a	3C.2a
REFERENCE VIRUSES															
A/Texas/50/2012		3C.1	2012-04-15	E5/E2	2560	1280	160	640	80	640	320	320	160	320	160
A/Samara/73/2013		3C.3	2013-03-12	C1/SIAT4	1280	640	640	640	80	320	320	320	640	160	320
A/Stockholm/6/2014		3C.3a	2014-02-06	SIAT1/SIAT2	160	80	640	320	160	160	80	160	160	160	160
A/Stockholm/6/2014	isolate 2	3C.3a	2014-02-06	E4/E1	640	80	160	320	80	640	40	160	320	40	80
A/Switzerland/9715293/2013		3C.3a	2013-12-06	SIAT1/SIAT2	40	40	320	160	80	80	40	80	80	40	40
A/Switzerland/9715293/2013	clone 123	3C.3a	2013-12-06	E4/E1	640	160	160	320	40	640	40	80	160	320	160
A/Netherlands/525/2014		3C.3b	2014-12-17	SIAT2/SIAT2	320	160	160	160	40	80	640	80	80	80	80
A/Hong Kong/146/2013		3C.2	2013-01-11	E6	2560	640	80	640	40	320	160	640	160	320	320
A/Hong Kong/5738/2014		3C.2a	2014-04-30	MDCK1/MDCK3	80	40	320	160	80	80	40	80	160	320	160
A/Hong Kong/4801/2014		3C.2a	2014-02-26	MDCK4/MDCK1	80	40	320	160	40	80	40	80	160	320	160
A/Hong Kong/4801/2014	isolate 1	3C.2a	2014-02-26	E6/E2	40	80	80	40	40	40	<	160	640	320	320
A/Georgia/532/2015	plaque 20	3C.2a	2015-03-09	SIAT2	160	80	320	160	40	40	40	80	160	320	640
TEST VIRUSES															
A/Iran/115843/2015		3C.3a	2015-09-23	SIAT2/SIAT1	80	40	640	160	80	80	40	80	160	80	160
A/Liaoning-Shuangta/1505/2015		3C.3a	2015-11-11	MDCK2/SIAT1/SIAT1	80	<	320	80	<	40	<	<	40	80	<
A/Austria/890287/2015		3C.3a	2015-11-18	SIAT1/SIAT6	80	40	320	160	80	80	40	80	160	160	80
A/Aix-en-Provence/2474/2015		3C.2a	2015-11-20	MDCK3/SIAT1	40	<	80	40	<	<	<	<	40	80	40
A/Austria/890744/2015		3C.3a	2015-11-20	SIAT1/SIAT5	80	40	320	160	80	80	40	80	80	80	80
A/Austria/891959/2015		3C.3a	2015-11-30	SIAT1/SIAT3	40	<	160	80	<	40	<	<	40	40	<
A/Austria/892085/2015		3C.3a	2015-11-30	SIAT1/SIAT4	160	80	640	320	160	160	160	160	160	160	160
A/Finland/547/2015		3C.3a	2015-11-30	SIAT1/SIAT1	80	40	320	160	40	80	40	40	160	80	40
A/Finland/547/2015		3C.3a	2015-11-30	SIAT1/SIAT2	80	40	320	160	80	80	40	80	80	160	80
A/Nordrhein-Westfalen/104/2015		3C.3a	2015-12-03	C2/SIAT1	160	40	320	160	80	80	40	80	160	160	40
A/Finland/552/2015		3C.2a	2015-12-03	SIAT1/SIAT2	40	<	80	40	<	<	<	40	80	40	<
A/Austria/893502/2015		3C.3a	2015-12-07	SIAT1/SIAT2	40	<	160	80	<	40	<	<	40	40	<
A/Berlin/173/2015		3C.3a	2015-12-09	C2/SIAT1	40	<	160	80	<	<	<	<	40	40	<
A/Austria/893755/2015		3C.3a	2015-12-09	SIAT1/SIAT1	40	<	160	80	<	40	<	<	40	40	<
A/Lyon/2608/2015		3C.2a	2015-12-14	MDCK2/SIAT1	40	<	80	40	40	<	<	40	80	80	80

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

¹ < = <40

Vaccine
NH 2015-16

Vaccine
SH 2016

Sequences in phylogenetic trees

Table 9-6. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2016-01-28, 2016-02-05

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre											
				Post-infection ferret antisera											
				A/Texas	A/Samara	A/Stock	A/Stock	A/Switz	A/Switz	A/Neth	A/HK	A/HK	A/HK	A/HK	A/Georgia
				50/12	73/13	6/14	6/14	9715293/13	9715293/13	525/14	146/13	5738/14	4801/14	4801/14	532/15
	Passage history			Egg	SIAT	SIAT	Egg	SIAT	Egg	SIAT	Egg	MDCK	MDCK	Egg	SIAT
	Ferret number			F36/12 ¹	F35/15 ¹	F14/14 ¹	F20/14 ¹	F18/15 ¹	F32/14 ¹	F23/15 ¹	F10/15 ¹	F30/14 ¹	F43/15 ¹	F12/15 ¹	F33/15 ¹
	Genetic group			3C.1	3C.3	3C.3a	3C.3a	3C.3a	3C.3a	3C.3b	3C.2	3C.2a	3C.2a	3C.2a	3C.2a
REFERENCE VIRUSES															
A/Texas/50/2012		3C.1	2012-04-15	E5/E2	2560	640	160	640	40	640	320	320	160	320	160
A/Samara/73/2013		3C.3	2013-03-12	C1/SIAT3	1280	640	640	320	160	320	320	320	640	160	320
A/Stockholm/6/2014		3C.3a	2014-02-06	SIAT1/SIAT2	160	80	640	320	160	160	80	160	320	160	160
A/Stockholm/6/2014	isolate 2	3C.3a	2014-02-06	E4/E1	640	80	160	320	80	640	80	160	320	40	80
A/Switzerland/9715293/2013		3C.3a	2013-12-06	SIAT1/SIAT2	80	40	320	160	80	80	40	160	160	80	80
A/Switzerland/9715293/2013	clone 123	3C.3a	2013-12-06	E4/E1	640	160	160	320	80	640	80	320	320	80	160
A/Netherlands/525/2014		3C.3b	2014-12-17	SIAT2/SIAT2	640	160	320	160	80	160	640	160	80	160	80
A/Hong Kong/146/2013		3C.2	2013-01-11	E6	2560	640	80	640	40	640	320	640	320	160	320
A/Hong Kong/5738/2014		3C.2a	2014-04-30	MDCK1/MDCK2	160	40	320	160	80	80	40	80	160	320	160
A/Hong Kong/4801/2014		3C.2a	2014-02-26	MDCK4/MDCK2	80	40	160	80	40	40	80	80	320	80	80
A/Hong Kong/4801/2014	isolate 1	3C.2a	2014-02-26	E6/E2	80	80	80	80	80	40	80	320	640	320	320
A/Georgia/532/2015	plaq 20	3C.2a	2015-03-09	SIAT1/SIAT3	160	80	160	160	40	80	40	80	160	320	320
TEST VIRUSES															
A/Yunnan-Hongta/11097/2015		3C.3b	2015-10-05	MDCK2/SIAT1/SIAT2	640	160	160	160	40	80	640	80	80	160	80
A/Norway/2665/2015		3C.3a	2015-11-09	SIAT1	80	80	640	320	80	80	40	80	160	160	80
A/Yunnan-Wenshan/1893/2015		3C.2a	2015-11-09	MDCK1/SIAT2/SIAT3	160	160	320	160	160	160	40	160	160	160	160
A/Liaoning-Shuangta/1491/2015		3C.3a	2015-11-10	MDCK2/SIAT1/SIAT2	80	40	320	160	80	80	40	40	80	80	80
A/Stockholm/58/2015		3C.3a	2015-11-17	MDCK0/SIAT1	80	40	320	320	80	160	40	80	160	160	80
A/Stockholm/65/2015		3C.3a	2015-11-20	MDCK0/SIAT1	160	80	640	320	160	160	80	160	320	160	160
A/Stockholm/63/2015		3C.2a	2015-11-23	MDCK0/SIAT2	80	40	160	80	40	40	<	80	80	160	160
A/Stockholm/61/2015		3C.3a	2015-11-23	MDCK0/SIAT1	160	80	640	320	160	160	80	160	160	160	80
A/Thuringen/102/2015		3C.3a	2015-12-04	C1/SIAT1	80	40	320	320	80	80	40	80	160	160	80
A/Norway/2877/2015		ngs?	2015-12-08	SIAT1	80	<	320	160	80	40	40	40	80	80	80
A/Nord Pas de Calais/2518/2015		3C.3a	2015-12-18	MDCK1/SIAT1	80	40	320	320	80	80	40	80	160	160	80
A/Lisboa/75/2015		ngs?	2015-12-25	SIAT2/SIAT1	40	<	320	160	40	80	40	<	80	80	40
A/Lisboa/76/2015		ngs?	2015-12-25	SIAT1/SIAT1	40	<	160	80	<	40	<	<	40	40	<
A/Padova/1/2015		ngs?	2015-12-28	SIAT1/SIAT2	<	<	<	<	640	<	<	<	<	<	<
A/Slovenia/3203/2015		3C.3a	2015-12-29	SIATx/SIAT2	40	<	160	80	<	40	40	<	40	40	40
A/Roma/29/2015		3C.2a	2015-12-30	SIAT1/SIAT2	40	<	160	40	40	<	<	40	80	80	160
A/Switzerland/16094175/2016		3C.3a	2016-01-05	SIAT1	80	40	640	320	80	80	40	80	160	160	80

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

¹ < = <40

Vaccine
2015-16

Vaccine
SH 2016

Sequences in phylogenetic trees

Table 9-7. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2016-02-12

Haemagglutination inhibition titre																
Viruses	Other information	Collection date	Passage history	Post-infection ferret antisera												
				A/Texas	A/Samara	A/Stock	A/Stock	A/Switz	A/Switz	A/Neth	A/HK	A/HK	A/HK	A/HK	A/Georgia	
				50/12	73/13	6/14	6/14	9715293/13	9715293/13	525/14	146/13	5738/14	4801/14	4801/14	532/15	
				Egg	SIAT	SIAT	Egg	SIAT	Egg	SIAT	Egg	MDCK	MDCK	Egg	SIAT	
				F36/12 ¹	F35/15 ¹	F14/14 ¹	F20/14 ¹	F18/15 ¹	F32/14 ¹	F23/15 ¹	F10/15 ¹	F30/14 ¹	F43/15 ¹	F12/15 ¹	F33/15 ¹	
				3C.1	3C.3	3C.3a	3C.3a	3C.3a	3C.3a	3C.3b	3C.2	3C.2a	3C.2a	3C.2a	3C.2a	
REFERENCE VIRUSES																
A/Texas/50/2012	3C.1	2012-04-15	E5/E2	5120	640	80	640	40	320	320	320	80	320	40	160	
A/Samara/73/2013	3C.3	2013-03-12	C1/SIAT3	2560	640	640	640	160	640	320	640	320	640	320	320	
A/Stockholm/6/2014	3C.3a	2014-02-06	SIAT1/SIAT2	80	40	320	160	160	80	40	80	80	160	80	80	
A/Stockholm/6/2014	isolate 2 3C.3a	2014-02-06	E4/E1	640	80	160	320	80	640	80	40	160	320	40	80	
A/Switzerland/9715293/2013	3C.3a	2013-12-06	SIAT1/SIAT2	80	40	320	160	80	80	40	40	160	160	80	80	
A/Switzerland/9715293/2013	clone 123 3C.3a	2013-12-06	E4/E1	640	160	160	320	80	640	80	80	160	320	80	160	
A/Netherlands/525/2014	3C.3b	2014-12-17	SIAT2/SIAT3	320	160	320	160	80	160	1280	80	80	160	80	80	
A/Hong Kong/146/2013	3C.2	2013-01-11	E6	2560	640	80	640	40	640	320	640	320	640	160	320	
A/Hong Kong/5738/2014	3C.2a	2014-04-30	MDCK1/MDCK2/SIAT2	80	40	160	160	40	40	<	80	160	320	160	160	
A/Hong Kong/4801/2014	3C.2a	2014-02-26	MDCK4/MDCK2	160	80	320	160	40	80	40	80	160	320	160	160	
A/Hong Kong/4801/2014	isolate 1 3C.2a	2014-02-26	E6/E2	80	80	80	80	80	40	40	<	320	640	320	320	
A/Georgia/532/2015	plaq 20 3C.2a	2015-03-09	SIAT1/SIAT3	160	80	160	80	80	40	<	80	160	320	160	640	
TEST VIRUSES																
A/Zambia/04-00420/2015	ngs?	2015-10-12	MDCK2/SIAT1	<	<	<	<	320	<	<	<	<	<	<	<	
A/Zambia/04-00313/2015	ngs?	2015-10-15	MDCK2/SIAT1	<	<	<	<	320	<	<	<	<	<	<	40	
A/Zambia/09-00003/2015	ngs?	2015-11-11	MDCK2/SIAT1	<	<	<	<	640	<	<	<	<	<	<	40	
A/Zambia/09-00006/2015	ngs?	2015-11-11	MDCK2/SIAT1	<	<	<	<	640	<	<	<	<	<	<	80	
A/Antalya/5181/2015	ngs?	2015-12-29	MDCK1/SIAT1	160	40	320	160	80	80	40	80	160	160	80	80	
A/Cankiri/2395/2015	ngs?	2015-12-30	SIAT1/SIAT1	80	40	320	160	80	80	40	80	80	160	80	80	
A/Ankara/2396/2015	ngs?	2015-12-30	SIAT1/SIAT1	160	80	640	320	160	160	40	80	160	160	160	80	
A/Turkey/TR-31/2016	ngs?	2016-01-15	SIAT1	160	40	320	160	80	80	40	80	160	160	160	160	

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

¹ < = <40Vaccine
NH 2015-16Vaccine
SH 2016

Sequences in phylogenetic trees

Table 9-8. Summary of HI reactivity of cell-propagated A(H3N2) Test viruses by clade

HI Table	Test Virus genetic group	No of Test Viruses by genetic group	Antiserum to Virus Host Genetic group	Number of test viruses reacting within 4-fold of the antiserum homologous titre												
				A/Perth	A/Texas	A/Samara	A/Stock	A/Stock	A/Switz	A/Switz	A/Neth	A/HK	A/HK	A/HK	A/HK	A/Georgia
				16/09	50/12	73/13	6/14	6/14	9715293/13	9715293/13	525/14	146/13	5738/14	4801/14	4801/14	532/15
				Egg	Egg	Cell	Cell	Egg	Cell	Egg	Cell	Egg	Cell	Cell	Egg	Cell
				3C.1	3C.3	3C.3a	3C.3a	3C.3a	3C.3a	3C.3b	3C.2	3C.2a	3C.2a	3C.2a	3C.3b	
9-1	3C.2a	4		0	0	1	4	4	4	0	0	4		4	0	
	3C.3a	0		0	0	0	0	0	0	0	0	0		0	0	
	3C.3b	6		1	0	6	6	6	6	1	6	0	6		5	0
	3C.3	3		0	0	2	3	3	3	0	0	0	3		3	0
9-2	3C.2a	12			0	4	12	8	12	0	0	12		9	5	
	3C.3a	3			0	0	3	3	3	0	0	0	3		1	0
	3C.3b	6			0	5	6	5	6	0	1	0	6		2	0
	3C.3	9			5	8	9	9	9	7	7	7	8		7	5
9-3	3C.2a	8			0	2	5	7	8	0	0	1	8		2	1
	3C.3a	1			0	0	1	1	1	0	0	0	1		0	0
	3C.3b	0			0	0	0	0	0	0	0	0	0		0	0
	3C.3	0			0	0	0	0	0	0	0	0	0		0	0
9-4	3C.2a	5			0	0	2	2	5	0	0	0	5	1	0	
	3C.3a	3			0	1	3	3	3	0	0	0	3	3	2	
	3C.3b	0			0	0	0	0	0	0	0	0	0	0	0	
	3C.3	1			0	0	1	1	1	0	1	0	1	0	0	
9-5	3C.2a	3			0	0	0	0	3	0	0	0	3	3	1	0
	3C.3a	12			0	0	2	12	12	1	0	1	12	8	7	1
	3C.3b	0			0	0	0	0	0	0	0	0	0	0	0	0
	3C.3	0			0	0	0	0	0	0	0	0	0	0	0	0
9-6	3C.2a	3			0	1	3	2	3	1	0	1	3	3	2	3
	3C.3a	9			0	0	9	9	9	3	0	2	9	8	9	8
	3C.3b	1			1	1	1	1	1	0	1	0	1	1	1	1
	3C.3	0			0	0	0	0	0	0	0	0	0	0	0	0
Totals			Number within 4-fold													
	3C.2a	35		0	0	8	26	23	35	1	0	2	35	7	18	9
	3C.3a	28		0	0	1	18	28	28	4	0	3	28	19	19	9
	3C.3b	13		1	1	12	13	13	13	1	8	0	13	1	3	1
	3C.3	13		0	5	10	13	13	13	7	8	7	12	0	7	5
Totals			Percentage within 4-fold													
	3C.2a	35		0.0	0.0	22.9	74.3	65.7	100.0	2.9	0.0	5.7	100.0	63.6	51.4	30
	3C.3a	28		0.0	0.0	3.6	64.3	100.0	100.0	14.3	0.0	10.7	100.0	79.2	67.9	36
	3C.3b	13		16.7	7.7	92.3	100.0	92.3	100.0	7.7	61.5	0.0	100.0	100.0	23.1	7.7
	3C.3	12		0.0	38.5	76.9	100.0	100.0	100.0	53.8	61.5	53.8	92.3	0.0	53.8	41.7

Vaccine
NH2015-16

Vaccine
SH2016

Table 10-1. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2016-01-13 and 2016-01-25

Viruses	Collection Date	Passage History	Neutralisation titre ¹												HA1 substitutions for 3C.2a viruses compared to A/Hong Kong/4801/2014: Egg adaptation HA substitutions compared to the corresponding cell isolate		
			Post-infection ferret antisera														
			A/HK 4801/14 Egg F12/15 3C.2a	A/HK 7295/14 MDCK F02/15 3C.2a	A/Sth Afr R2665/15 SIAT F50/15 3C.2a	A/Sth Afr R2651/15 SIAT F51/15 3C.2a	A/Switz 9715923/13 SIAT NIB F18/15 3C.3a	A/Neth 525/14 SIAT F23/15 3C.3b									
			2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read			
REFERENCE VIRUSES																Additional information	
A/Hong Kong/4801/2014	3C.2a	2014-02-26	E6/E2	640	718	1280	1193	5120	4495	1280	1612	80	72	80	113		T160K, L194P
A/Hong Kong/7295/2014	3C.2a	2014-08-07	MDCK3	80	95	320	283	320	282	640	618	80	67	40	53		none
A/South Africa/R2665/2015	3C.2a	2015-05-25	MDCK2/SIAT1	160	191	1280	988	20480	19017	2560	2290	160	186	40	56		R142K, Q197R
A/South Africa/R2651/2016	3C.2a	2015-05-25	MDCK2/SIAT1	320	256	1280	977	1280	1558	1280	1775	160	130	80	75		Q197R, I214T
A/Switzerland/ 9715293/2013	3C.3a	2013-12-06	SIAT1/SIAT2	80	78	160	208	160	153	80	62	160	137	80	90		
A/Netherlands/525/2014	3C.3b	2014-12-17	SIAT2/SIAT1	80	62	80	88	40	56	40	50	40	59	2560	2360		
TEST VIRUSES																	
A/Saudi Arabia/503/2015	3C.2a	2015-09-01	SIAT1	640	491	640	760	2560	2762	640	609	80	116	80	63	none	
A/Saudi Arabia/5786/2015	3C.2a	2015-09-01	SIAT1	320	272	640	597	2560	2062	640	640	160	192	40	40	none	
A/England/333/2015	3C.2a	2015-09-04	MDCK2/SIAT1	160	155	320	278	640	608	160	182	40	49	40	45	T160X	
A/England/334/2015	3C.2a	2015-09-06	MDCK2/SIAT1	160	235	640	509	1280	1280	1280	1849	40	57	40	55	T160X	
A/Saudi Arabia/78018/2015	3C.2a	2015-09-21	SIAT1	320	320	1280	1015	2560	3266	1280	1512	160	209	80	69	S262R, K310N	
A/Saudi Arabia/78279/2015	3C.2a	2015-09-22	SIAT1	320	338	640	828	640	870	640	613	160	136	80	68	R142G	
A/Saudi Arabia/78063/2015	3C.2a	2015-09-22	SIAT1	160	150	640	540	1280	1280	2560	2133	80	95	40	47	P21Q, S262R	
A/Iran/105051/2015	3C.2a	2015-10-02	SIAT2/SIAT1	160	129	640	714	640	717	640	862	80	80	40	49	R142G	
A/Norway/2629/2015	3C.2a	2015-10-22	SIAT2	80	90	320	457	1280	1006	640	557	80	88	<	0	N171K, I214T	
A/Belgium/2015G0760/2015	3C.2a	2015-11-04	SIAT1/SIAT1	160	128	640	565	1280	1536	640	837	80	93	40	40	none	
A/Hessen/49/2015	3C.2a	2015-11-11	C2/SIAT1	80	69	320	452	640	686	640	845	40	52	<	10	N121D	
A/Slovenia/2975/2015	3C.2a	2015-11-14	MDCKx/SIAT1	80	80	320	272	320	437	320	277	51	40	40	40	T160X, R261L	
A/Slovenia/2974/2015	3C.2a	2015-11-15	MDCKx/SIAT1	80	71	320	293	640	486	640	506	40	40	20	40	R261L	
A/Brandenburg/48/2015	3C.2a	2015-11-16	C2/SIAT1	80	68	640	487	320	428	640	512	40	58	40	40	R142G	
A/Finland/551/2015	3C.2a	2015-11-30	SIAT1/SIAT1	160	159	640	512	640	721	320	417	80	74	80	61	N171K, I214T	
A/Finland/552/2015	3C.2a	2015-12-03	SIAT1/SIAT1	80	70	640	743	640	596	640	630	80	62	40	40	S144R, S203N	
A/Iran/115843/2015	3C.3a	2015-09-23	SIAT2/SIAT1	80	70	80	89	160	126	80	73	80	91	80	66		
				Vaccine SH 2016													

¹ Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation

Table 10-2. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2016-02-08

Viruses	Passage history Ferret number Genetic group	Collection Date	Passage History	Neutralisation titre ¹												HA1 substitutions for 3C.2a viruses compared to A/Hong Kong/4801/2014: Egg adaptation HA substitutions compared to the corresponding cell isolate
				Post-infection ferret antisera												
				A/HK 4801/14		A/HK 7295/14		A/Sth Afr R2665/15		A/Sth Afr R2651/15		A/Switz 9715923/13		A/Neth 525/14		
				Egg F12/15		MDCK F02/15		SIAT F50/15		SIAT F51/15		SIAT NIB F18/15		SIAT F23/15		
				3C.2a		3C.2a		3C.2a		3C.2a		3C.3a		3C.3b		
				2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	
REFERENCE VIRUSES																
A/Hong Kong/4801/2014	3C.2a	2014-02-26	E6/E2	640	718	1280	1193	5120	4495	1280	1612	80	72	80	113	T160K, L194P
A/Hong Kong/7295/2014	3C.2a	2014-08-07	MDCK3	80	95	320	283	320	282	640	618	80	67	40	53	none
A/South Africa/R2665/2015	3C.2a	2015-05-25	MDCK2/SIAT1	160	191	1280	988	20480	19017	2560	2290	160	186	40	56	R142K, Q197R
A/South Africa/R2651/2016	3C.2a	2015-05-25	MDCK2/SIAT1	320	256	1280	977	1280	1558	1280	1775	160	130	80	75	Q197R, I214T
A/Switzerland/ 9715293/2013	3C.3a	2013-12-06	SIAT1/SIAT2	80	78	160	208	160	153	80	62	160	137	80	90	
A/Netherlands/525/2014	3C.3b	2014-12-17	SIAT2/SIAT1	80	62	80	88	40	56	40	50	40	59	2560	2360	
TEST VIRUSES																
A/Montana/28/2015	3C.2a	2015-09-24	SIAT3/SIAT1	80	62	320	330	320	464	640	560	20	20	20	20	T160X, R261Q S199T T160X, N171K,I406V N144K, K326R Y94H T160X Y94H Y94H Y94H R142K, Q197R, M168V Y94H E50K, Y94H E50K, Y94H E50K, Y94H T160X, N171K
A/Yunnan-Hongta/11097/2015	3C.3b	2015-10-05	MDCK2/SIAT1/SIAT1	40	40	80	71	40	40	80	65	40	40	1280	1629	
A/Yunnan-Wenshan/1893/2015	3C.2a	2015-11-09	MDCK1/SIAT2/SIAT3	80	86	320	264	320	427	160	160	20	20	40	40	
A/Jiangsu-Hailing/11782/2015	3C.2a	2015-11-09	MDCK1/SIAT1/SIAT1	40	51	160	203	320	256	320	253	40	43	40	47	
A/Shanghai-Pudongxin/11809/2015	3C.2a	2015-11-10	MDCK2/SIAT1/SIAT1	40	42	160	144	160	200	320	312	40	40	20	20	
A/Liaoning-Shuangta/1491/2015	3C.3a	2015-11-10	MDCK2/SIAT1/SIAT2	40	40	80	84	80	92	80	75	80	70	40	50	
A/Austria/890287/2015	3C.3a	2015-11-18	SIAT1/SIAT1	40	40	40	59	80	107	80	67	80	72	40	49	Y94H
A/Aix-en-Provence/2474/2015	3C.2a	2015-11-20	MDCK3/SIAT1	80	109	320	320	640	533	320	264	80	70	80	63	T160X
A/Austria/890744/2015	3C.3a	2015-11-20	SIAT1/SIAT1	20	20	40	55	80	67	40	45	40	57	40	40	Y94H
A/Austria/892085/2015	3C.3a	2015-11-30	SIAT1/SIAT1	40	40	40	47	80	106	80	80	80	71	40	41	Y94H
A/Finland/547/2015	3C.3a	2015-11-30	SIAT1/SIAT2	20	20	40	40	80	63	40	51	40	50	20	20	Y94H
A/Austria/892551/2015	3C.2a	2015-12-02	SIAT1/SIAT1	80	110	640	500	>10240	>10240	2560	2987	80	78	40	42	R142K, Q197R, M168V
A/Nordrhein-Westfalen/104/2015	3C.3a	2015-12-03	C2/SIAT1	20	20	40	49	80	63	40	40	80	72	40	47	Y94H
A/Austria/893502/2015	3C.3a	2015-12-07	SIAT1/SIAT1	40	40	40	47	20	20	<	10	40	47	40	52	E50K, Y94H
A/Berlin/173/2015	3C.3a	2015-12-09	C2/SIAT1	20	20	20	20	40	40	40	40	40	40	40	45	E50K, Y94H
A/Austria/893755/2015	3C.3a	2015-12-09	SIAT1/SIAT1	40	46	40	57	40	40	20	20	40	41	40	56	E50K, Y94H
A/Lyon/2608/2015	3C.2a	2015-12-14	MDCK2/SIAT1	40	40	80	101	160	145	160	208	20	20	<	0	T160X, N171K
				Vaccine SH 2016												

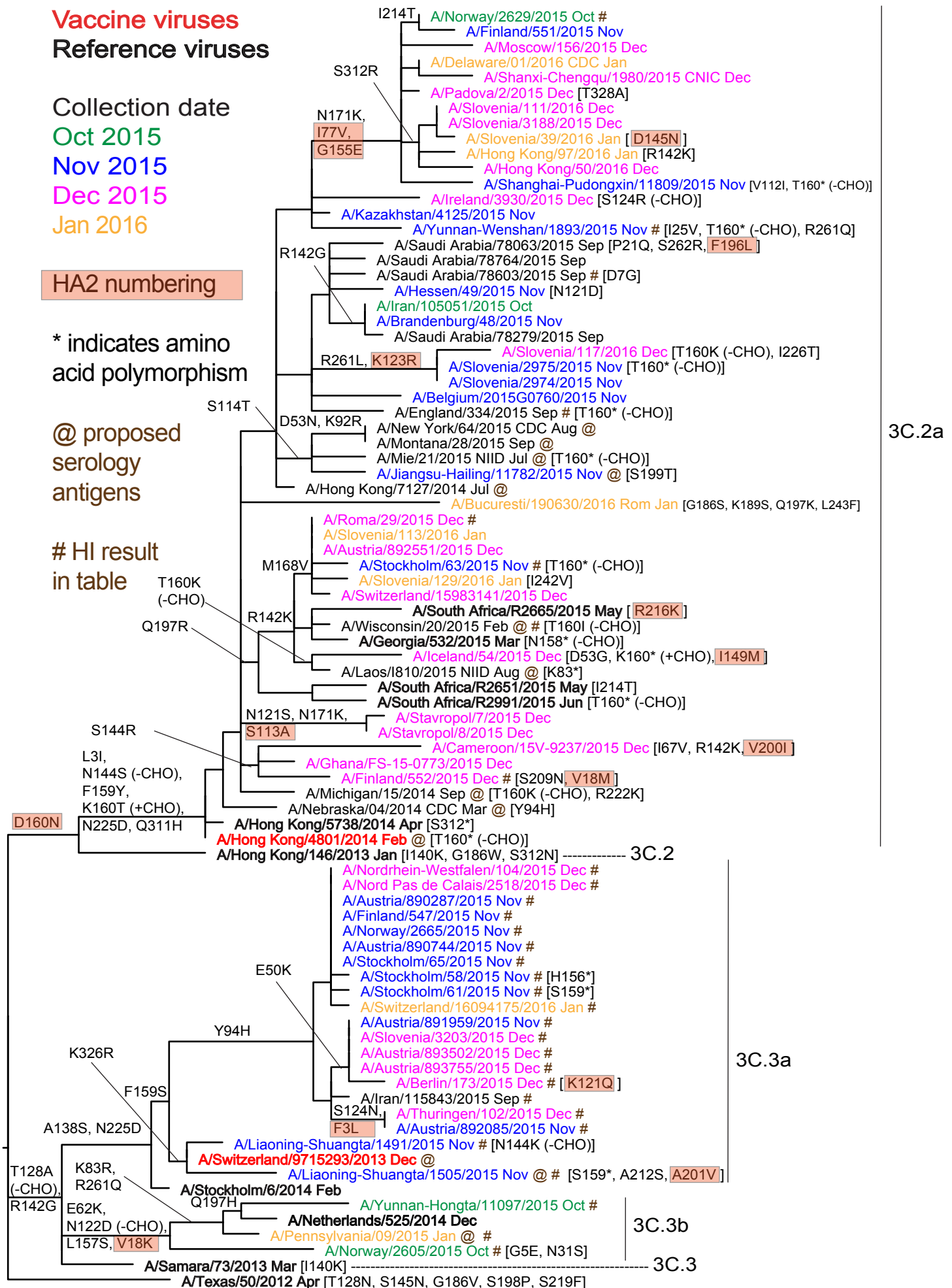
¹ Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation

Table 10-3. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2016-02-10

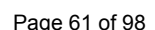
Viruses	Collection Date	Passage History	Neutralisation titre ¹												HA1 substitutions for 3C.2a viruses compared to A/Hong Kong/4801/2014: Egg adaptation HA substitutions compared to the corresponding cell isolate	
			Post-infection ferret antisera													
			A/HK 4801/14	A/HK 7295/14		A/Sth Afr R2665/15		A/Sth Afr R2651/15		A/Switz 9715923/13		A/Neth 525/14				
			Egg	MDCK		SIAT		SIAT		SIAT		SIAT				
			F12/15	F02/15		F50/15		F51/15		NIB F18/15		F23/15				
Passage history	Ferret number	Genetic group	3C.2a		3C.2a		3C.2a		3C.2a		3C.3a		3C.3b			
			2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read		
REFERENCE VIRUSES																
A/Hong Kong/4801/2014	3C.2a	2014-02-26	E6/E2	640	718	1280	1193	5120	4495	1280	1612	80	72	80	113	T160K, L194P
A/Hong Kong/7295/2014	3C.2a	2014-08-07	MDCK3	80	95	320	283	320	282	640	618	80	67	40	53	none
A/South Africa/R2665/2015	3C.2a	2015-05-25	MDCK2/SIAT1	160	191	1280	988	20480	19017	2560	2290	160	186	40	56	R142K, Q197R
A/South Africa/R2651/2016	3C.2a	2015-05-25	MDCK2/SIAT1	320	256	1280	977	1280	1558	1280	1775	160	130	80	75	Q197R, I214T
A/Switzerland/ 9715293/2013	3C.3a	2013-12-06	SIAT1/SIAT2	80	78	160	208	160	153	80	62	160	137	80	90	
A/Netherlands/525/2014	3C.3b	2014-12-17	SIAT2/SIAT1	80	62	80	88	40	56	40	50	40	59	2560	2360	
TEST VIRUSES																
Additional information																
A/Stockholm/58/2015	3C.3a	2015-11-17	MDCK0/SIAT1	80	73	160	130	320	359	160	167	160	147	80	73	Y94H
A/Stockholm/63/2015	3C.2a	2015-11-23	MDCK0/SIAT1	80	110	160	213	2560	2534	160	224	80	89	80	77	R142K, T160X, M168V, Q197R
A/Stockholm/61/2015	3C.3a	2015-11-23	MDCK0/SIAT1	80	72	160	147	320	262	80	112	80	72	80	102	Y94H, F159X
A/Slovenia/111/2015	3C.2a	2015-12-15	MDCKx/SIAT1	640	640	1280	1513	5120	5225	2560	2804	160	233	80	74	N171K, S312R, I406V, G484E
A/Iceland/54/2015	3C.2a	2015-12-21	MDCK0/SIAT1	160	137	320	430	>10240	>10240	2560	1991	80	64	40	53	R142K, T160X, Q197R
A/Slovenia/117/2015	3C.2a	2015-12-22	SIATx/SIAT1	1280	971	1280	1189	10240	8828	2560	2377	160	152	160	152	T160K
A/Slovenia/3188/2015	3C.2a	2015-12-26	MDCKx/SIAT1	320	288	1280	1246	5120	4004	2560	3446	160	182	40	58	N171K, S312R, I406V, G484E
A/Slovenia/3203/2016	3C.3a	2015-12-29	MDCKx/SIAT1	80	102	80	75	80	62	40	58	80	77	80	70	E50K, Y94H
A/Hong Kong/50/2016	3C.2a	2015-12-30	MDCK1/SIAT1	160	194	640	649	1280	1831	1280	960	80	88	40	55	N171K, S312R, I406V, G484E
A/Hong Kong/97/2016	3C.2a	2016-01-01	MDCK1/SIAT1	160	160	640	528	10240	11815	1280	1493	80	72	40	51	R142K, N171K, S312R, I406V, G484E
A/Slovenia/39/2016	3C.2a	2016-01-06	MDCKx/SIAT1	40	58	160	147	320	421	320	436	80	57	40	40	N171K, S312R, I406V, G484E
A/Slovenia/113/2016	3C.2a	2016-01-07	MDCKx/SIAT1	640	915	2560	2346	>10240	>10240	2560	3328	640	801	320	340	R142K, M168V, Q197R
A/Slovenia/129/2016	3C.2a	2016-01-12	SIATx/SIAT1	640	670	2560	1958	>10240	>10240	2560	2925	320	405	160	143	R142K, M168V, Q197R
				Vaccine SH 2016												

¹ Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation

Figure 7. Phylogenetic comparison of A(H3N2) HA genes



Vaccine virus: **Reference virus:** **Genetic group:** **HA1/HA2 boundary:** **Potential N-linked glycosylation site**



	310	320	330	340	350	360	370	380	390	400
A/Texas/50/2012	TYGACPRYVKQSTLKLATGMRNVPEKQTR	IGIFGAIAGFIENGWEGMVDGWYGF	FRHQNSEGRGQAADLKSTQA	IDDQINGKLNRLIGKTN	EFHQIEKEFS					
A/Samara/73/2013										
A/Norway/2605/2015					K					
A/Netherlands/525/2014					K					
A/Stockholm/6/2014										
A/Switzerland/9715293/2013			R							
A/Thuringen/102/2015										
A/Slovenia/3203/2015			L							
A/Stockholm/58/2015										
A/Hong Kong/146/2013		N								
A/Hong Kong/4801/2014		H								
A/Hong Kong/5738/2014		HX								
A/Ghana/FS-15-0773/2015		H								
A/Stavropol/7/2015		H								
A/South Africa/R2991/2015		H								
A/South Africa/R2651/2015		H								
A/Iceland/54/2015		H								
A/Georgia/532/2015		H								
A/South Africa/R2665/2015		H								
A/Slovenia/129/2016		H								
A/Roma/29/2015		H								
A/Bucuresti/190630/2016		H								
A/Jiangsu-Hailing/11782/2015		H								
A/Montana/28/2015		H								
A/Belgium/201560760/2015		H								
A/Slovenia/117/2016		H								
A/Iran/105051/2015		H								
A/Ireland/3930/2015		H								
A/Hong Kong/97/2016		HR								
A/Finland/551/2015		H								
	410	420	430	440	450	460	470	480	490	500
A/Texas/50/2012	EEVGRIQDLEKYVEDTKIDLWSYNAELLVALENQHTIDLT	SEMKNLF	EKTKKQLRENAEDMGN	GCFKIYHKCDNACIGSIRNGT	YDHDVYRDEALNNRF					
A/Samara/73/2013										
A/Norway/2605/2015										
A/Netherlands/525/2014										
A/Stockholm/6/2014										
A/Switzerland/9715293/2013										
A/Thuringen/102/2015										
A/Slovenia/3203/2015										
A/Stockholm/58/2015										
A/Hong Kong/146/2013									N	
A/Hong Kong/4801/2014									N	
A/Hong Kong/5738/2014									N	
A/Ghana/FS-15-0773/2015									N	
A/Stavropol/7/2015			A						N	
A/South Africa/R2991/2015									N	
A/South Africa/R2651/2015									N	
A/Iceland/54/2015								M	N	
A/Georgia/532/2015									N	
A/South Africa/R2665/2015									N	
A/Slovenia/129/2016									N	
A/Roma/29/2015									N	
A/Bucuresti/190630/2016									N	
A/Jiangsu-Hailing/11782/2015									N	
A/Montana/28/2015									N	
A/Belgium/201560760/2015									N	
A/Slovenia/117/2016					R				N	
A/Iran/105051/2015									N	
A/Ireland/3930/2015									N	
A/Hong Kong/97/2016		V							E	N
A/Finland/551/2015		V							E	N
	510	520	530	540	550					
A/Texas/50/2012	QIKGV	ELKSGYKDWILWISFAISCFLLCVALLGFIMWACQKGNIRCNICI								
A/Samara/73/2013										
A/Norway/2605/2015										
A/Netherlands/525/2014										
A/Stockholm/6/2014										
A/Switzerland/9715293/2013										
A/Thuringen/102/2015										
A/Slovenia/3203/2015										
A/Stockholm/58/2015										
A/Hong Kong/146/2013										
A/Hong Kong/4801/2014										
A/Hong Kong/5738/2014										
A/Ghana/FS-15-0773/2015										
A/Stavropol/7/2015										
A/South Africa/R2991/2015										
A/South Africa/R2651/2015										
A/Iceland/54/2015										
A/Georgia/532/2015										
A/South Africa/R2665/2015									K	
A/Slovenia/129/2016										
A/Roma/29/2015										
A/Bucuresti/190630/2016										
A/Jiangsu-Hailing/11782/2015										
A/Montana/28/2015										
A/Belgium/201560760/2015										
A/Slovenia/117/2016										
A/Iran/105051/2015										
A/Ireland/3930/2015										
A/Hong Kong/97/2016										
A/Finland/551/2015										

Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

Figure 9A. Amino acid substitutions associated with the HA of 3C.2a viruses relative to A/Texas/50/2012

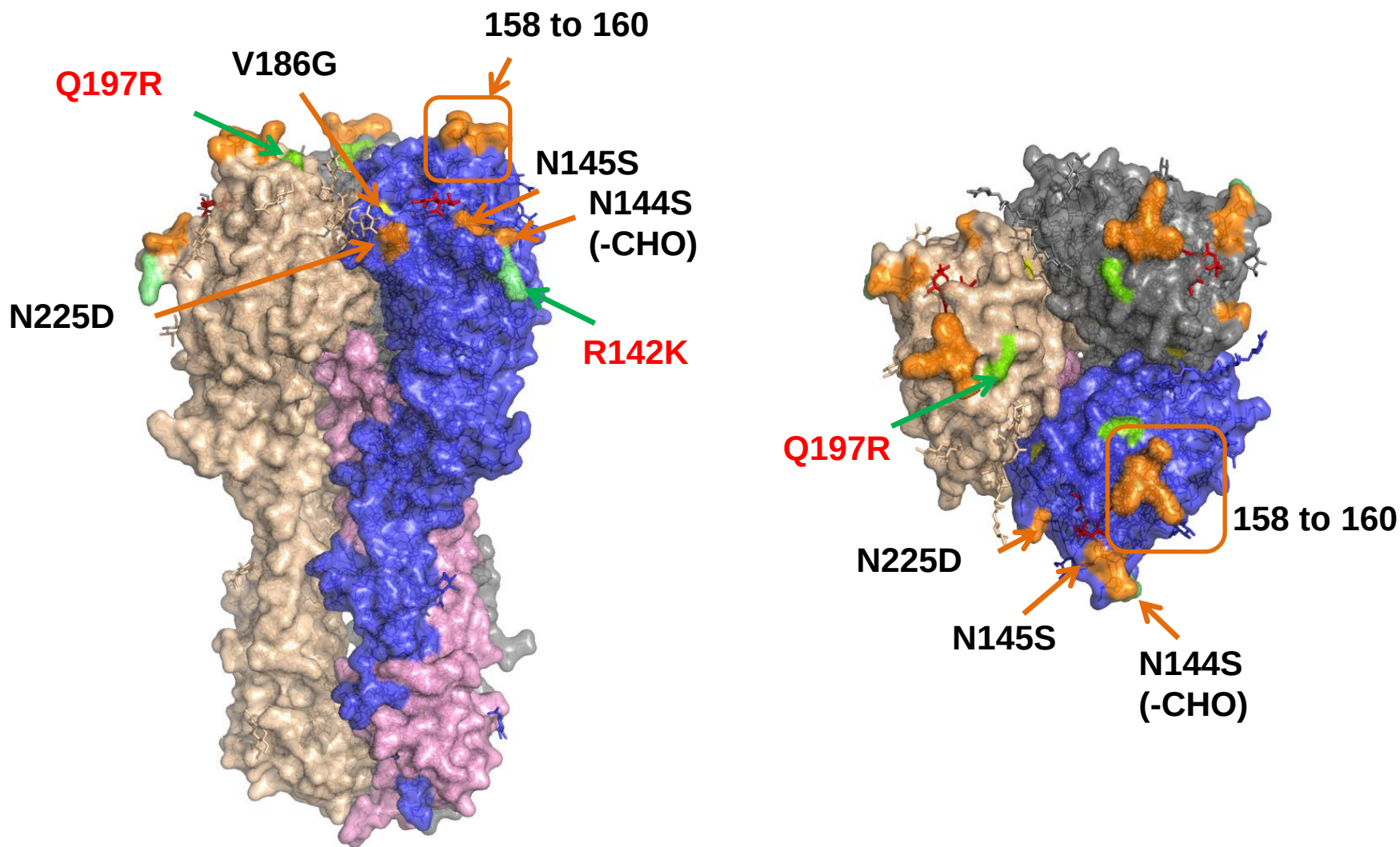


Figure 9B. Amino acid substitutions associated with the HA of 3C.3a viruses relative to A/Texas/50/2012

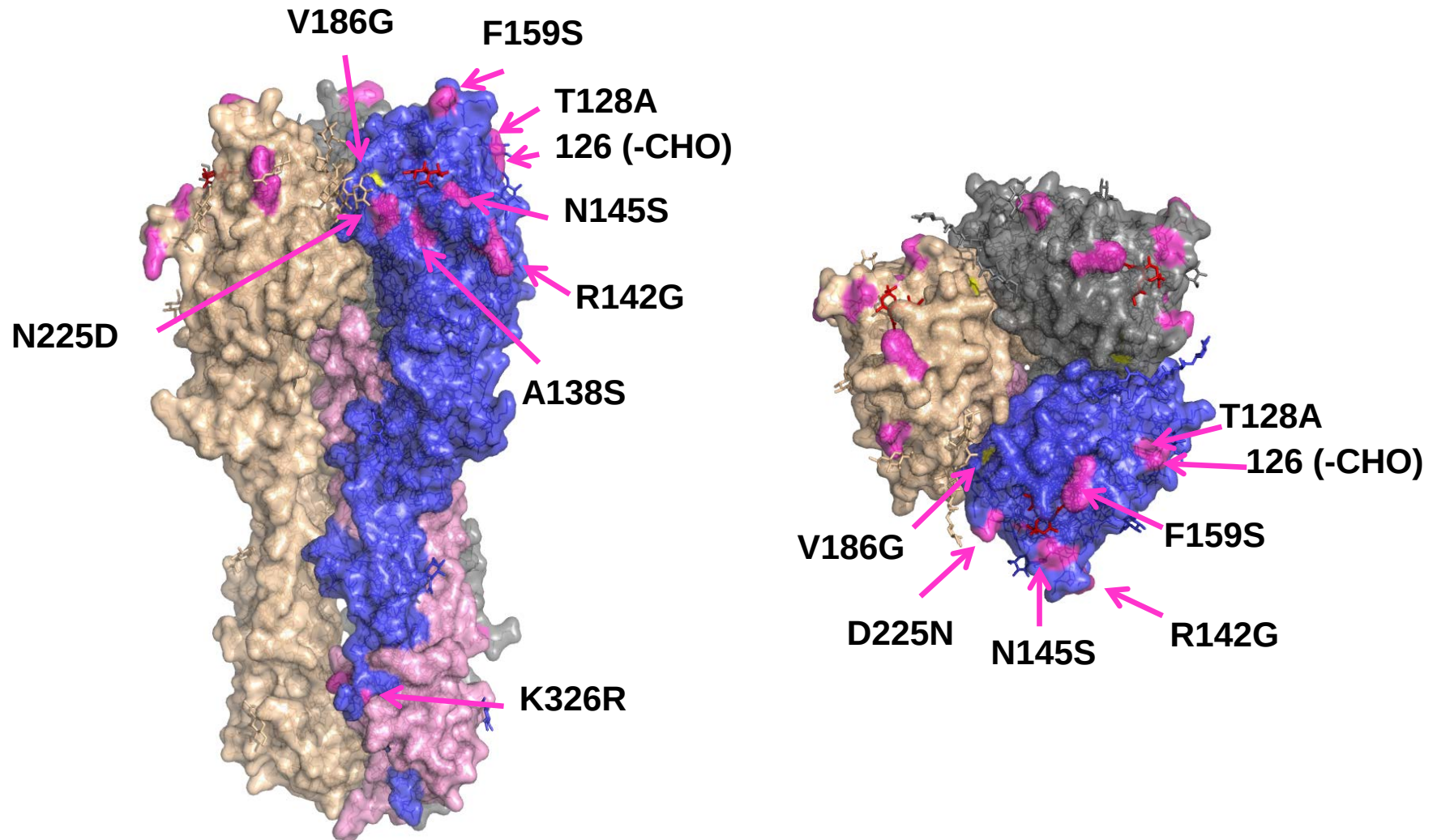


Figure 10. Phylogenetic comparison of A(H3N2) NA genes

Vaccine viruses

Reference viruses

Collection date

Oct 2015

Nov 2015

Dec 2015

Jan 2016

* indicates amino acid polymorphism

@ proposed serology antigens

HI result in table

D151*: If D151N (+CHO)

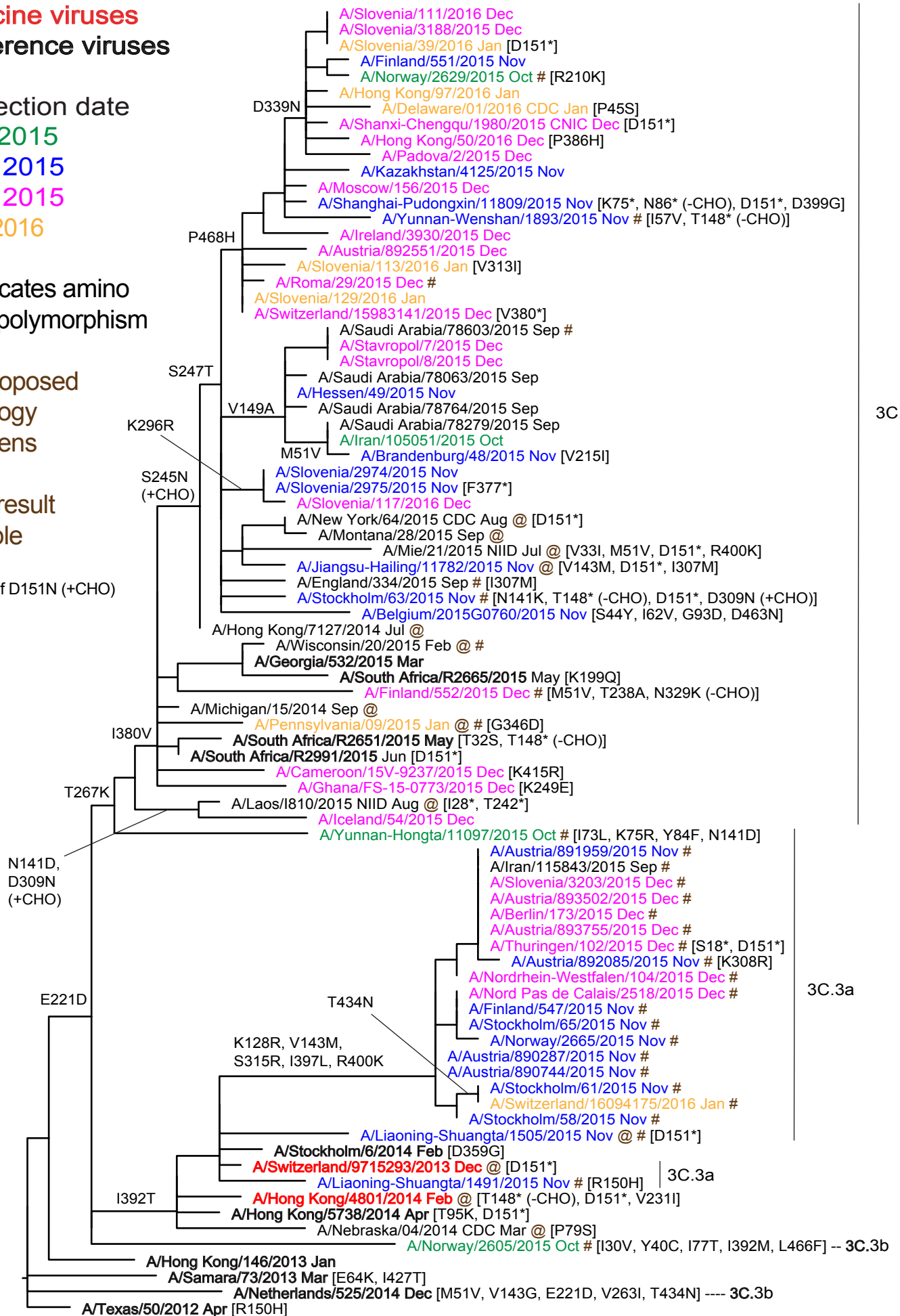


Figure 11. NA protein alignment for A(H3N2) viruses

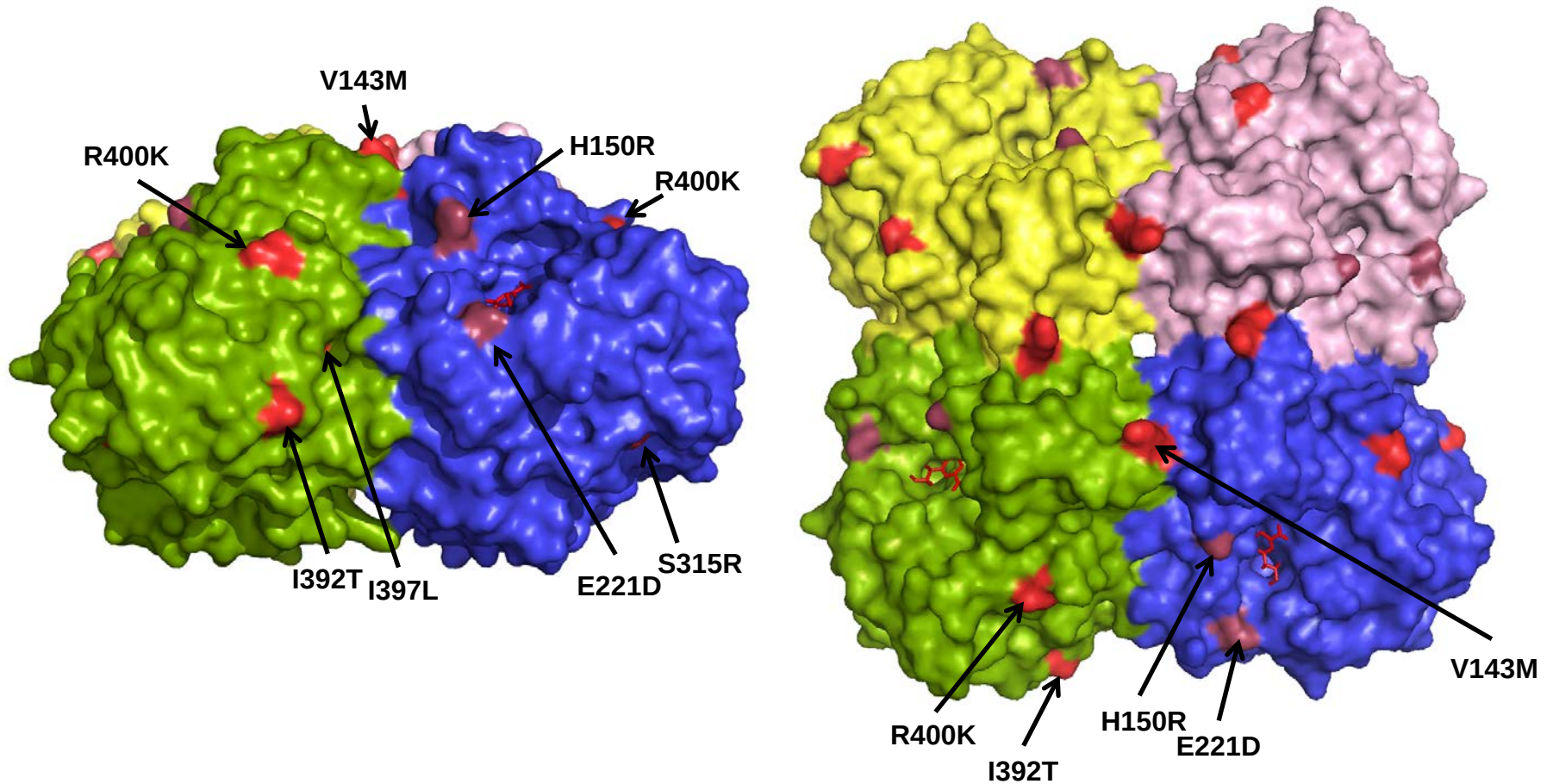
Vaccine virus: Reference virus: Genetic group: Potential N-linked glycosylation site

	10	20	30	40	50	60	70	80	90	100		
A/Texas/50/2012	M	N	P	N	Q	K	I	I	T	G	S	3C.1
A/Samara/73/2013	M	N	P	N	Q	K	I	I	T	G	S	3C.3
A/Norway/2605/2015	.	.	V	.	.	C	.	.	T	.	.	3C.3b
A/Netherlands/525/2014	.	.	.	.	.	V	.	.	.	.	.	3C.3b
A/Stockholm/6/2014	.	.	.	.	.	.	.	.	.	.	.	3C.3
A/Switzerland/9715293/2013	.	.	.	.	.	.	.	.	.	.	.	3C.3a
A/Thuringen/102/2015	.	.	X	.	.	.	.	.	.	.	.	3C.3a
A/Slovenia/3203/2015	.	.	.	.	.	.	.	.	.	.	.	3C.3a
A/Stockholm/58/2015	.	.	.	.	.	.	.	.	.	.	.	3C.3a
A/Hong Kong/146/2013	.	.	.	.	.	.	.	.	.	.	.	3C.2
A/Hong Kong/4801/2014	.	.	.	.	.	.	.	.	.	.	.	3C.2a
A/Hong Kong/5738/2014	.	.	.	.	.	.	.	.	.	K	.	3C.2a
A/Ghana/FS-15-0773/2015	.	.	.	.	.	.	.	.	.	.	.	3C.2a
A/Stavropol/7/2015	.	.	.	.	.	.	.	.	.	.	.	3C.2a
A/South Africa/R2991/2015	.	.	.	.	.	.	.	.	.	.	.	3C.2a
A/South Africa/R2651/2015	.	.	.	.	.	.	.	.	.	.	.	3C.2a
A/Iceland/54/2015	.	.	.	S	.	.	.	.	.	.	.	3C.2a
A/Georgia/532/2015	.	.	.	.	.	.	.	.	.	.	.	3C.2a
A/South Africa/R2665/2015	.	.	.	.	.	.	.	.	.	.	.	3C.2a
A/Slovenia/129/2016	.	.	.	.	.	.	.	.	.	.	.	3C.2a
A/Roma/29/2015	.	.	.	.	.	.	.	.	.	.	.	3C.2a
A/Hong Kong/7127/2014	.	.	.	.	.	.	.	.	.	.	.	3C.2a
A/Jiangsu-Hailing/11782/2015	.	.	.	.	.	.	.	.	.	.	.	3C.2a
A/Montana/28/2015	.	.	.	.	.	.	.	.	.	.	.	3C.2a
A/Belgium/20150760/2015	.	.	.	.	Y	.	.	.	.	D	.	3C.2a
A/Slovenia/117/2016	.	.	.	.	.	.	V	.	.	.	.	3C.2a
A/Iran/105051/2015	.	.	.	.	.	.	V	.	.	.	.	3C.2a
A/Ireland/3930/2015	.	.	.	.	.	.	.	.	.	.	.	3C.2a
A/Hong Kong/97/2016	.	.	.	.	.	.	.	.	.	.	.	3C.2a
A/Finland/551/2015	.	.	.	.	.	.	.	.	.	.	.	3C.2a

	110	120	130	140	150	160	170	180	190	200		
A/Texas/50/2012	S	K	D	N	S	I	R	L	S	A	G	3C.2a
A/Samara/73/2013	S	K	D	N	S	I	R	L	S	A	G	3C.2a
A/Norway/2605/2015	.	.	.	.	.	.	.	.	.	.	.	3C.2a
A/Netherlands/525/2014	.	.	.	.	G	.	.	.	.	.	.	3C.2a
A/Stockholm/6/2014	.	.	.	.	.	.	.	.	.	.	.	3C.2a
A/Switzerland/9715293/2013	.	.	.	.	.	.	.	.	.	.	.	3C.2a
A/Thuringen/102/2015	.	.	.	R	.	.	M	.	.	R	.	3C.2a
A/Slovenia/3203/2015	.	.	.	R	.	.	M	.	.	R	.	3C.2a
A/Stockholm/58/2015	.	.	.	R	.	.	M	.	.	R	.	3C.2a
A/Hong Kong/146/2013	.	.	.	.	.	.	.	.	.	R	.	3C.2a
A/Hong Kong/4801/2014	.	.	.	.	.	X	R	.	.	.	.	3C.2a
A/Hong Kong/5738/2014	.	.	.	.	.	.	R	.	.	.	.	3C.2a
A/Ghana/FS-15-0773/2015	.	.	.	.	.	.	R	.	.	.	.	3C.2a
A/Stavropol/7/2015	.	.	.	.	.	.	A	R	.	.	.	3C.2a
A/South Africa/R2991/2015	.	.	.	.	.	.	.	R	.	.	.	3C.2a
A/South Africa/R2651/2015	.	.	.	.	.	.	X	R	.	.	.	3C.2a
A/Iceland/54/2015	.	.	.	.	D	.	.	R	.	.	.	3C.2a
A/Georgia/532/2015	.	.	.	.	.	.	.	R	.	.	.	3C.2a
A/South Africa/R2665/2015	.	.	.	.	.	.	.	R	.	.	Q	3C.2a
A/Slovenia/129/2016	.	.	.	.	.	.	.	R	.	.	.	3C.2a
A/Roma/29/2015	.	.	.	.	.	.	.	R	.	.	.	3C.2a
A/Hong Kong/7127/2014	.	.	.	.	.	.	.	R	.	.	.	3C.2a
A/Jiangsu-Hailing/11782/2015	.	.	.	.	M	.	.	R	.	.	.	3C.2a
A/Montana/28/2015	.	.	.	.	.	.	.	R	.	.	.	3C.2a
A/Belgium/20150760/2015	.	.	.	.	.	.	.	R	.	.	.	3C.2a
A/Slovenia/117/2016	.	.	.	.	.	.	.	R	.	.	.	3C.2a
A/Iran/105051/2015	.	.	.	.	.	.	.	A	R	.	.	3C.2a
A/Ireland/3930/2015	.	.	.	.	.	.	.	R	.	.	.	3C.2a
A/Hong Kong/97/2016	.	.	.	.	.	.	.	R	.	.	.	3C.2a
A/Finland/551/2015	.	.	.	.	.	.	.	R	.	.	.	3C.2a

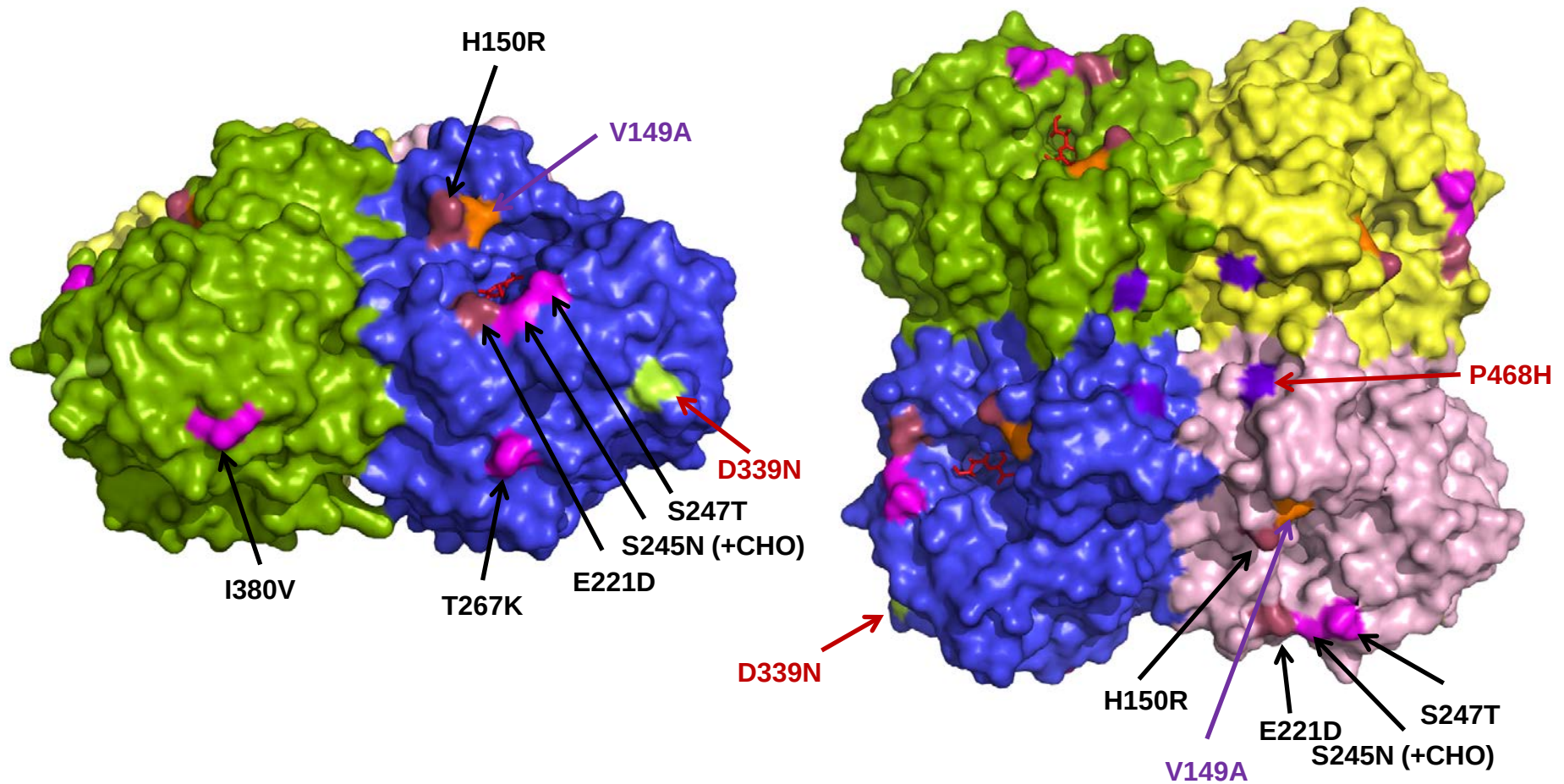
	210	220	230	240	250	260	270	280	290	300		
A/Texas/50/2012	A	T	A	S	F	I	Y	N	G	R	L	3C.2a
A/Samara/73/2013	A	T	A	S	F	I	Y	N	G	R	L	3C.2a
A/Norway/2605/2015	.	.	.	.	.	.	.	.	.	.	.	3C.2a
A/Netherlands/525/2014	.	.	.	.	.	.	.	.	I	.	.	3C.2a
A/Stockholm/6/2014	.	.	.	.	.	.	.	.	.	.	.	3C.2a
A/Switzerland/9715293/2013	.	.	.	.	.	.	.	.	.	.	.	3C.2a
A/Thuringen/102/2015	.	.	.	.	.	.	.	.	.	.	.	3C.2a
A/Slovenia/3203/2015	.	.	.	.	.	.	.	.	.	.	.	3C.2a
A/Stockholm/58/2015	.	.	.	.	.	.	.	.	.	.	.	3C.2a
A/Hong Kong/146/2013	.	.	.	.	.	.	.	.	.	.	.	3C.2a
A/Hong Kong/4801/2014	.	.	.	.	.	.	.	.	.	.	.	3C.2a
A/Hong Kong/5738/2014	.	.	.	.	.	.	.	.	.	.	.	3C.2a
A/Ghana/FS-15-0773/2015	.	.	.	.	.	.	.	E	.	K	.	3C.2a
A/Stavropol/7/2015	.	.	.	.	.	N	T	.	.	K	.	3C.2a
A/South Africa/R2991/2015	.	.	.	.	.	.	.	.	.	K	.	3C.2a
A/South Africa/R2651/2015	.	.	.	.	.	.	.	.	.	K	.	3C.2a
A/Iceland/54/2015	.	.	.	.	.	.	.	.	.	K	.	3C.2a
A/Georgia/532/2015	.	.	.	.	.	.	.	.	.	K	.	3C.2a
A/South Africa/R2665/2015	.	.	.	.	.	.	.	.	.	K	.	3C.2a
A/Slovenia/129/2016	.	.	.	.	.	N	T	.	.	K	.	3C.2a
A/Roma/29/2015	.	.	.	.	.	N	T	.	.	K	.	3C.2a
A/Hong Kong/7127/2014	.	.	.	.	.	N	T	.	.	K	.	3C.2a
A/Jiangsu-Hailing/11782/2015	.	.	.	.	.	N	T	.	.	K	.	3C.2a
A/Montana/28/2015	.	.	.	.	.	N	T	.	.	K	.	3C.2a
A/Belgium/20150760/2015	.	.	.	.	.	N	T	.	.	K	.	3C.2a
A/Slovenia/117/2016	.	.	.	.	.	N	T	.	.	K	.	3C.2a
A/Iran/105051/2015	.	.	.	.	.	N	T	.	.	K	.	3C.2a
A/Ireland/3930/2015	.	.	.	.	.	N	T	.	.	K	.	3C.2a
A/Hong Kong/97/2016	.	.	.	.	.	N	T	.	.	K	.	3C.2a
A/Finland/551/2015	.	.	.	.	.	N	T	.	.	K	.	3C.2a

Figure 12. N2 NA amino acid substitutions in 3C.2a and 3C.3a viruses
a) Recent 3C.3a viruses compared with the NA of A/Texas/50/2012



Residue 128 (K128R) is on the underside of the NA tetramer.

b) 3C.2a viruses compared with the NA of A/Texas/50/2012



Influenza B virus analyses.

Influenza B viruses represented 12% of the samples received with collection dates after 2015-08-31 (**Table 1**). B/Victoria lineage viruses predominated over those of the B/Yamagata lineage at a ratio of approximately 3.5:1.

B/Victoria lineage.

Influenza B viruses of the B/Victoria lineage were received from 19 countries or administrative regions in three WHO Regions (**Table 1**): African (Cameroon and Ghana), European (Belgium, Croatia, England, Estonia, France, Germany, Hungary, Iceland, Ireland, Israel, the Netherlands, Norway, Poland, Spain, Switzerland and Ukraine) and Western Pacific (Hong Kong SAR). The collection details and summary results for viruses that were recovered and assayed by HI are shown in **Table 11**.

HI results for the propagated viruses are shown in **Tables 12-1 to 12-4**. Test viruses in each table are sorted by date of collection and those below the red horizontal lines in the tables are those with collection dates after 2015-08-31. The HA genetic group is indicated for those viruses that have been sequenced and those included in phylogenetic trees are highlighted. The results for two egg-propagated viruses received from the USA are shown in **Table 12-1**. Only two of the 58 cell-propagated test viruses (inclusive of a virus from Algeria with a collection date in 2012) were recognised at titres within 4-fold of the homologous titre with post-infection ferret antisera raised against the recommended vaccine virus, B/Brisbane/60/2008. Similarly, the cell culture-propagated test viruses were poorly recognised by post-infection ferret antisera raised against other egg-propagated reference viruses: B/Malta/636714/2011, B/Johannesburg/3964/2012 and B/South Australia/81/2012, with the same two exceptions. In contrast, all of the test viruses showed reactivity within 4-fold of the titres with the corresponding homologous virus for antisera raised against viruses genetically closely related to B/Brisbane/60/2008 but propagated in cells. These antisera were raised against B/Paris/1762/2009, B/Hong Kong/514/2009 and B/Odessa/3886/2010, viruses that are considered to be surrogate cell-propagated antigens representing the egg-propagated B/Brisbane/60/2008 prototype strain. An antiserum raised against cell-propagated B/Formosa/V2367/2012 failed to recognise six (10%) test viruses within 4-fold of the homologous titre. The two cell culture-propagated viruses with collection dates after 2015-08-31 recognised by the antiserum raised against egg-propagated B/Brisbane/60/2008 vaccine virus showed polymorphism at the glycosylation site at residues 197-199, frequently lost on egg-adaptation (**Table 12-3**).

Phylogenetic analysis of the HA gene of representative B/Victoria lineage viruses is shown in **Figure 13** and an alignment of the HA glycoproteins of a selection of these viruses is shown in **Figure 14**. All recently circulating viruses analysed carried HA genes that fell into genetic group 1A, the B/Brisbane/60/2008 genetic group. **Figure 15** shows a phylogenetic tree for the NA gene and **Figure 16** shows a corresponding

amino acid sequence alignment for a selection of NA glycoproteins. In the NA gene phylogeny, a small group of reassortant viruses with HA genes from the B/Yamagata lineage but NA genes from the B/Victoria lineage are shown (**Figure 15**).

B/Yamagata lineage.

Influenza B viruses of the B/Yamagata lineage were received from 11 countries or administrative regions in four WHO Regions (**Table 1**): African (Cameroon and Ghana), Eastern Mediterranean (Iran), European (Belgium, Germany, Iceland, Italy, Norway, Portugal and Switzerland) and Western Pacific (Hong Kong SAR). The collection details and summary results for viruses that were recovered and assayed by HI are shown in **Table 13**.

The results of HI analyses for propagated viruses of the B/Yamagata lineage are shown in **Tables 14-1 to 14-6**. Test viruses in each table are sorted by date of collection and those below the red horizontal lines in **Tables 14-4 to 14-6** are viruses with collection dates after 2015-08-31. The clade into which the HA gene falls is shown, where known, and viruses for which gene sequences are included in phylogenetic trees are highlighted. A summary of the results for four of the post-infection ferret antisera is shown in **Table 14-7**.

Post-infection ferret antiserum raised against the egg-propagated vaccine virus B/Phuket/3073/2013, recommended for use in the Northern Hemisphere for 2015/2016, recognised 100% of test viruses collected after 2015-08-31 at titres within 4-fold of the titre with the homologous virus, 87% within 2-fold (98% and 88% respectively for all test viruses). However, the ferret antiserum raised against a cell-propagated cultivar of B/Phuket/3073/2013 recognised only 67% of the recently collected test viruses at titres within 4-fold, and 27% within 2-fold, of its titre with the homologous virus (72% and 37% respectively for all test viruses). These antisera recognised the test viruses better than an antiserum raised against egg-propagated B/Massachusetts/02/2012, which only recognised 60% of recently collected test viruses (39% of all test viruses) at titres within 4-fold of the titre of the antiserum for the homologous virus. The antiserum raised against cell culture-propagated B/Massachusetts/02/2012 recognised a third of recently collected test viruses at a titre within 4-fold of its homologous titre (45% of all test viruses). Test viruses were recognised somewhat better by an antiserum raised against cell culture-propagated B/Estonia/55669/2011, another virus from the B/Massachusetts/02/2012 clade (clade 2), with 14 of the 15 test viruses collected after 2015-08-31 being recognised by the antiserum at a titre within 4-fold of the titre of the antiserum for the homologous virus (thirteen were within 2-fold of the homologous titre); however, the homologous titre of this antiserum was very low.

Antisera raised against other viruses from the B/Phuket/3073/2013 clade (clade 3) propagated in eggs, those raised against B/Wisconsin/1/2010, B/Stockholm/12/2011 and B/Hong Kong/3417/2014, also recognised the test viruses collected since 2015-

08-31 well, recognising 100%, 80% and 100%, respectively, at titres within 4-fold of the titres of the antisera for their homologous viruses, and 90%, 66% and 100%, respectively, at titres within 2-fold of the homologous titres.

Figure 17 shows a phylogenetic analysis of the HA gene of representative B/Yamagata lineage viruses and an amino acid sequence alignment of a selection of the HA glycoproteins is shown in **Figure 18**. Reassortant viruses with HA genes from the B/Yamagata lineage but NA genes from the B/Victoria lineage are indicated, falling within two distinct HA gene phylogenetic clusters.

Figure 19 shows a phylogenetic tree for NA genes of the same viruses represented in the HA phylogeny and **Figure 20** shows a corresponding amino acid sequence alignment for selected NA glycoproteins. All HA and NA genes of recently collected viruses fell into the B/Phuket/3073/2013 clade (clade 3).

Antiviral Analysis.

Sensitivity to the neuraminidase inhibitors zanamivir and oseltamivir was assessed for 44 influenza B viruses collected after 2015-08-31 (**Table 15**). All viruses showed normal inhibition (no reduced susceptibility) by both antivirals.

Conclusion.

Viruses of the B/Victoria lineage have increased in frequency. Most viruses of the B/Victoria lineage were antigenically similar to viruses genetically closely related to B/Brisbane/60/2008. Viruses of the B/Yamagata lineage fell into genetic clade 3, the B/Phuket/3073/2013 genetic clade, and were recognised well by antisera raised against egg-propagated vaccine/reference viruses from genetic clade 3.

Table 11. Summary of influenza B (Victoria-lineage) viruses analysed, collected after 2015-08-31

MONTH	B Victoria lineage				
Country	Number received	Number propagated¹	≤2 fold²	4 fold²	≥8 fold²
OCTOBER					
Norway	1	1	1		
United Kingdom	1	1	1		
NOVEMBER					
Belgium	6	in process		5	
France	3	in process			
Germany	1	1	1		
Netherlands	2	in process		1	
Norway	1	1	1		
Ukraine	1	1	1		
DECEMBER					
Estonia	4	in process	1		
France	15	in process	11	1	
Germany	1	1	1		
Ghana	1	in process			
Iceland	1	1	1		
Ireland	1	1	1		
Israel	9	9	9		
Spain	4	4	4		
Switzerland	4	in process	2		
Ukraine	1	1	1		
JANUARY					
Croatia	2	in process			
Estonia	1	1	1		
Germany	3	3	3		
Hong Kong	3	3	2	1	
Hungary	3	in process			
Ireland	1	1	1		
	70	51	43	8	0
17 Countries			84.3%	15.7%	0.0%

1. Propagated to sufficient titre to perform HI assay

2. Compared to homologous titres for ferret antisera raised against at least one of three tissue-culture grown B/Brisbane/60/2008-like equivalents (B/Paris/1762/2008, B/Hong Kong/514/2009 & B/Odessa/3886/2010)

Table 12-1. Antigenic analyses of influenza B viruses (Victoria lineage) 2015-09-16, 2015-09-23, 2015-11-18, 2015-11-25, 2015-12-08

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
				Post infection ferret antisera									
				B/Bris 60/08	B/Mal 2506/04	B/Bris 60/08	B/Paris 1762/09	B/Malta 636714/11	B/Jhb 3964/12	B/For V2367/12	B/Sth Aus 81/12	B/HK 514/09	B/Odessa 3886/10
				Egg Sh 539/540 /543/544 ^{1,3}	Egg F37/11 ²	Egg F22/12 ²	MDCK F07/11 ²	Egg F29/13 ²	Egg F01/13 ²	MDCK F04/13 ⁴	Egg F41/13 ²	MDCK F19/13 ²	MDCK F19/11 ²
	Passage history												
	Ferret number												
	Genetic group			1A		1A	1A	1A	1A	1A	1A	1B	1B
REFERENCE VIRUSES													
B/Malaysia/2506/2004		2004-12-06	E3/E6	1280	640	80	<	160	160	80	160	20	<
B/Brisbane/60/2008	1A	2008-08-04	E4/E5	640	80	160	40	320	320	320	640	80	40
B/Paris/1762/2009	1A	2009-02-09	C2/MDCK2	640	<	10	40	20	20	40	40	40	40
B/Hong Kong/514/2009	1B	2009-10-11	MDCK1/MDCK2	320	<	10	40	40	10	40	40	40	80
B/Odessa/3886/2010	1B	2010-03-19	C2/MDCK2	320	<	10	20	40	20	40	40	40	80
B/Malta/636714/2011	1A	2011-03-07	E4/E1	640	80	160	40	320	320	320	640	80	40
B/Johannesburg/3964/2012	1A	2012-08-03	E1/E2	2560	640	1280	160	1280	1280	1280	1280	320	160
B/Formosa/V2367/2012	1A	2012-08-06	MDCK1/MDCK3	1280	40	160	40	160	160	160	320	40	40
B/South Australia/81/2012	1A	2012-11-28	E4/E1	640	80	160	40	320	160	320	640	80	40
TEST VIRUSES													
B/Kaliningrad/14/2015	1A	unknown	C1/MDCK1	640	<	<	40	20	<	20	20	40	40
B/Texas/11/2015	1A	2015-02-25	C1/MDCK1	1280	10	10	80	80	40	160	80	80	80
B/Iceland/48/2015	1A	2015-03-25	MDCK/MDCK1	320	<	<	40	40	<	40	40	40	40
B/Wyoming/24/2015	1A	2015-04-11	C1/MDCK1	1280	10	<	40	80	20	80	80	80	80
B/Indiana/25/2015	1A	2015-04-27	E2/E2	640	160	80	20	320	160	80	320	40	20
B/Hawaii/24/2014	1A	2015-07-16	E4/E2	160	20	40	40	160	80	80	160	20	20
B/England/345/2015	1A	2015-10-08	SIAT1/MDCK1	640	10	<	40	40	<	40	40	40	80
B/Norway/2622/2015	1A	2015-10-21	MDCK1/MDCK1	640	<	<	40	40	<	40	80	40	80

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

¹ < = <40; ² < = <10; ³ hyperimmune sheep serum; ⁴ < = <20

Vaccine
NH 2015-16[#]
SH 2016

[#] B/Victoria-lineage virus recommended for use in quadravalent vaccines

Sequences in phylogenetic trees

197/199 NEX

197/199 NEN

Table 12-2. Antigenic analyses of influenza B viruses (Victoria lineage) 2016-01-13, 2016-01-19

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
				Post-infection ferret antisera									
				B/Bris	B/Mal	B/Bris	B/Paris	B/Malta	B/Jhb	B/For	B/Sth Aus	B/HK	B/Odessa
				60/08	2506/04	60/08	1762/09	636714/11	3964/12	V2367/12	81/12	514/09	3886/10
				Egg	Egg	Egg	MDCK	Egg	Egg	MDCK	Egg	MDCK	MDCK
Passage history				Sh 539/540 /543/544 ^{1,3}	F37/11 ²	F22/12 ²	F07/11 ²	F29/13 ²	F01/13 ²	F04/13 ⁴	F41/13 ²	F09/13 ²	F19/11 ²
Ferret number													
Genetic group				1A		1A	1A	1A	1A	1A	1A	1B	1B
REFERENCE VIRUSES													
B/Malaysia/2506/2004		2004-12-06	E3/E7	320	320	20	<	80	80	40	80	10	<
B/Brisbane/60/2008	1A	2008-08-04	E4/E5	640	40	160	40	320	320	160	640	40	40
B/Paris/1762/2009	1A	2009-02-09	C2/MDCK2	640	<	<	40	40	40	40	40	20	40
B/Malta/636714/2011	1A	2011-03-07	E4/E1	640	40	160	40	320	320	160	640	40	40
B/Johannesburg/3964/2012	1A	2012-08-03	E1/E2	2560	160	640	80	1280	1280	640	1280	80	40
B/Formosa/V2367/2012	1A	2012-08-06	MDCK1/MDCK3	640	20	80	40	160	160	160	320	20	20
B/South Australia/81/2012	1A	2012-11-28	E4/E2	640	40	160	40	160	320	160	640	40	20
B/Hong Kong/514/2009	1B	2009-10-11	MDCK3	640	<	<	40	40	20	20	40	80	40
B/Odessa/3886/2010	1B	2010-03-19	MDCK2/MDCK4	640	<	<	40	40	40	40	40	40	80
TEST VIRUSES													
B/Belgium/2015G0748/2015	1A	2015-11-03	SIAT1/MDCK1	320	<	<	20	20	20	40	40	20	20
B/Belgium/2015G0753/2015	1A	2015-11-03	SIAT1/MDCK1	640	<	<	20	20	20	40	40	20	20
B/Belgium/2015G0762/2015	1A	2015-11-10	SIAT1/MDCK1	640	<	<	40	40	40	40	40	20	40
B/Belgium/2015G0764/2015	1A	2015-11-10	SIAT1/MDCK1	640	<	<	40	40	40	40	40	20	40
B/Netherlands/2914/2015	1A	2015-11-10	MDCK1/MDCK1	640	<	<	40	20	20	40	40	20	20
B/Belgium/2015G0769/2015	1A	2015-11-18	SIAT1/MDCK1	640	<	<	20	40	40	40	40	20	20
B/Sachsen/37/2015	1A	2015-11-25	C2/MDCK1	640	<	<	40	40	40	40	80	40	40
B/Marseille/2589/2015	1A	2015-12-03	MDCK2/MDCK1	640	<	<	40	40	40	40	80	40	40
B/Marseille/2590/2015	1A	2015-12-09	MDCK2/MDCK1	640	<	<	40	40	40	40	80	40	40
B/Lyon/2583/2015	1A	2015-12-11	MDCK2/MDCK1	640	<	<	40	40	20	40	40	20	20
B/Lyon/14.12.R13/2015	1A	2015-12-13	MDCK2/MDCK1	640	<	<	40	40	40	40	80	40	40
B/Marseille/2643/2015	1A	2015-12-13	MDCK2/MDCK1	640	<	<	40	40	40	40	80	40	40
B/Marseille/2645/2015	1A	2015-12-13	MDCK2/MDCK1	640	<	<	40	80	80	80	160	40	40
B/Thuringen/43/2015	1A	2015-12-14	C2/MDCK1	640	<	<	40	40	40	40	80	40	40
B/Lyon/2585/2015	1A	2015-12-15	MDCK2/MDCK1	640	<	<	40	40	40	40	80	40	40
B/Lyon/17.12.R76/2015	1A	2015-12-17	MDCK2/MDCK1	640	<	<	40	40	40	40	80	40	40
B/Lyon/2658/2015	1A	2015-12-22	MDCK2/MDCK1	640	<	<	40	40	40	40	40	40	40

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

¹ < = <40; ² < = <10; ³ hyperimmune sheep serum; ⁴ < = <20

Vaccine
NH 2015-16[#]
SH 2016

[#] B/Victoria-lineage virus recommended for use in quadravalent vaccines

Sequences in phylogenetic trees

Table 12-3. Antigenic analyses of influenza B viruses (Victoria lineage) 2016-01-26

Haemagglutination inhibition titre														
Viruses	Other information	Collection date	Passage history	Post-infection ferret antisera										
				B/Bris	B/Mal	B/Bris	B/Paris	B/Malta	B/Jhb	B/For	B/Sth Aus	B/HK	B/Odessa	
				60/08	2506/04	60/08	1762/09	636714/11	3964/12	V2367/12	81/12	514/09	3886/10	
				Passage history	Egg	Egg	Egg	MDCK	Egg	Egg	MDCK	Egg	MDCK	MDCK
				Ferret number	Sh 539/540 /543/544 ^{1,3}	F37/11 ^{1,2}	F22/12 ²	F07/11 ^{1,2}	F29/13 ^{1,2}	F01/13 ^{1,2}	F04/13 ^{1,4}	F41/13 ^{1,2}	F09/13 ^{1,2}	F19/11 ^{1,2}
Genetic group	1A		1A	1A	1A	1A	1A	1A	1A	1B	1B			
REFERENCE VIRUSES														
B/Malaysia/2506/2004		2004-12-06	E3/E7	640	320	20	<	40	160	40	80	10	<	
B/Brisbane/60/2008	1A	2008-08-04	E4/E5	640	40	160	40	320	320	160	640	40	40	
B/Paris/1762/2009	1A	2009-02-09	C2/MDCK2	640	<	<	40	20	40	40	40	20	40	
B/Malta/636714/2011	1A	2011-03-07	E4/E1	640	40	80	40	320	320	160	640	40	20	
B/Johannesburg/3964/2012	1A	2012-08-03	E1/E2	2560	320	640	160	1280	1280	1280	1280	160	80	
B/Formosa/V2367/2012	1A	2012-08-06	MDCK1/MDCK3	1280	20	80	40	160	160	160	320	40	20	
B/South Australia/81/2012	1A	2012-11-28	E4/E2	640	40	80	20	160	320	160	640	40	40	
B/Hong Kong/514/2009	1B	2009-10-11	MDCK3	640	<	<	40	40	20	40	40	40	80	
B/Odessa/3886/2010	1B	2010-03-19	C2/MDCK2	1280	<	<	40	80	40	80	80	40	80	
TEST VIRUSES														
B/Yunnan-Mengzi/1977/2015	1A	2015-11-03	MDCK2/MDCK1	640	<	<	40	40	40	40	80	20	40	
B/Shanghai-Pudongxin/11802/2015	1A	2015-11-10	MDCK3/MDCK1	640	<	<	40	40	40	40	80	40	40	
B/Israel/Q-501/2015	1A	2015-12-15	C1/MDCK1	640	<	<	40	20	40	40	40	20	40	
B/Israel/Q-503/2015	1A	2015-12-15	C2/MDCK1	1280	40	40	40	160	320	160	320	40	40	
B/Israel/Q-476/2015	1A	2015-12-15	C2/MDCK1	640	<	<	40	40	40	40	80	20	40	
B/Israel/Q-540/2015	1A	2015-12-22	C2/MDCK1	640	<	<	40	40	40	80	80	40	40	
B/Israel/Q-567/2015	1A	2015-12-22	C2/MDCK1	640	<	<	40	40	40	40	40	20	40	
B/Israel/Q-578/2015	1A	2015-12-22	C2/MDCK1	640	<	<	40	40	40	40	40	20	40	
B/Switzerland/16034416/2015	1A	2015-12-25	MDCK1	640	<	<	40	40	40	40	80	20	40	
B/Switzerland/16034436/2015	1A	2015-12-28	MDCK1	640	<	<	40	40	20	40	40	20	40	
B/Estonia/97395/2015	1A	2015-12-30	MDCK2	640	<	<	40	40	40	40	80	40	40	
B/Iceland/56/2015	1A	2015-12-30	MDCK0/MDCK1	640	<	<	40	20	20	40	40	20	40	
B/Hong Kong/58/2016	1A	2016-01-02	MDCK1/MDCK1	640	<	<	40	40	40	40	80	40	40	
B/Estonia/97415/2016	1A	2016-01-04	MDCK1	640	<	<	40	40	40	40	80	40	40	
B/Hong Kong/57/2016	1A	2016-01-04	MDCK1/MDCK1	640	20	80	20	320	320	160	320	40	20	
B/Hong Kong/62/2016	1A	2016-01-05	MDCK1/MDCK1	640	<	<	40	40	40	40	80	40	40	

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

¹ < = <40; ² < = <10; ³ hyperimmune sheep serum; ⁴ < = <20

[#] B/Victoria-lineage virus recommended for use in quadrivalent vaccines

Sequences in phylogenetic trees

Vaccine
NH 2015-16[#]
SH 2016

Table 12-4. Antigenic analyses of influenza B viruses (Victoria lineage) 2016-02-10

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
				Post-infection ferret antisera									
				B/Bris 60/08 Egg	B/Mal 2506/04 Egg	B/Bris 60/08 Egg	B/Paris 1762/09 MDCK	B/Malta 636714/11 Egg	B/Jhb 3964/12 Egg	B/For V2367/12 MDCK	B/Sth Aus 81/12 Egg	B/HK 514/09 MDCK	B/Odessa 3886/10 MDCK
				Sh 539/540 /543/544 ^{1,3}	F37/11 ²	F22/12 ²	F07/11 ²	F29/13 ²	F01/13 ¹	F04/13 ⁴	F41/13 ²	F09/13 ²	F19/11 ²
	Passage history			1A		1A	1A	1A	1A	1A	1A	1B	1B
REFERENCE VIRUSES													
B/Malaysia/2506/2004		2004-12-06	E3/E7	640	320	20	<	80	160	80	160	20	<
B/Brisbane/60/2008	1A	2008-08-04	E4/E5	640	40	160	40	160	160	160	640	40	40
B/Paris/1762/2009	1A	2009-02-09	C2/MDCK2	640	<	<	80	20	40	40	80	40	40
B/Malta/636714/2011	1A	2011-03-07	E4/E1	640	80	160	40	640	320	320	640	80	40
B/Johannesburg/3964/2012	1A	2012-08-03	E1/E2	5120	320	1280	160	1280	1280	1280	1280	320	160
B/Formosa/V2367/2012	1A	2012-08-06	MDCK1/MDCK3	1280	20	80	40	160	160	320	320	40	20
B/South Australia/81/2012	1A	2012-11-28	E4/E2	1280	80	160	40	320	320	320	640	80	40
B/Hong Kong/514/2009	1B	2009-10-11	MDCK3	640	<	<	80	40	40	40	40	40	80
B/Odessa/3886/2010	1B	2010-03-19	C2/MDCK2	640	<	<	40	40	40	80	80	40	80
TEST VIRUSES													
B/Algeria/G229-12/2012		2012-12-11	Cx/MDCK1	1280	<	<	80	40	40	80	80	40	80
B/Norway/2726/2015	1A	2015-11-14	MDCK1	1280	<	<	80	80	80	80	80	40	40
B/Odessa/72/2016	ngs?	2015-11-18	SIAT2/MDCK1	640	10	<	80	40	40	40	40	40	40
B/Castilla La Mancha/1807/2015	1A	2015-12-03	SIAT1/MDCK1	1280	<	<	80	40	40	80	40	40	80
B/Israel/A-6494/2015	ngs?	2015-12-14	C1/MDCK1	1280	<	<	80	80	80	80	80	40	80
B/Israel/A-6460/2015	ngs?	2015-12-14	C1/MDCK1	1280	<	<	80	40	40	80	80	40	40
B/Castilla La Mancha/1808/2015	1A	2015-12-14	SIAT1/MDCK1	1280	<	<	40	40	40	80	40	20	40
B/Castilla La Mancha/1809/2015	1A	2015-12-14	SIAT1/MDCK1	640	10	<	80	40	40	80	80	40	80
B/Castilla La Mancha/1820/2015	1A	2015-12-14	SIAT1/MDCK1	1280	<	<	80	40	40	80	40	40	40
B/Israel/A-6501/2015	ngs?	2015-12-15	C1/MDCK1	640	<	<	80	40	40	80	80	40	40
B/Ireland/71062/2015	ngs?	2015-12-15	MDCK1/MDCK1	1280	<	<	80	40	40	80	80	40	40
B/Bretagne/2531/2015	1A	2015-12-16	MDCK1/MDCK1	640	<	<	80	40	<	40	80	20	40
B/Ukraine/6913/2015	ngs?	2015-12-18	C1/MDCK1	1280	<	<	80	40	40	80	80	40	40
B/Nord Pas de Calais/2578/2015	1A	2015-12-23	MDCK1/MDCK1	640	<	<	80	40	<	40	80	20	40
B/Bretagne/2597/2015	1A	2015-12-29	MDCK1/MDCK1	640	<	<	80	40	<	40	80	20	40
B/Nordrhein-Westfalen/1/2016	1A	2016-01-04	C2/MDCK1	640	<	<	80	40	40	40	80	20	40
B/Schleswig-Holstein/1/2016	1A	2016-01-05	C2/MDCK1	1280	<	<	80	40	40	80	80	40	40
B/Thuringen/1/2016	1A	2016-01-11	C2/MDCK1	1280	<	<	80	40	40	80	80	40	40
B/Ireland/3154/2016	1A	2016-01-14	MDCK1/MDCK1	640	<	<	80	40	40	80	40	40	40

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

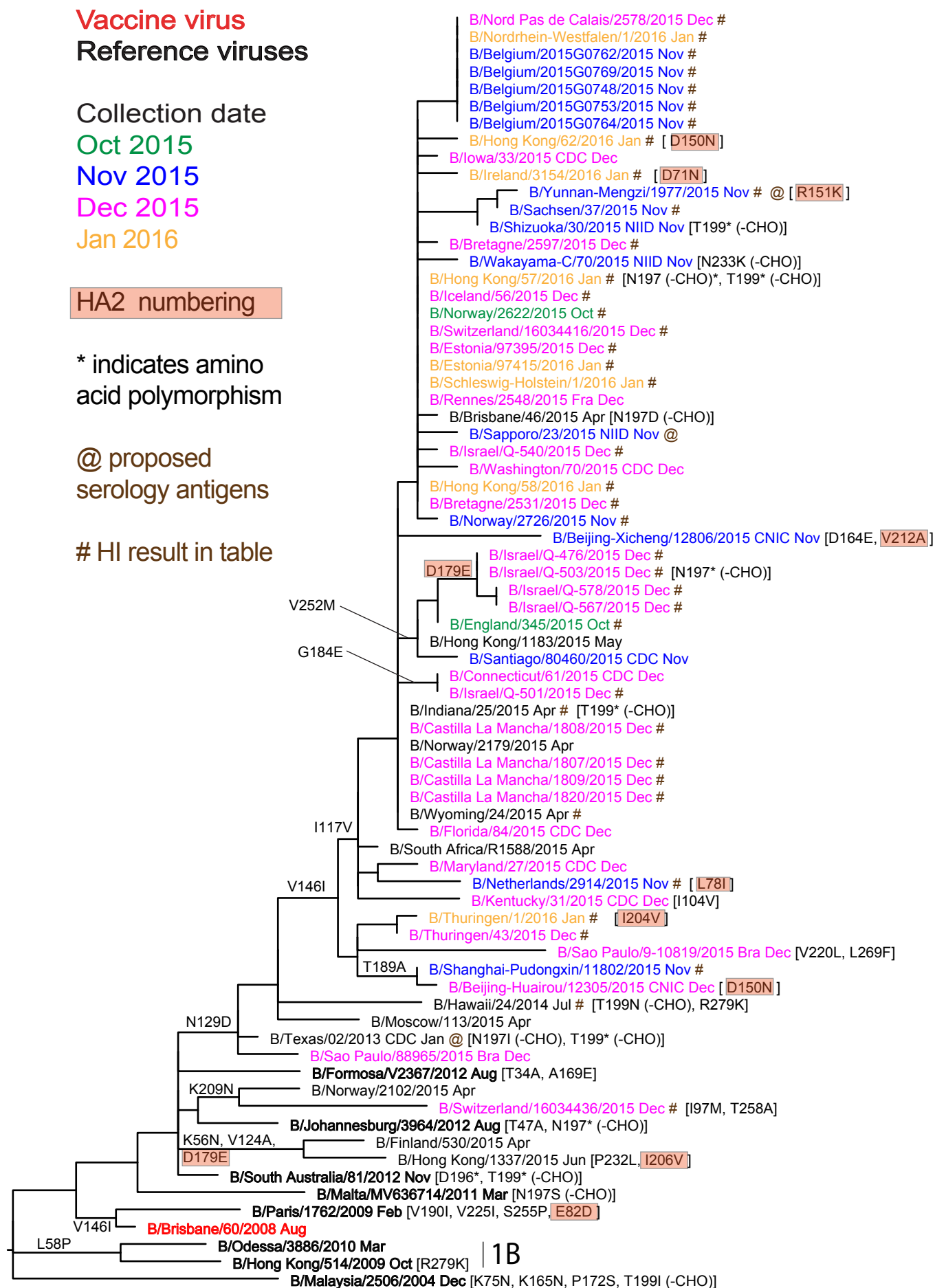
¹ < = <40; ² < = <10; ³ hyperimmune sheep serum; ⁴ < = <20

Vaccine
NH 2015-16[#]
SH 2016

[#] B/Victoria-lineage virus recommended for use in quadravalent vaccines

Sequences in phylogenetic trees

Figure 13. Phylogenetic comparison of influenza B (Victoria-lineage) HA genes

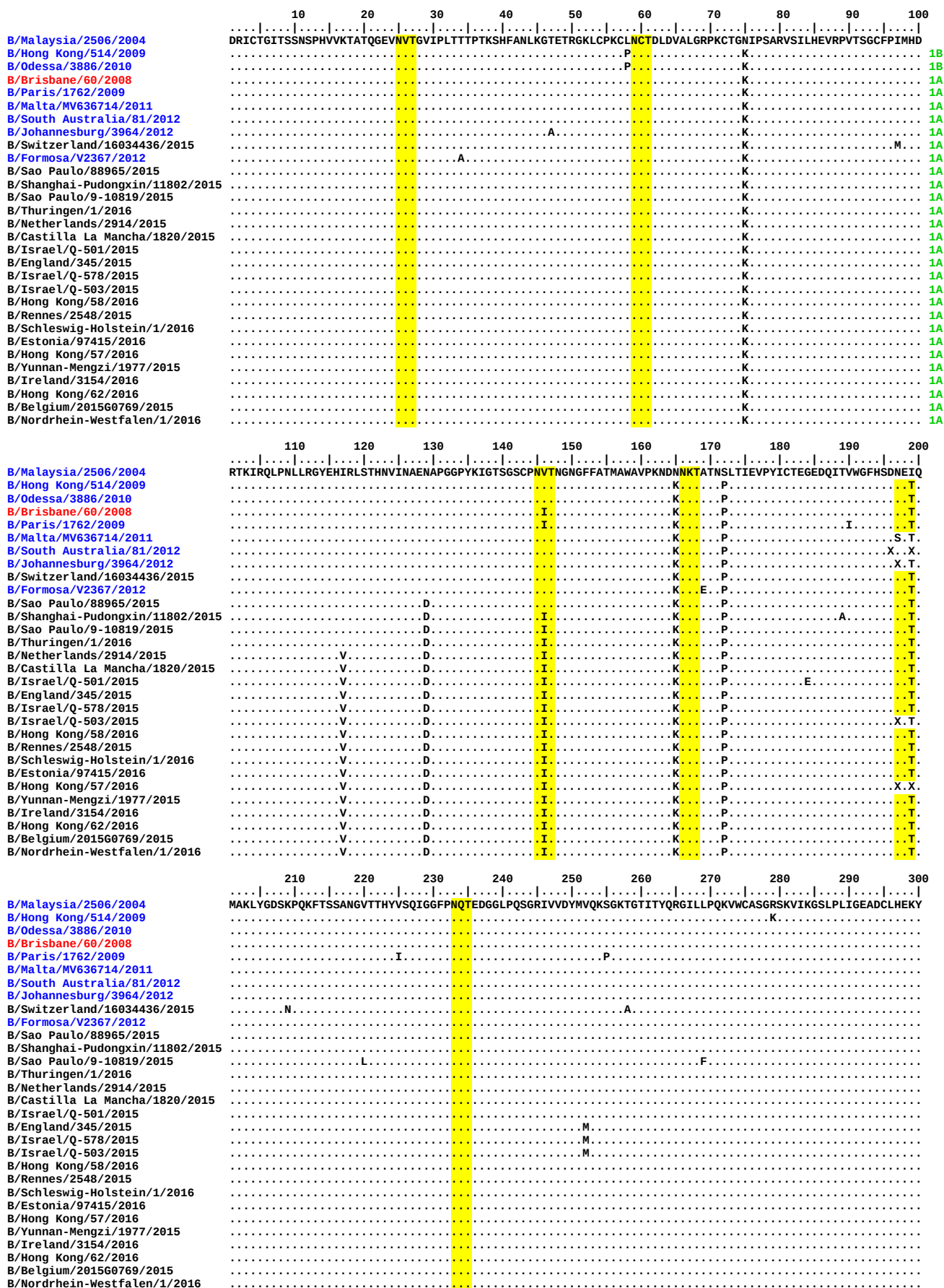


1A

1B

Figure 14. HA protein alignment for B (Victoria-lineage) viruses

Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site



	310	320	330	340	350	360	370	380	390	400
B/Malaysia/2506/2004	GGLN	KSKPY	YTGEHAKA	IGNCPI	WVKTP	LKLAN	GT	KYRPP	AKLLK	RGFFG
B/Hong Kong/514/2009										
B/Odessa/3886/2010										
B/Brisbane/60/2008										
B/Paris/1762/2009										
B/Malta/MV636714/2011										
B/South Australia/81/2012										
B/Johannesburg/3964/2012										
B/Switzerland/16034436/2015										
B/Formosa/V2367/2012										
B/Sao Paulo/88965/2015										
B/Shanghai-Pudongxin/11802/2015										
B/Sao Paulo/9-10819/2015										
B/Thuringen/1/2016										
B/Netherlands/2914/2015										
B/Castilla La Mancha/1820/2015										
B/Israel/Q-501/2015										
B/England/345/2015										
B/Israel/Q-578/2015										
B/Israel/Q-503/2015										
B/Hong Kong/58/2016										
B/Rennes/2548/2015										
B/Schleswig-Holstein/1/2016										
B/Estonia/97415/2016										
B/Hong Kong/57/2016										
B/Yunnan-Mengzi/1977/2015										
B/Ireland/3154/2016										
B/Hong Kong/62/2016										
B/Belgium/201560769/2015										
B/Nordrhein-Westfalen/1/2016										

	410	420	430	440	450	460	470	480	490	500
B/Malaysia/2506/2004	SLSE	LEVKN	LQRL	SGAM	DELH	NEIL	ELDE	KVDL	RADT	ISSQ
B/Hong Kong/514/2009										
B/Odessa/3886/2010										
B/Brisbane/60/2008										
B/Paris/1762/2009										
B/Malta/MV636714/2011										
B/South Australia/81/2012										
B/Johannesburg/3964/2012										
B/Switzerland/16034436/2015										
B/Formosa/V2367/2012										
B/Sao Paulo/88965/2015										
B/Shanghai-Pudongxin/11802/2015										
B/Sao Paulo/9-10819/2015										
B/Thuringen/1/2016										
B/Netherlands/2914/2015										
B/Castilla La Mancha/1820/2015										
B/Israel/Q-501/2015										
B/England/345/2015										
B/Israel/Q-578/2015										
B/Israel/Q-503/2015										
B/Hong Kong/58/2016										
B/Rennes/2548/2015										
B/Schleswig-Holstein/1/2016										
B/Estonia/97415/2016										
B/Hong Kong/57/2016										
B/Yunnan-Mengzi/1977/2015										
B/Ireland/3154/2016										
B/Hong Kong/62/2016										
B/Belgium/201560769/2015										
B/Nordrhein-Westfalen/1/2016										

	510	520	530	540	550	560	570
B/Malaysia/2506/2004	AGTF	DAGE	FSLP	TFDSL	NNTA	ASLND	DGLD
B/Hong Kong/514/2009							
B/Odessa/3886/2010							
B/Brisbane/60/2008							
B/Paris/1762/2009							
B/Malta/MV636714/2011							
B/South Australia/81/2012							
B/Johannesburg/3964/2012							
B/Switzerland/16034436/2015							
B/Formosa/V2367/2012							
B/Sao Paulo/88965/2015							
B/Shanghai-Pudongxin/11802/2015							
B/Sao Paulo/9-10819/2015							
B/Thuringen/1/2016							
B/Netherlands/2914/2015							
B/Castilla La Mancha/1820/2015							
B/Israel/Q-501/2015							
B/England/345/2015							
B/Israel/Q-578/2015							
B/Israel/Q-503/2015							
B/Hong Kong/58/2016							
B/Rennes/2548/2015							
B/Schleswig-Holstein/1/2016							
B/Estonia/97415/2016							
B/Hong Kong/57/2016							
B/Yunnan-Mengzi/1977/2015							
B/Ireland/3154/2016							
B/Hong Kong/62/2016							
B/Belgium/201560769/2015							
B/Nordrhein-Westfalen/1/2016							

Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

Figure 15. Phylogenetic comparison of influenza B (Victoria-lineage) NA genes

Vaccine virus

Reference viruses

Collection date

Oct 2015

Nov 2015

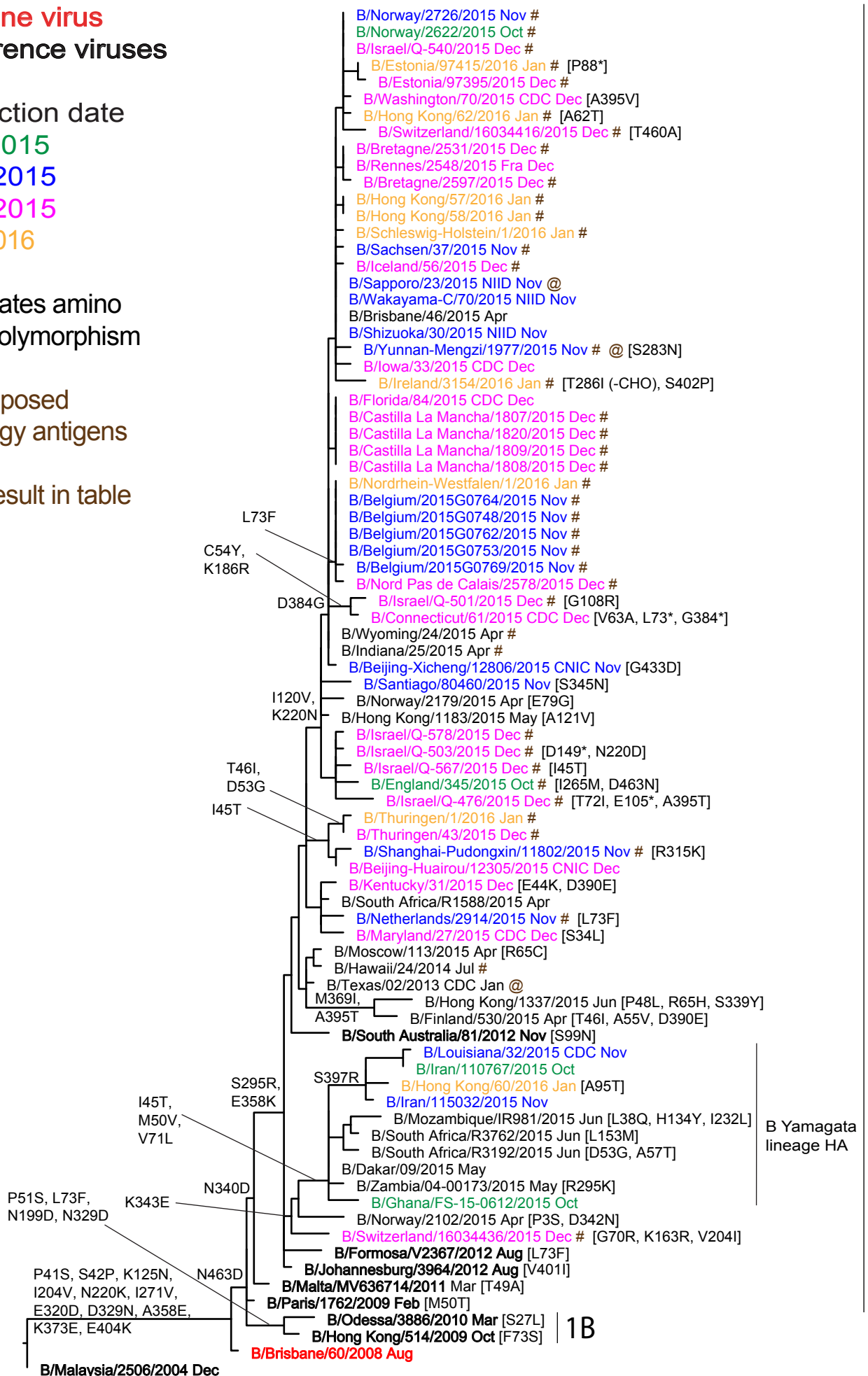
Dec 2015

Jan 2016

* indicates amino acid polymorphism

@ proposed serology antigens

HI result in table



1A

1B

B Yamagata lineage HA

Figure 16. NA protein alignment for B (Victoria-lineage) viruses

Vaccine virus: Reference virus: Genetic group: Potential N-linked glycosylation site

	10	20	30	40	50	60	70	80	90	100
B/Malaysia/2506/2004	M	L	P	S	T	I	Q	T	L	T
B/Hong Kong/514/2009	M	L	P	S	T	I	Q	T	L	T
B/Odessa/3886/2010	M	L	P	S	T	I	Q	T	L	T
B/Brisbane/60/2008	M	L	P	S	T	I	Q	T	L	T
B/Paris/1762/2009	M	L	P	S	T	I	Q	T	L	T
B/Malta/MV636714/2011	M	L	P	S	T	I	Q	T	L	T
B/South Australia/81/2012	M	L	P	S	T	I	Q	T	L	T
B/Johannesburg/3964/2012	M	L	P	S	T	I	Q	T	L	T
B/Switzerland/16034436/2015	M	L	P	S	T	I	Q	T	L	T
B/Formosa/V2367/2012	M	L	P	S	T	I	Q	T	L	T
B/Shanghai-Pudongxin/11802/2015	M	L	P	S	T	I	Q	T	L	T
B/Thuringen/1/2016	M	L	P	S	T	I	Q	T	L	T
B/Netherlands/2914/2015	M	L	P	S	T	I	Q	T	L	T
B/Castilla La Mancha/1820/2015	M	L	P	S	T	I	Q	T	L	T
B/Israel/Q-501/2015	M	L	P	S	T	I	Q	T	L	T
B/England/345/2015	M	L	P	S	T	I	Q	T	L	T
B/Israel/Q-578/2015	M	L	P	S	T	I	Q	T	L	T
B/Israel/Q-503/2015	M	L	P	S	T	I	Q	T	L	T
B/Hong Kong/58/2016	M	L	P	S	T	I	Q	T	L	T
B/Rennes/2548/2015	M	L	P	S	T	I	Q	T	L	T
B/Schleswig-Holstein/1/2016	M	L	P	S	T	I	Q	T	L	T
B/Estonia/97415/2016	M	L	P	S	T	I	Q	T	L	T
B/Hong Kong/57/2016	M	L	P	S	T	I	Q	T	L	T
B/Yunnan-Mengzi/1977/2015	M	L	P	S	T	I	Q	T	L	T
B/Ireland/3154/2016	M	L	P	S	T	I	Q	T	L	T
B/Hong Kong/62/2016	M	L	P	S	T	I	Q	T	L	T
B/Belgium/201560769/2015	M	L	P	S	T	I	Q	T	L	T
B/Nordrhein-Westfalen/1/2016	M	L	P	S	T	I	Q	T	L	T
B/Ghana/FS-15-0612/2015	M	L	P	S	T	I	Q	T	L	T
B/Hong Kong/60/2016	M	L	P	S	T	I	Q	T	L	T
B/Iran/110767/2015	M	L	P	S	T	I	Q	T	L	T
B/Malaysia/2506/2004	H	R	F	G	E	T	K	G	N	S
B/Hong Kong/514/2009	H	R	F	G	E	T	K	G	N	S
B/Odessa/3886/2010	H	R	F	G	E	T	K	G	N	S
B/Brisbane/60/2008	H	R	F	G	E	T	K	G	N	S
B/Paris/1762/2009	H	R	F	G	E	T	K	G	N	S
B/Malta/MV636714/2011	H	R	F	G	E	T	K	G	N	S
B/South Australia/81/2012	H	R	F	G	E	T	K	G	N	S
B/Johannesburg/3964/2012	H	R	F	G	E	T	K	G	N	S
B/Switzerland/16034436/2015	H	R	F	G	E	T	K	G	N	S
B/Formosa/V2367/2012	H	R	F	G	E	T	K	G	N	S
B/Shanghai-Pudongxin/11802/2015	H	R	F	G	E	T	K	G	N	S
B/Thuringen/1/2016	H	R	F	G	E	T	K	G	N	S
B/Netherlands/2914/2015	H	R	F	G	E	T	K	G	N	S
B/Castilla La Mancha/1820/2015	H	R	F	G	E	T	K	G	N	S
B/Israel/Q-501/2015	H	R	F	G	E	T	K	G	N	S
B/England/345/2015	H	R	F	G	E	T	K	G	N	S
B/Israel/Q-578/2015	H	R	F	G	E	T	K	G	N	S
B/Israel/Q-503/2015	H	R	F	G	E	T	K	G	N	S
B/Hong Kong/58/2016	H	R	F	G	E	T	K	G	N	S
B/Rennes/2548/2015	H	R	F	G	E	T	K	G	N	S
B/Schleswig-Holstein/1/2016	H	R	F	G	E	T	K	G	N	S
B/Estonia/97415/2016	H	R	F	G	E	T	K	G	N	S
B/Hong Kong/57/2016	H	R	F	G	E	T	K	G	N	S
B/Yunnan-Mengzi/1977/2015	H	R	F	G	E	T	K	G	N	S
B/Ireland/3154/2016	H	R	F	G	E	T	K	G	N	S
B/Hong Kong/62/2016	H	R	F	G	E	T	K	G	N	S
B/Belgium/201560769/2015	H	R	F	G	E	T	K	G	N	S
B/Nordrhein-Westfalen/1/2016	H	R	F	G	E	T	K	G	N	S
B/Ghana/FS-15-0612/2015	H	R	F	G	E	T	K	G	N	S
B/Hong Kong/60/2016	H	R	F	G	E	T	K	G	N	S
B/Iran/110767/2015	H	R	F	G	E	T	K	G	N	S
B/Malaysia/2506/2004	L	L	K	K	Y	G	E	A	Y	T
B/Hong Kong/514/2009	L	L	K	K	Y	G	E	A	Y	T
B/Odessa/3886/2010	L	L	K	K	Y	G	E	A	Y	T
B/Brisbane/60/2008	L	L	K	K	Y	G	E	A	Y	T
B/Paris/1762/2009	L	L	K	K	Y	G	E	A	Y	T
B/Malta/MV636714/2011	L	L	K	K	Y	G	E	A	Y	T
B/South Australia/81/2012	L	L	K	K	Y	G	E	A	Y	T
B/Johannesburg/3964/2012	L	L	K	K	Y	G	E	A	Y	T
B/Switzerland/16034436/2015	L	L	K	K	Y	G	E	A	Y	T
B/Formosa/V2367/2012	L	L	K	K	Y	G	E	A	Y	T
B/Shanghai-Pudongxin/11802/2015	L	L	K	K	Y	G	E	A	Y	T
B/Thuringen/1/2016	L	L	K	K	Y	G	E	A	Y	T
B/Netherlands/2914/2015	L	L	K	K	Y	G	E	A	Y	T
B/Castilla La Mancha/1820/2015	L	L	K	K	Y	G	E	A	Y	T
B/Israel/Q-501/2015	L	L	K	K	Y	G	E	A	Y	T
B/England/345/2015	L	L	K	K	Y	G	E	A	Y	T
B/Israel/Q-578/2015	L	L	K	K	Y	G	E	A	Y	T
B/Israel/Q-503/2015	L	L	K	K	Y	G	E	A	Y	T
B/Hong Kong/58/2016	L	L	K	K	Y	G	E	A	Y	T
B/Rennes/2548/2015	L	L	K	K	Y	G	E	A	Y	T
B/Schleswig-Holstein/1/2016	L	L	K	K	Y	G	E	A	Y	T
B/Estonia/97415/2016	L	L	K	K	Y	G	E	A	Y	T
B/Hong Kong/57/2016	L	L	K	K	Y	G	E	A	Y	T
B/Yunnan-Mengzi/1977/2015	L	L	K	K	Y	G	E	A	Y	T
B/Ireland/3154/2016	L	L	K	K	Y	G	E	A	Y	T
B/Hong Kong/62/2016	L	L	K	K	Y	G	E	A	Y	T
B/Belgium/201560769/2015	L	L	K	K	Y	G	E	A	Y	T
B/Nordrhein-Westfalen/1/2016	L	L	K	K	Y	G	E	A	Y	T
B/Ghana/FS-15-0612/2015	L	L	K	K	Y	G	E	A	Y	T
B/Hong Kong/60/2016	L	L	K	K	Y	G	E	A	Y	T
B/Iran/110767/2015	L	L	K	K	Y	G	E	A	Y	T

[illegible]

Table 13. Summary of influenza B (Yamagata-lineage) viruses analysed, collected after 2015-08-31

MONTH	B Yamagata lineage				
Country	Number received	Number propagated¹	≤2 fold²	4 fold²	≥8 fold²
OCTOBER					
Germany	1	1	1		
Ghana	1	1	1		
Iran	1	1	1		
NOVEMBER					
Belgium	2	2	2		
Iran	2	2	1	1	
Norway	2	2	1	1	
DECEMBER					
Portugal	1	1	1		
Switzerland	2	2	2		
JANUARY					
Hong Kong	2	2	2		
Iceland	1	1	1		
	15	15	13	2	0
9 Countries			86.7%	13.3%	0.0%

1. Propagated to sufficient titre to perform HI assay

2. Reduction in HI titre with ferret antiserum raised against B/Phuket/3073/2013 (egg-grown vaccine virus)

Table 14-1. Antigenic analyses of influenza B viruses (Yamagata lineage) 2015-09-09

					Haemagglutination inhibition titre										
Viruses	Other information	Passage history	Collection date	Passage history	Post-infection ferret antisera										
					B/Phuket	B/Fl	B/Bris	B/Estonia	B/Mass	B/Mass	B/Wis	B/Stock	B/Phuket	B/Phuket	B/HK
					3073/13	4/06	3/07	55669/11	02/12	02/12	1/10	12/11	3073/13	3073/13	3417/14
					Egg	Egg	Egg	MDCK	MDCK	Egg	Egg	Egg	MDCK	Egg	Egg
					Ferret number	SH614 ^{1,3}	F1/10 ⁻¹	F38/14 ⁻²	F32/12 ⁻²	F15/13 ⁻²	F42/14 ⁻²	F10/13 ⁻²	F06/15 ⁻¹	F35/14 ⁻²	F36/14 ⁻²
Genetic Group	3	1	2	2	2	2	3	3	3	3	3				
REFERENCE VIRUSES															
B/Florida/4/2006	1	2006-12-15	E7/E1	2560	640	1280	80	160	1280	160	160	40	320	320	
B/Brisbane/3/2007	2	2007-09-03	E2/E3	1280	320	640	40	160	640	80	160	20	160	160	
B/Estonia/55669/2011	2	2011-03-14	MDCK2/MDCK3	1280	80	160	80	320	160	40	40	40	80	320	
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK4	5120	640	640	320	640	1280	640	160	320	640	640	
B/Massachusetts/02/2012	2	2012-03-13	E3/E3	1280	320	640	80	160	1280	160	160	20	320	320	
B/Wisconsin/1/2010	3	2010-02-20	E3/E3	2560	320	320	20	40	320	320	160	40	160	320	
B/Stockholm/12/2011	3	2011-03-28	E4/E1	2560	160	160	20	40	160	80	160	40	160	320	
B/Phuket/3073/2013	3	2013-11-21	MDCK2/MDCK2	5120	160	160	160	320	320	320	160	640	640	320	
B/Phuket/3073/2013	3	2013-11-21	E4/E2	2560	160	320	20	40	320	320	160	80	320	320	
B/Hong Kong/3417/2014	3	2014-06-04	E4/E1	1280	80	80	10	40	80	80	20	20	80	160	
TEST VIRUSES															
B/Paris/1553/2015	3	2015-03-15	MDCK1/MDCK1	5120	160	160	80	80	160	320	40	320	320	320	
B/Lorraine/1556/2015	3	2015-03-17	MDCK1/MDCK1	5120	160	160	80	80	160	320	80	320	160	320	
B/Norway/1564/2015	3	2015-03-17	MDCK1/MDCK1	5120	160	160	160	160	320	320	40	320	320	320	
B/Champagne Ardenne/1563/2015	3	2015-03-18	MDCK1/MDCK1	2560	160	160	160	160	160	320	80	320	320	320	
B/Guyane/1751/2015	3	2015-03-19	MDCK1/MDCK1	2560	160	160	80	80	160	160	80	160	160	320	
B/Franche Comte/1596/2015	3	2015-03-23	MDCK1/MDCK1	5120	320	320	160	160	320	320	160	320	320	320	
B/Antalya/1168/2015	3	2015-03-27	MDCK1/MDCK1	5120	80	160	80	160	80	160	80	160	160	320	
B/Istanbul/1141/2015	3	2015-03-27	MDCK1/MDCK1	5120	160	160	160	160	320	320	160	320	320	320	
B/Paris/1632/2015	3	2015-03-30	MDCK1/MDCK1	2560	160	160	80	80	160	160	80	160	160	320	
B/Bretagne/1659/2015	3	2015-03-31	MDCK1/MDCK1	2560	160	160	40	80	80	160	40	80	160	160	
B/Paris/1669/2015	3	2015-04-01	MDCK1/MDCK1	5120	320	320	160	160	320	320	80	160	320	320	
B/Bretagne/1668/2015	3	2015-04-04	MDCK1/MDCK1	5120	160	160	80	80	160	160	80	160	160	320	
B/Nord Pas de Calais/1681/2015	3	2015-04-07	MDCK1/MDCK1	5120	160	160	80	80	160	160	40	160	160	320	
B/Pays de Loire/1692/2015	3	2015-04-09	MDCK1/MDCK1	5120	160	160	80	160	160	320	80	160	320	320	
B/Brisbane/47/2015	3	2015-04-14	MDCK3/MDCK1	5120	160	160	160	320	320	160	40	640	320	320	
B/Victoria/503/2015	3	2015-04-25	MDCK1/MDCK1	2560	160	160	80	80	160	160	20	160	160	320	
B/Victoria/503/2015	3	2015-04-25	E2/E1	1280	40	20	20	40	20	40	<	40	80	160	
B/Norway/2343/2015	3	2015-05-02	MDCK1/MDCK1	5120	160	160	160	160	320	160	40	640	320	320	
B/Norway/2444/2015	3	2015-05-30	MDCK1/MDCK1	5120	160	160	160	160	160	160	20	320	320	320	
B/South Africa/R3183/2015	3	2015-06-10	E1/E1	640	<	20	<	10	20	40	<	10	20	80	
B/Norway/2538/2015	3	2015-06-22	MDCK1/MDCK1	5120	160	160	80	160	160	320	20	320	320	320	
B/Norway/2545/2015	3	2015-06-24	MDCK1/MDCK1	5120	160	160	80	160	160	160	40	320	320	320	
B/Norway/2552/2015	3	2015-06-25	MDCK1/MDCK1	5120	160	160	160	320	320	320	20	640	320	320	

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

¹ < = <40; ² < = <10; ³ hyperimmune sheep serum; ⁴ RDE serum pre-absorbed with TRBC

Vaccine
NH 2015-16
SH 2016[#]

[#] B/Yamagata-lineage virus recommended for use in quadrivalent vaccines

Sequences in phylogenetic trees

Table 14-2. Antigenic analyses of influenza B viruses (Yamagata lineage) 2015-09-15

Haemagglutination inhibition titre																
Viruses		Other information	Passage history	Collection date	Passage history	Post-infection ferret antisera										
						B/Phuket	B/FI	B/Bris	B/Estonia	B/Mass	B/Mass	B/Wis	B/Stock	B/Phuket	B/Phuket	B/HK
						3073/13	4/06	3/07	55669/11	02/12	02/12	1/10	12/11	3073/13	3073/13	3417/14
						Egg	Egg	Egg	MDCK	MDCK	Egg	Egg	Egg	MDCK	Egg	Egg
						SH614 ^{1,3}	F1/10 ⁻¹	F38/14 ⁻²	F32/12 ⁻²	F15/13 ⁻²	F42/14 ⁻²	F10/13 ⁻²	F06/15 ⁻¹	F35/14 ⁻²	F36/14 ⁻²	St Jude's F715/14 ^{-2,4}
Genetic Group						3	1	2	2	2	3	3	3	3	3	
REFERENCE VIRUSES																
B/Florida/4/2006	1	2006-12-15	E7/E1	1280	320	640	40	80	640	160	160	40	160	320		
B/Brisbane/3/2007	2	2007-09-03	E2/E3	1280	320	640	40	80	640	80	160	20	160	160		
B/Estonia/55669/2011	2	2011-03-14	MDCK2/MDCK3	1280	80	80	80	320	40	40	20	40	80	160		
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK3	1280	320	640	160	320	640	160	80	80	160	640		
B/Massachusetts/02/2012	2	2012-03-13	E3/E3	1280	160	320	40	160	320	80	80	20	80	160		
B/Wisconsin/1/2010	3	2010-02-20	E3/E3	2560	160	160	10	20	160	160	80	20	160	160		
B/Stockholm/12/2011	3	2011-03-28	E4/E1	1280	160	80	10	20	80	80	160	20	80	160		
B/Phuket/3073/2013	3	2013-11-21	MDCK2/MDCK2	5120	160	160	160	160	160	160	80	320	320	320		
B/Phuket/3073/2013	3	2013-11-21	E4/E2	2560	160	160	20	40	160	160	80	40	160	160		
B/Hong Kong/3417/2014	3	2014-06-04	E4/E1	1280	80	80	10	20	40	80	20	10	40	320		
TEST VIRUSES																
B/Iceland/28/2015	3	2015-02-05	MDCKx/MDCK1	1280	80	40	20	40	40	40	20	80	80	160		
B/Hungary/437/2015	3	2015-02-09	MDCK3/SIAT2	2560	160	80	40	80	80	160	40	160	160	160		
B/Iceland/33/2015	3	2015-02-09	MDCKx/MDCK1	1280	80	80	20	40	40	40	40	20	80	160		
B/Sapporo/2/2015	3	2015-02-16	MDCK2/MDCK1	2560	80	80	40	40	40	80	40	80	80	320		
B/Hyogo/3210/2015	3	2015-02-18	MDCK2/MDCK1	2560	80	80	40	40	40	80	40	80	80	160		
B/Iceland/42/2015	3	2015-03-03	MDCKx/MDCK1	2560	80	80	40	40	80	80	40	80	80	320		
B/Iceland/44/2015	3	2015-03-12	MDCKx/MDCK1	1280	80	40	20	40	40	40	40	80	80	160		
B/Iceland/47/2015	3	2015-03-16	MDCKx/MDCK1	2560	160	160	40	80	160	160	80	160	160	320		
B/Brisbane/47/2015	3	2015-04-14	MDCK3/SIAT2	2560	160	80	40	80	80	80	40	160	80	160		
B/Jordan/30324/2015	2	2015-05-09	MDCK1/MDCK1	1280	80	80	80	320	80	40	20	80	80	160		
B/South Africa/R3192/2015	3	2015-06-08	E1/E1	<	40	<	<	<	<	40	<	<	<	<		
B/Mozambique/IR1010/2015	3	2015-06-11	MDCK1	1280	80	40	10	20	40	40	20	20	40	160		
B/Jordan/30356/2015	3	2015-06-15	MDCK1/MDCK1	1280	40	40	10	20	20	40	20	40	40	160		
B/Jordan/20606/2015	3	2015-07-06	MDCK1/MDCK1	2560	160	80	40	80	160	160	40	160	160	320		

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

¹ < = <40; ² < = <10; ³ hyperimmune sheep serum; ⁴ RDE serum pre-absorbed with TRBC

[#] B/Yamagata-lineage virus recommended for use in quadrivalent vaccines

Sequences in phylogenetic trees

Vaccine
NH 2015-16
SH 2016[#]

Table 14-3. Antigenic analyses of influenza B viruses (Yamagata lineage) 2015-09-24, 2015-09-29, 2015-10-20

					Haemagglutination inhibition titre										
					Post-infection ferret antisera										
Viruses					B/Phuket	B/FI	B/Bris	B/Estonia	B/Mass	B/Mass	B/Wis	B/Stock	B/Phuket	B/Phuket	B/HK
	Other information		Collection	Passage	3073/13	4/06	3/07	55669/11	02/12	02/12	1/10	12/11	3073/13	3073/13	3417/14
	Passage history	date	history	Egg	Egg	Egg	MDCK	MDCK	Egg	Egg	Egg	Egg	MDCK	Egg	Egg
	Ferret number			SH614 ^{1,3}	F1/10 ¹	F38/14 ²	F32/12 ²	F15/13 ²	F42/14 ²	F10/13 ²	F06/15 ¹	F35/14 ²	F36/14 ²	St Jude's F715/14 ^{2,4}	
	Genetic Group				3	1	2	2	2	2	3	3	3	3	3
REFERENCE VIRUSES															
B/Florida/4/2006		1	2006-12-15	E7/E1	1280	320	640	40	80	640	160	160	40	160	160
B/Brisbane/3/2007		2	2007-09-03	E2/E3	1280	320	640	40	80	640	160	160	20	160	160
B/Estonia/55669/2011		2	2011-03-14	MDCK2/MDCK3	1280	80	80	80	320	80	40	<	40	80	160
B/Massachusetts/02/2012		2	2012-03-13	MDCK1/C2/MDCK3	1280	320	320	160	320	640	160	80	80	160	320
B/Massachusetts/02/2012		2	2012-03-13	E3/E3	1280	320	640	40	160	640	160	160	20	160	160
B/Wisconsin/1/2010		3	2010-02-20	E3/E3	2560	160	160	20	40	160	320	80	40	160	320
B/Stockholm/12/2011		3	2011-03-28	E4/E1	1280	160	80	10	40	160	80	160	40	80	160
B/Phuket/3073/2013		3	2013-11-21	MDCK2/MDCK2	5120	160	160	160	160	160	160	80	320	320	320
B/Phuket/3073/2013		3	2013-11-21	E4/E2	2560	160	160	20	40	160	160	160	40	160	320
B/Hong Kong/3417/2014		3	2014-06-04	E4/E1	1280	80	80	10	40	80	80	<	20	80	160
TEST VIRUSES															
B/California/12/2015		3	2015-02-04	C2 /MDCK1	2560	160	160	80	80	160	160	<	160	160	320
B/California/12/2015		3	2015-02-04	E4/E2	640	40	40	<	20	20	40	<	<	40	80
B/Laos/99/2015		2	2015-02-13	C1/C1/MDCK1	1280	80	80	160	320	160	80	<	80	80	160
B/Serbia/1753/2015		3	2015-03-10	C1/MDCK1	2560	80	80	40	40	80	80	<	160	160	160
B/Maryland/12/2015		3	2015-03-14	E3/E2	640	40	40	<	20	40	20	<	10	40	80
B/Serbia/2064/2015		3	2015-03-17	C2/MDCK1	2560	80	160	80	160	160	160	40	320	320	160
B/Norway/1859/2015		3	2015-03-28	MDCK1	2560	160	80	40	80	160	160	<	160	160	320
B/Iowa/23/2015		3	2015-04-06	C1 /MDCK1	2560	160	80	40	80	80	160	<	80	160	320
B/Serbia/2756/2015		3	2015-04-09	MDCK1	2560	80	80	20	40	80	80	<	80	80	160
B/Norway/1921/2015		3	2015-04-13	MDCK1	2560	160	80	40	80	160	160	<	160	160	320
B/Jordan/20365/2015		3	2015-04-13	MDCK1/MDCK1	2560	160	80	40	80	80	160	<	160	160	160
B/Jordan/30236/2015		3	2015-04-21	MDCK1/MDCK1	2560	160	80	40	80	80	160	<	160	160	160
B/Jordan/30264/2015		3	2015-04-21	MDCK1/MDCK1	2560	80	80	40	80	80	160	<	80	160	160
B/Iceland/6544/2015		3	2015-05-05	MDCK1	2560	80	80	40	40	80	160	40	160	160	160
B/Zambia/04-00173/2015		3	2015-05-27	MDCK2/MDCK1	1280	40	80	40	80	40	80	40	80	160	160
B/Mozambique/IR981/2015		3	2015-06-08	MDCK2	2560	80	80	40	80	80	80	<	80	160	160
B/Mozambique/IR1062/2015		3	2015-06-23	MDCK2	2560	80	80	20	40	80	80	<	80	80	160
B/Zambia/04-00314/2015		3	2015-07-23	MDCK1	1280	40	40	20	40	40	80	<	80	40	80

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

¹ < = <40; ² < = <10; ³ hyperimmune sheep serum; ⁴ RDE serum pre-absorbed with TRBC

[#] B/Yamagata-lineage virus recommended for use in quadrivalent vaccines

Sequences in phylogenetic trees

Vaccine
NH 2015-16
SH 2016[#]

Table 14-4. Antigenic analyses of influenza B viruses (Yamagata lineage) 2015-11-25, 2015-12-08, 2015-12-15

Viruses	Other information	Passage history	Collection date	Passage history	Haemagglutination inhibition titre										
					Post-infection ferret antisera										
					B/Phuket	B/FI	B/Bris	B/Estonia	B/Mass	B/Mass	B/Wis	B/Stock	B/Phuket	B/Phuket	B/HK
					3073/13	4/06	3/07	55669/11	02/12	02/12	1/10	12/11	3073/13	3073/13	3417/14
					Egg	Egg	Egg	MDCK	MDCK	Egg	Egg	Egg	MDCK	Egg	Egg
	Ferret number				SH614 ^{1,3}	F1/10 ¹	F38/14 ²	F32/12 ²	F15/13 ²	F42/14 ²	F10/13 ²	F06/15 ¹	F35/14 ²	F36/14 ²	St Jude's F715/14 ^{2,4}
	Genetic Group				3	1	2	2	2	2	3	3	3	3	3
REFERENCE VIRUSES															
B/Florida/4/2006		1	2006-12-15	E7/E1	1280	320	640	80	160	640	160	160	20	160	320
B/Brisbane/3/2007		2	2007-09-03	E2/E3	1280	320	320	40	80	640	80	80	20	160	160
B/Estonia/55669/2011		2	2011-03-14	MDCK2/MDCK3	1280	80	80	40	80	80	40	20	40	80	160
B/Massachusetts/02/2012		2	2012-03-13	MDCK1/C2/MDCK3	2560	640	640	160	320	640	320	80	80	320	640
B/Massachusetts/02/2012		2	2012-03-13	E3/E4	2560	640	1280	80	160	1280	320	160	40	320	320
B/Wisconsin/1/2010		3	2010-02-20	E3/E3	2560	160	160	10	40	160	160	80	20	160	160
B/Stockholm/12/2011		3	2011-03-28	E4/E1	1280	160	80	10	20	80	80	80	20	80	160
B/Phuket/3073/2013		3	2013-11-21	MDCK2/MDCK2	5120	160	320	160	160	320	320	80	640	640	320
B/Phuket/3073/2013		3	2013-11-21	E4/E2	2560	160	160	20	40	320	320	80	40	160	160
B/Hong Kong/3417/2014		3	2014-06-04	E4/E1	1280	80	80	10	40	80	80	20	20	80	160
TEST VIRUSES															
B/Moscow/5/2015		3	Unknown	C1/MDCK1	2560	80	80	320	320	320	160	160	640	640	160
B/Berlin/50/2015		3	2015-10-12	C2/MDCK1	1280	<	40	10	40	40	40	<	80	80	80
B/Norway/2692/2015		3	2015-11-12	MDCK1	2560	80	80	20	40	80	80	40	80	80	160

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

¹ < = <40; ² < = <10; ³ hyperimmune sheep serum; ⁴ RDE serum pre-absorbed with TRBC

Vaccine
NH 2015-16
SH 2016^y

^y B/Yamagata-lineage virus recommended for use in quadravalent vaccines

Sequences in phylogenetic trees

Table 14-5. Antigenic analyses of influenza B viruses (Yamagata lineage) 2015-01-12, 2015-01-19

Viruses	Haemagglutination inhibition titre													
	Post-infection ferret antisera													
	Passage history	Collection date	Passage History	B/Phuket ^{1,3}	B/FI ¹	B/Bris ²	B/Estonia ²	B/Mass ²	B/Mass ²	B/Wis ²	B/Stock ¹	B/Phuket ²	B/Phuket ²	B/HK ⁴
				3073/13	4/06	3/07	55669/11	02/12	02/12	1/10	12/11	3073/13	3073/13	3417/14
				Egg	Egg	Egg	MDCK	MDCK	Egg	Egg	Egg	MDCK	Egg	Egg
				SH614	F1/10	F38/14	F32/12	F05/15	F42/14	F10/13	F06/15	F35/14	F36/14	St Jude's F715/14
Genetic Group				3	1	2	2	2	2	3	3	3	3	3
REFERENCE VIRUSES														
B/Florida/4/2006	1	2006-12-15	E7/E1	2560	640	640	80	160	1280	160	320	40	320	320
B/Brisbane/3/2007	2	2007-09-03	E2/E3	640	160	320	20	40	320	40	80	10	80	80
B/Estonia/55669/2011	2	2011-03-14	MDCK2/MDCK3	640	80	80	80	160	40	40	20	40	40	160
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK3	1280	320	320	80	320	640	160	80	40	160	320
B/Massachusetts/02/2012	2	2012-03-13	E3/E3	640	160	320	20	80	640	80	80	10	80	80
B/Wisconsin/1/2010	3	2010-02-20	E3/E3	1280	80	80	10	<	80	80	80	20	160	80
B/Stockholm/12/2011	3	2011-03-28	E4/E1	1280	80	80	10	<	80	40	80	20	80	80
B/Phuket/3073/2013	3	2013-11-21	MDCK2/MDCK2	2560	160	160	80	160	320	160	80	320	320	160
B/Phuket/3073/2013	3	2013-11-21	E4/E2	2560	160	160	10	<	160	160	80	20	160	160
B/Hong Kong/3417/2014	3	2014-06-04	E4/E1	640	80	40	<	<	40	40	20	10	40	160
TEST VIRUSES														
B/Egypt/4125/2015	3	2014-06-17	MDCKx/MDCK1	1280	160	80	10	<	80	160	40	20	80	160
B/Egypt/3802/2014	3	2014-12-20	MDCKx/MDCK1	1280	80	80	10	20	80	80	40	20	80	160
B/Egypt/4500/2015	3	2014-12-30	MDCKx/MDCK1	2560	80	80	80	160	160	160	80	160	160	160
B/Egypt/3716/2015	3	2015-01-19	MDCKx/MDCK1	1280	80	80	20	20	80	80	40	20	80	160
B/Iran/91653/2015	3	2015-01-22	SIAT1/MDCK1	1280	40	40	20	40	40	80	40	80	80	160
B/Egypt/3722/2015	3	2015-01-22	MDCKx/MDCK1	1280	160	80	10	<	80	160	40	10	80	160
B/Egypt/4290/2015	2	2015-01-29	MDCKx/MDCK1	320	160	160	40	80	160	40	40	<	80	160
B/Egypt/3743/2015	3	2015-02-11	MDCKx/MDCK1	2560	320	160	40	40	320	160	80	40	320	160
B/Egypt/4417/2015	3	2015-02-11	MDCKx/MDCK1	1280	80	80	10	<	80	80	40	10	80	160
B/Egypt/4563/2015	2	2015-02-21	MDCKx/MDCK1	640	320	320	80	160	320	80	40	10	80	320
B/Egypt/4959/2015	3	2015-03-14	MDCKx/MDCK1	1280	160	80	10	<	80	80	40	10	80	160
B/Ghana/FS-15-0612/2015	3	2015-10-09	Cx/MDCK1	2560	40	80	80	80	80	80	40	160	160	160
B/Iran/110767/2015	3	2015-10-19	SIAT1/MDCK1	2560	80	80	80	80	160	160	80	160	160	160
B/Iran/115032/2015	3	2015-11-02	SIAT1/MDCK1	2560	80	80	80	80	160	160	40	160	160	160
B/Iran/115032/2015	3	2015-11-02	E1/E1	640	40	40	<	<	40	40	20	10	40	80
B/Belgium/2015G0771/2015	3	2015-11-18	SIAT1/MDCK1	1280	80	80	20	40	40	80	40	80	80	160
B/Belgium/2015G0775 /2015	3	2015-11-20	SIAT1/MDCK1	1280	80	40	20	20	40	40	20	40	80	160

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

¹ < = <40; ² < = <10; ³ hyperimmune sheep serum; ⁴ RDE serum pre-absorbed with TRBCVaccine
NH 2015-16
SH 2016[#][#] B/Yamagata-lineage virus recommended for use in quadrivalent vaccines

Sequences in phylogenetic trees

Table 14-6. Antigenic analyses of influenza B viruses (Yamagata lineage) 2016-01-26, 2016-02-10

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre										
				Post-infection ferret antisera										
				B/Phuket	B/FI	B/Bris	B/Estonia	B/Mass	B/Mass	B/Wis	B/Stock	B/Phuket	B/Phuket	B/HK
				3073/13	4/06	3/07	55669/11	02/12	02/12	1/10	12/11	3073/13	3073/13	3417/14
	Passage history			Egg	Egg	Egg	MDCK	MDCK	Egg	Egg	Egg	MDCK	Egg	Egg
	Ferret number			SH614 ^{1,3}	F1/10 ¹	F38/14 ²	F32/12 ²	F05/15 ²	F42/14 ²	F10/13 ²	F06/15 ¹	F35/14 ²	F36/14 ²	St Jude's F715/14 ^{2,4}
	Genetic Group			3	1	2	2	2	2	3	3	3	3	3
REFERENCE VIRUSES														
B/Florida/4/2006	1	2006-12-15	E7	1280	640	640	80	160	1280	160	320	40	320	320
B/Brisbane/3/2007	2	2007-09-03	E2/E3	640	160	160	20	40	320	40	80	10	80	80
B/Estonia/55669/2011	2	2011-03-14	MDCK2/MDCK3	640	80	80	80	160	80	40	20	40	80	160
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK3	1280	320	320	80	320	640	160	80	40	160	320
B/Massachusetts/02/2012	2	2012-03-13	E3/E3	640	160	320	20	40	320	80	80	10	80	80
B/Wisconsin/1/2010	3	2010-02-20	E3/E3	1280	80	80	10	<	80	80	80	20	80	80
B/Stockholm/12/2011	3	2011-03-28	E4/E1	1280	160	80	10	<	160	80	80	20	80	160
B/Phuket/3073/2013	3	2013-11-21	MDCK2/MDCK2	5120	160	160	80	160	320	160	80	640	320	160
B/Phuket/3073/2013	3	2013-11-21	E4/E3	2560	320	160	20	20	320	320	160	40	320	160
B/Hong Kong/3417/2014	3	2014-06-04	E4/E1	640	80	40	10	<	40	40	20	10	40	160
TEST VIRUSES														
B/Norway/2760/2015	3	2015-11-23	MDCK1	2560	80	80	40	<	80	80	<	80	80	160
B/Lisboa/151/2015	ngs?	2015-12-17	SIAT1/MDCK1	5120	80	160	160	160	160	160	<	320	320	160
B/Switzerland/15976242/2015	3	2015-12-22	MDCK1	2560	80	80	40	40	160	160	40	160	160	160
B/Switzerland/16034257/2015	3	2015-12-30	MDCK1	2560	80	80	40	40	80	160	40	160	160	160
B/Hong Kong/61/2016	3	2016-01-03	MDCK1/MDCK1	2560	80	80	80	80	160	160	80	160	320	160
B/Hong Kong/60/2016	3	2016-01-04	MDCK1/MDCK1	2560	80	80	40	40	160	160	40	160	160	160
B/Iceland/3/2016	3	2016-01-11	MDCK0/MDCK1	2560	80	80	40	40	160	160	40	160	160	160

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

¹ < = <40; ² < = <10; ³ hyperimmune sheep serum; ⁴ RDE serum pre-absorbed with TRBC

Vaccine
NH 2015-16
SH 2016[#]

[#] B/Yamagata-lineage virus recommended for use in quadravalent vaccines

Sequences in phylogenetic trees

Table 14-7. Summary of HI reactivity of influenza B/Yamagata lineage Test viruses

HI Table	Fold reduction compared to homologous titre	Antiserum to Virus Host Genetic group	No of test viruses	Number of test viruses reacting within 2-fold and 4-fold of the antiserum homologous titre											No of test viruses
				All test viruses					Test viruses collected after 2015-08-31						
				B/Mass	B/Wis	B/Phuket	B/Phuket	B/HK	B/Mass	B/Wis	B/Phuket	B/Phuket	B/HK		
				02/12	1/10	3073/13	3073/13	3417/14	02/12	1/10	3073/13	3073/13	3417/14		
				Egg	Egg	Cell	Egg	Egg	Egg	Egg	Egg	Egg			
				2	3	3	3	3	2	3	3	3	3		
14-1	2-fold		23	0	21	12	21	23							
	4-fold			7	21	20	22	23							
14-2	2-fold		14	2	7	4	11	13							
	4-fold			6	14	10	13	13							
14-3	2-fold		18	0	9	8	15	18							
	4-fold			5	16	16	18	18							
14-4	2-fold		3	0	2	1	3	3	0	1	0	2	2	2	
	4-fold			1	3	1	3	3	0	2	0	2	2		
14-5	2-fold		17	2	17	4	16	17	0	6	3	5	6	6	
	4-fold			6	17	6	17	17	2	6	4	6	6		
14-6	2-fold		7	5	7	1	6	7	5	7	1	6	7	7	
	4-fold			7	7	6	7	7	7	7	6	7	7		
Total No			82											15	
Number reacting within 2-fold				9	63	30	72	81	5	14	4	13	15		
Number reacting within 4-fold				32	78	59	80	81	9	15	10	15	15		
Percentage reacting within 2-fold				10.98	76.83	36.59	87.80	98.78	33.33	93.33	26.67	86.67	100.00		
Percentage reacting within 4-fold				39.02	95.12	71.95	97.56	98.78	60.00	100.00	66.67	100.00	100.00		

For use in quadrivalent vaccines

Vaccine
NH2015-16
SH2016[#]

Vaccine
NH2015-16
SH2016[#]

Results for antisera raised against five vaccine/reference viruses are shown

Figure 17. Phylogenetic comparison of influenza B (Yamagata-lineage) HA genes

Vaccine virus
Reference viruses

Collection date
Sep/Oct 2015
Nov 2015
Dec 2015
Jan 2016

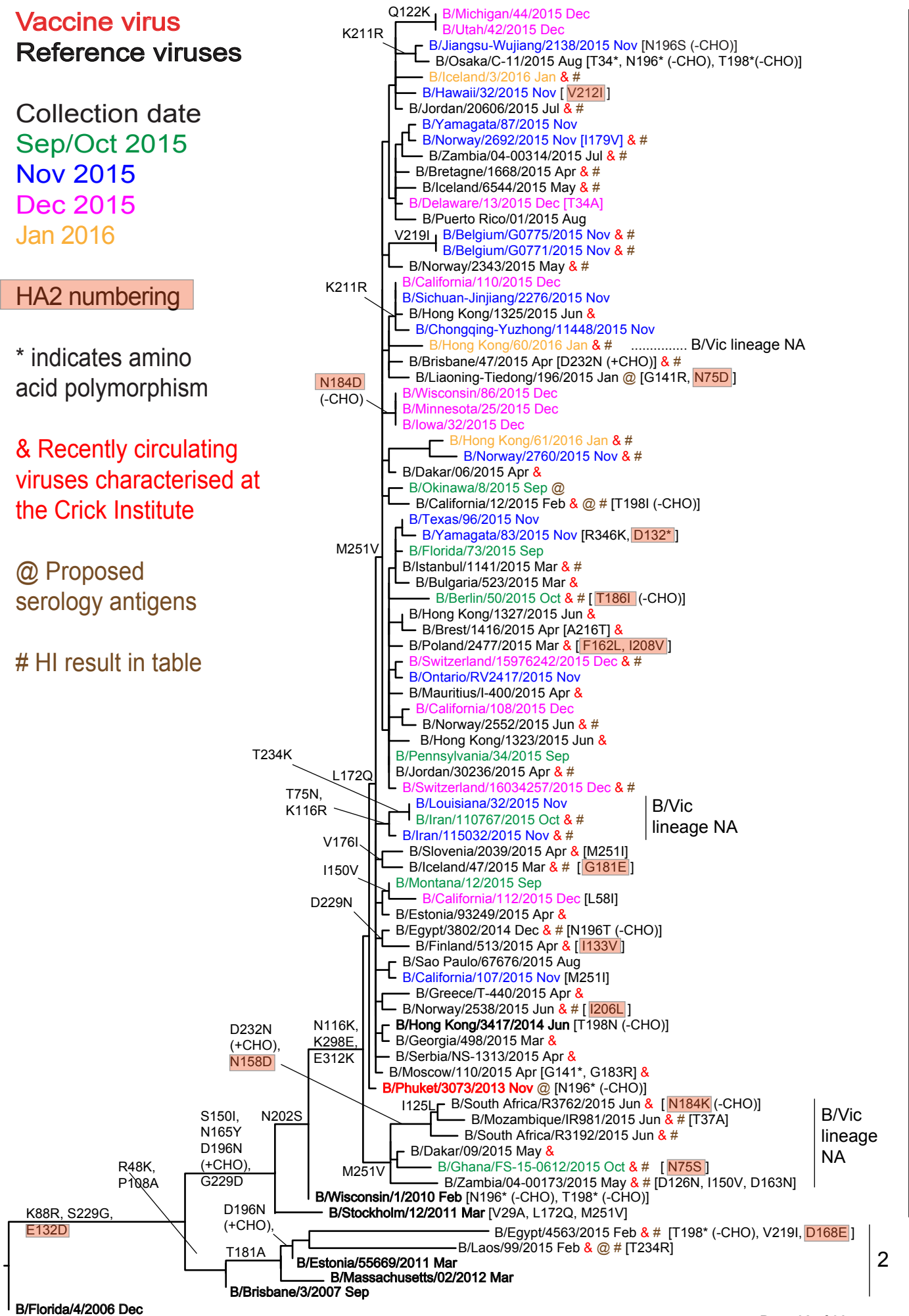
HA2 numbering

* indicates amino acid polymorphism

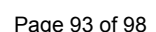
& Recently circulating viruses characterised at the Crick Institute

@ Proposed serology antigens

HI result in table



Vaccine virus: **Reference virus:** **Genetic group:** **HA1/HA2 boundary:** **Potential N-linked glycosylation site**



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Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

Figure 19. Phylogenetic comparison of influenza B (Yamagata-lineage) NA genes

Vaccine virus
Reference viruses

Collection date
Sep/Oct 2015
Nov 2015
Dec 2015
Jan 2016

* indicates amino
acid polymorphism

& Recently circulating
viruses characterised at
the Crick Institute

@ Proposed
serology antigens

HI result in table

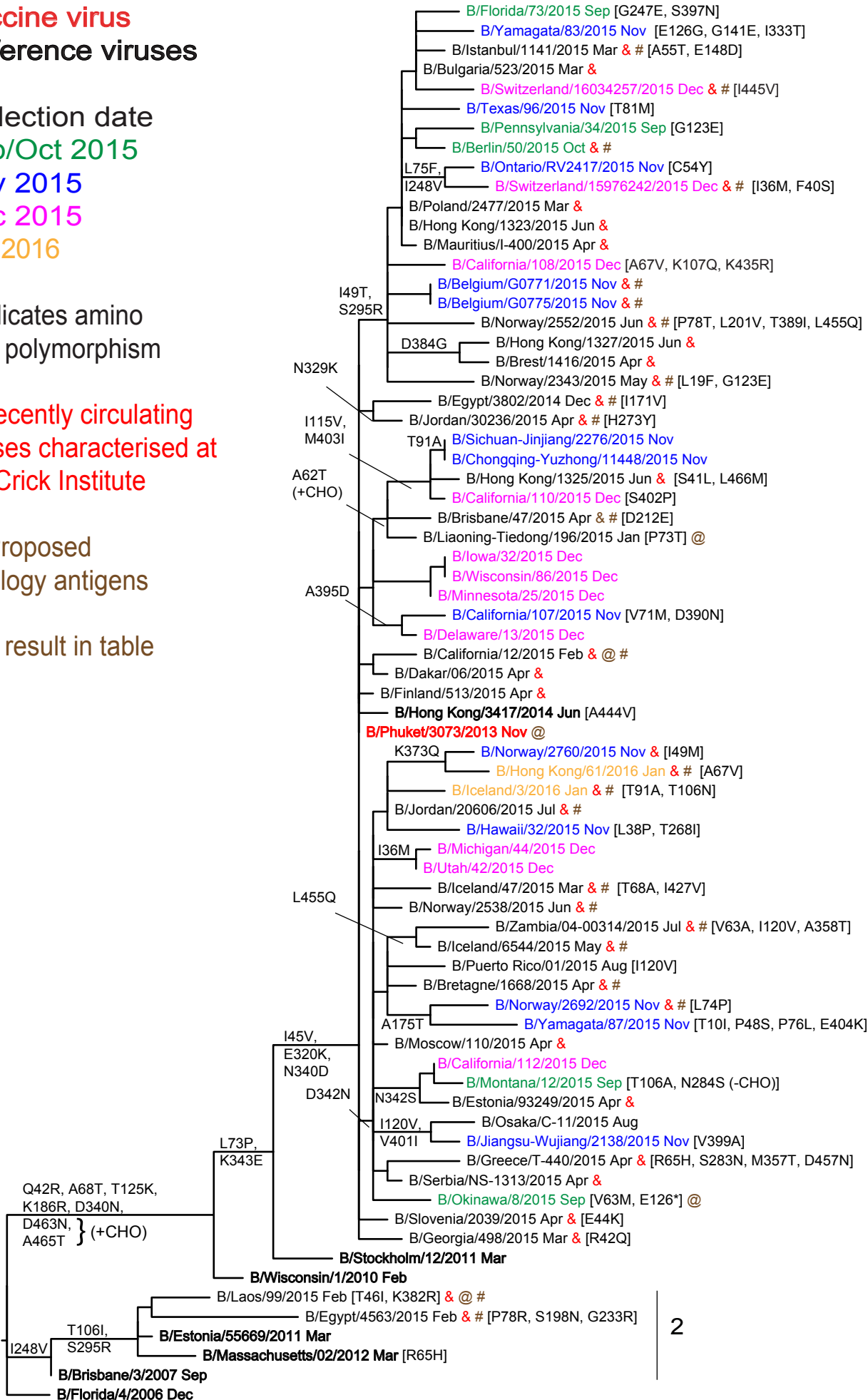


Figure 20. NA protein alignment for B (Yamagata-lineage) viruses

Vaccine virus: Reference virus: Genetic group: Potential N-linked glycosylation site

	10	20	30	40	50	60	70	80	90	100		
B/Florida/4/2006	M	L	P	S	T	I	Q	T	L	T	L	1
B/Brisbane/3/2007												2
B/Massachusetts/02/2012												2
B/Estonia/55669/2011												2
B/Egypt/4563/2015												2
B/Stockholm/12/2011												3
B/Wisconsin/1/2010												3
B/Phuket/3073/2013												3
B/Hong Kong/3417/2014												3
B/California/107/2015												3
B/California/112/2015												3
B/Montana/12/2015												3
B/Switzerland/16034257/2015												3
B/California/108/2015												3
B/Switzerland/15976242/2015												3
B/Berlin/50/2015 Oct												3
B/Yamagata/83/2015												3
B/Norway/2760/2015												3
B/Hong Kong/61/2016												3
B/Minnesota/25/2015												3
B/Hong Kong/1325/2015												3
B/California/110/2015												3
B/Belgium/60775/2015												3
B/Delaware/13/2015												3
B/Norway/2692/2015												3
B/Yamagata/87/2015												3
B/Hawaii/32/2015												3
B/Iceland/3/2016												3
B/Jiangsu-Wujiang/2138/2015												3
B/Michigan/44/2015												3

	110	120	130	140	150	160	170	180	190	200		
B/Florida/4/2006	H	R	F	G	E	T	K	G	N	S	A	2
B/Brisbane/3/2007												2
B/Massachusetts/02/2012												2
B/Estonia/55669/2011												2
B/Egypt/4563/2015												2
B/Stockholm/12/2011												3
B/Wisconsin/1/2010												3
B/Phuket/3073/2013												3
B/Hong Kong/3417/2014												3
B/California/107/2015												3
B/California/112/2015												3
B/Montana/12/2015												3
B/Switzerland/16034257/2015												3
B/California/108/2015												3
B/Switzerland/15976242/2015												3
B/Berlin/50/2015 Oct												3
B/Yamagata/83/2015												3
B/Norway/2760/2015												3
B/Hong Kong/61/2016												3
B/Minnesota/25/2015												3
B/Hong Kong/1325/2015												3
B/California/110/2015												3
B/Belgium/60775/2015												3
B/Delaware/13/2015												3
B/Norway/2692/2015												3
B/Yamagata/87/2015												3
B/Hawaii/32/2015												3
B/Iceland/3/2016												3
B/Jiangsu-Wujiang/2138/2015												3
B/Michigan/44/2015												3

	210	220	230	240	250	260	270	280	290	300	
B/Florida/4/2006	L	L	K	K	Y	G	E	A	Y	T	2
B/Brisbane/3/2007											2
B/Massachusetts/02/2012											2
B/Estonia/55669/2011											2
B/Egypt/4563/2015											2
B/Stockholm/12/2011											3
B/Wisconsin/1/2010											3
B/Phuket/3073/2013											3
B/Hong Kong/3417/2014											3
B/California/107/2015											3
B/California/112/2015											3
B/Montana/12/2015											3
B/Switzerland/16034257/2015											3
B/California/108/2015											3
B/Switzerland/15976242/2015											3
B/Berlin/50/2015 Oct											3
B/Yamagata/83/2015											3
B/Norway/2760/2015											3
B/Hong Kong/61/2016											3
B/Minnesota/25/2015											3
B/Hong Kong/1325/2015											3
B/California/110/2015											3
B/Belgium/60775/2015											3
B/Delaware/13/2015											3
B/Norway/2692/2015											3
B/Yamagata/87/2015											3
B/Hawaii/32/2015											3
B/Iceland/3/2016											3
B/Jiangsu-Wujiang/2138/2015											3
B/Michigan/44/2015											3

	310	320	330	340	350	360	370	380	390	400
B/Florida/4/2006	P	F	V	K	L	N	V	E	T	D
B/Brisbane/3/2007	A	E	I	R	L	M	C	T	E	T
B/Massachusetts/02/2012	Y	L	D	T	P	R	P	N	D	G
B/Estonia/55669/2011	S	I	T	G	P	C	E	S	D	G
B/Egypt/4563/2015	K	G	S	G	G	I	K	G	G	F
B/Stockholm/12/2011	V	H	Q	R	M	A	S	K	I	G
B/Wisconsin/1/2010	R	W	S	R	T	M	S	K	T	K
B/Phuket/3073/2013	R	M	G	M	G	L	Y	K	Y	D
B/Hong Kong/3417/2014	G	P	W	T	D	S	E	A	L	S
B/California/107/2015	G	V	M	.	.	.	.	.	.	.
B/California/112/2015	.	.	.	.	.	.	.	.	.	.
B/Montana/12/2015	.	.	.	.	.	.	.	.	.	.
B/Switzerland/16034257/2015	.	.	.	.	.	.	.	.	.	.
B/California/108/2015	.	.	.	.	.	.	.	.	.	.
B/Switzerland/15976242/2015	.	.	.	.	.	.	.	.	.	.
B/Berlin/50/2015 Oct	.	.	.	.	.	.	.	.	.	.
B/Yamagata/83/2015	.	.	.	.	.	.	.	.	.	.
B/Norway/2760/2015	.	.	.	.	.	.	.	.	.	.
B/Hong Kong/61/2016	.	.	.	.	.	.	.	.	.	.
B/Minnesota/25/2015	.	.	.	.	.	.	.	.	.	.
B/Hong Kong/1325/2015	.	.	.	.	.	.	.	.	.	.
B/California/110/2015	.	.	.	.	.	.	.	.	.	.
B/Belgium/G0775/2015	.	.	.	.	.	.	.	.	.	.
B/Delaware/13/2015	.	.	.	.	.	.	.	.	.	.
B/Norway/2692/2015	.	.	.	.	.	.	.	.	.	.
B/Yamagata/87/2015	.	.	.	.	.	.	.	.	.	.
B/Hawaii/32/2015	.	.	.	.	.	.	.	.	.	.
B/Iceland/3/2016	.	.	.	.	.	.	.	.	.	.
B/Jiangsu-Wujiang/2138/2015	.	.	.	.	.	.	.	.	.	.
B/Michigan/44/2015	.	.	.	.	.	.	.	.	.	.

	410	420	430	440	450	460
B/Florida/4/2006	V	S	M	E	E	P
B/Brisbane/3/2007	G	W	S	F	G	F
B/Massachusetts/02/2012	E	I	K	D	K	K
B/Estonia/55669/2011	C	D	V	P	C	I
B/Egypt/4563/2015	G	E	M	V	H	D
B/Stockholm/12/2011	G	G	K	T	T	W
B/Wisconsin/1/2010	H	S	A	A	T	Y
B/Phuket/3073/2013	C	L	M	G	S	Q
B/Hong Kong/3417/2014	L	L	W	D	T	V
B/California/107/2015	G	V	D	M	A	L
B/California/112/2015	.	.	.	.	.	.
B/Montana/12/2015	.	.	.	.	.	.
B/Switzerland/16034257/2015	.	.	.	.	.	.
B/California/108/2015	.	.	.	.	.	.
B/Switzerland/15976242/2015	.	.	.	.	.	.
B/Berlin/50/2015 Oct	.	.	.	.	.	.
B/Yamagata/83/2015	.	.	.	.	.	.
B/Norway/2760/2015	.	.	.	.	.	.
B/Hong Kong/61/2016	.	.	.	.	.	.
B/Minnesota/25/2015	.	.	.	.	.	.
B/Hong Kong/1325/2015	.	.	.	.	.	.
B/California/110/2015	.	.	.	.	.	.
B/Belgium/G0775/2015	.	.	.	.	.	.
B/Delaware/13/2015	.	.	.	.	.	.
B/Norway/2692/2015	.	.	.	.	.	.
B/Yamagata/87/2015	.	.	.	.	.	.
B/Hawaii/32/2015	.	.	.	.	.	.
B/Iceland/3/2016	.	.	.	.	.	.
B/Jiangsu-Wujiang/2138/2015	.	.	.	.	.	.
B/Michigan/44/2015	.	.	.	.	.	.

Table 15. Antiviral (NAI) testing of isolates with collection dates in 2015 and 2016

Month of Collection	Number of viruses tested ¹						
	A(H1N1)pdm09		A(H3N2)		Influenza B		Virus type/subtype
	NI	RI/HRI	NI	RI/HRI	NI	RI/HRI	Phenotype ²
2015							
January	86 (6)		205 (6)		81 (4)	1 ^b	
February	78 (10)		161 (14)		131 (5)	1 ^c	
March	84 (6)		60 (7)		100 (8)		
April	65 (32)		18 (3)		41 (5)		
May	20 (9)	1 ^a	15 (5)		8 (3)		
June	23 (12)		33 (0)		17 (5)		
July	2 (2)		8 (1)		2 (2)		
August	1 (1)		1 (10)		0		
Totals	359 (78)	1	501 (46)		380 (32)	2	1243 (156)
September	14 (14)		9 (9)				
October	18 (18)		3 (3)		4 (4)		
November	50 (50)		12 (12)		12 (12)		
December	101 (101)		9 (9)		21 (21)		
2016							
January	39 (39)		1 (1)		7 (7)		
Totals	222 (222)		34 (34)		44 (44)		300 (300)
Overall Totals	581 (300)	1	535 (80)		424 (76)	2	1543 (456)

¹ Numbers in parentheses indicate the number of viruses assayed since the September 2015 VCM

² Phenotype: NI (normal inhibition), RI (reduced inhibition), HRI (highly reduced inhibition) as defined in WHO Wkly. Epidemiol. Rec. (2012) 87, 369–374

Viruses were subjected to phenotypic testing against oseltamivir and zanamivir

a A(H1N1)pdm09 virus (A/Norway/2227/2015) carries H275Y

Not tested

b B/Yam virus (B/Ukraine/6295/2015) carries D197N

Fold difference = os 6 zan 4

c B/Yam virus (B/Extremadura/728/2015) carries D197G

Fold difference = os 4 zan 2

Outlier